SWAFL Desktop 1.1.0 — User Guide and Technical Reference

Volume 1 of 3: installing, every window and control, every layer and design, the weather, the virtual farm, the crops, the reports, and the governing equations behind every output

Naziru Halilu

Version 1.1.0

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About this manual

This is the manual of **SWAFL Desktop 1.1.0**, the Soil–Water–Atmosphere–Field–Loop decision-support workbench of the repository *Coupled Water Fluxes, Pore-Scale Dynamics, Precision Irrigation and Mulching Feedback Loops*. It replaces nothing: the manual of version 1.0 stays in the repository (`docs/SWAFL_Manual.pdf`) for readers of that version. Every picture in this manual was taken from version 1.1.0.

The three volumes.

Volume	Contents
1. User guide and technical reference (this volume)	Installing; the window, menus and every control; the clock; the two computing engines; every layer and the three irrigation designs with their equations; the dashboard; the weather data; the virtual farm; the crop library and the crop water requirement; the seasonal data; reports; workflows; validity limits; appendices with the symbols, the provenance ledger and the complete reference of the 134 numbered equations.
2. Screenshot atlas at the default settings	Every screen of the program: the window, the eight menus, every dialog, the Properties panel, the dashboard tabs, and every layer drawn from top to bottom, cut into page-size parts, each followed by the full text of every equation that the layer draws on.
3. Screenshot atlas by soil and by crop	Every layer drawn for each of the ten soils, the three irrigation designs and their virtual farms for each soil, and the crop water requirement for each crop of the library.

How this volume is organised.

1. Chapters 1–4 install the program and explain the window, every control and the clock.
2. Chapter 5 explains the two computing engines under every layer.
3. Chapters 6–10 document the layers of the Layers panel; chapters 11–13 the three irrigation designs.
4. Chapters 14–17 cover the dashboard, worked workflows, files and troubleshooting, and the validity limits.
5. Chapters 18–24 cover what is new in version 1.1: running a design and finding the most efficient inputs, weather data, the virtual farm, the crop library and the crop water requirement, the seasonal data, reports, and a list of every change from version 1.0.
6. The appendices give the symbols, the control values, the data sets, the provenance ledger, the references and the equation reference.

A note on honesty of the equations. SWAFL mixes three kinds of computation, and the manual labels them so that nothing is over-claimed. Every numbered equation carries its label directly beneath it:

Label	Meaning
[Literature]	A published relation implemented as written, cited to its source.
[Data-driven]	Obtained by interpolating or scaling the 10,190 Daisy simulations that stand behind the program, not computed from a closed-form law.

Label	Meaning
[Engineered]	A parameterisation chosen for the workbench (for drawing, for speed, or because no closed form exists), stated with its constants so that it can be audited, but not presented as a physical law.

Where the simulation model is named. The program itself shows results only. The simulation model behind them (Daisy, 10,190 runs in seventeen experiments on ten soils) is described here, in Chapter 5 and in the appendices, and not on the screens.

Units. Unless stated, depths are in mm, lengths in m, suction in cm of water (1 cm $\approx$ 0.0980665 kPa), flows in L s^{-1} or L h^{-1} as marked on the screen, times in min or h, and areas in ha.

Chapter 1

Installing and starting SWAFL

1.1 What you receive

SWAFL Desktop is a self-contained Electron program. The simulation engine is one HTML page (`app/engine/engine.html`) with a few script files beside it, which the program serves to itself on a local port, so **no internet connection is needed once it is installed**. The internet is used only if you ask for the online weather of a location (Chapter 19). Three ways of running it are provided.

Route	Who it suits	Steps
A. Build and install on Windows	Most users	Unzip `SWAFL_Desktop.zip`, double-click `BUILD_AND_INSTALL_SWAFL.bat`.
B. Windows program	Users who received `SWAFL.exe`	Double-click `SWAFL.exe`.
C. From source	Developers	`npm install`, then `npm start` (desktop) or `npm run web` (browser).

1.2 Route A — build and install, step by step

1. Install **Node.js** (the LTS version) from nodejs.org if it is not already installed. Close and reopen any terminal afterwards.
2. Unzip `SWAFL_Desktop.zip` to a short path outside cloud-synchronised folders, for example `C:\SWAFL_Desktop`. The zip contains a folder `SWAFL_Desktop`; the folder you work in is the one that contains `package.json`.
3. Double-click **`BUILD_AND_INSTALL_SWAFL.bat`**. It installs the packages, builds the program and puts **SWAFL** on the Desktop. The first build downloads Electron and takes a few minutes.
4. Start **SWAFL** from the Desktop. The title bar reads *SWAFL 1.1.0*. If Windows SmartScreen shows "Windows protected your PC", click **More info** and then **Run anyway** (the program is not code-signed).

`START_SWAFL.bat` starts the program from the source folder without building it, and `START_SWAFL_WEB.bat` opens the same program in the browser on `http://127.0.0.1:5199`.

1.3 Route C — from source

```
git clone https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops.git
cd  Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/Software/SWAFL_Desktop
npm install
npm start          # desktop window
npm run web        # the same program in a browser tab on http://127.0.0.1:5199
npm run get-app    # build SWAFL.exe, copy it to the Desktop and start it
```

1.4 Requirements

- Windows 10 or 11 (64-bit) for the built program; any current Chrome, Edge or Firefox for the browser route.
- A screen of at least 1280 × 720 px. The layout is designed for 1600 × 900; panels can be hidden (View menu) or the drawing shown alone (*Focus view*, F9).
- About 400 MB of free memory. The 20-year daily simulation is computed in the program; no server is involved.

1.5 First-run check

After start-up, verify three things: (i) the status bar at the bottom reads *Ready* and ends with *SWAFL 1.1.0*; (ii) the Results tab of the dashboard shows *On* followed by today's date with numbers below; (iii) clicking **Surface irrigation design** in the Layers panel opens the design with its inputs and a *Run the design* box. If the drawing area stays blank, see Chapter 16.

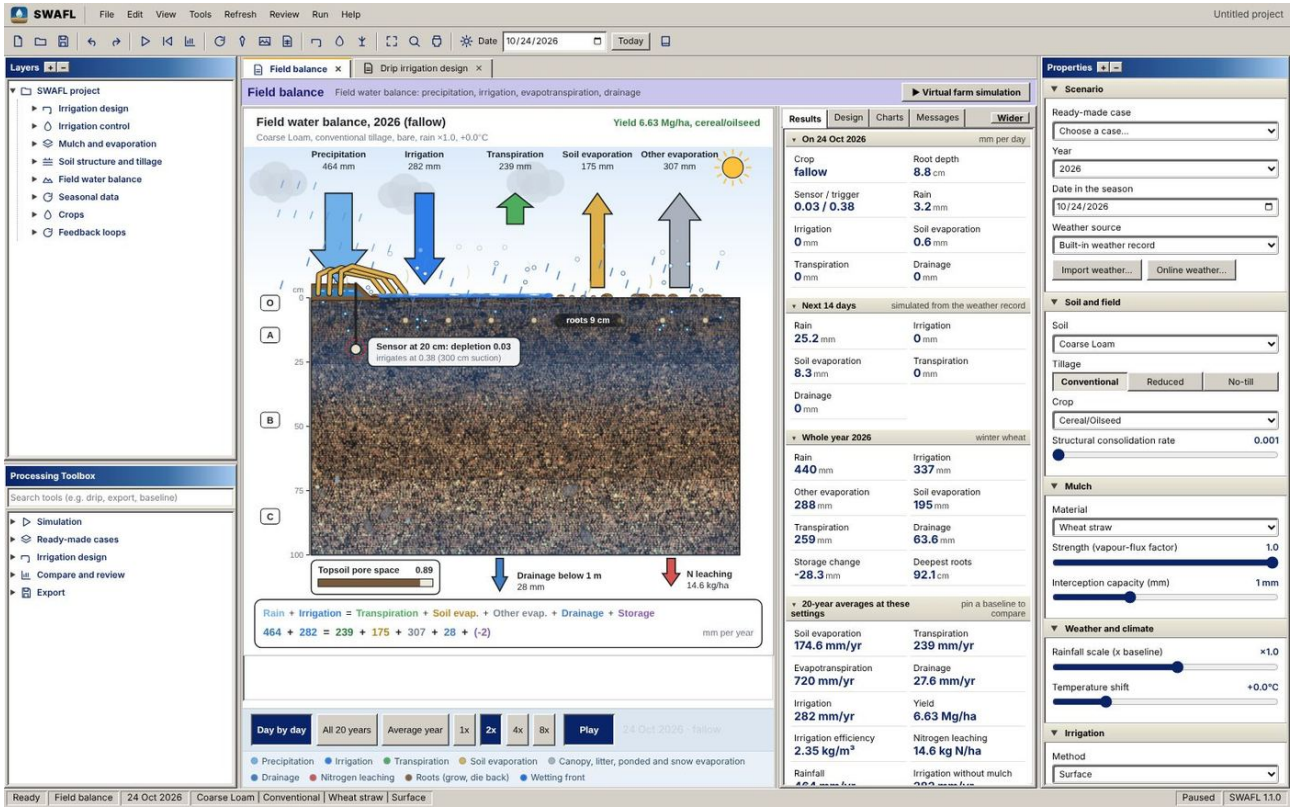


Figure 1.1: The workbench on first start: the Field balance layer for today's date. Left: Layers and Processing Toolbox. Centre: the drawing. Right of the drawing: the dashboard. Far right: Properties.

Chapter 2

The window: a guided tour

2.1 Anatomy

The window has eight regions, named here as they are named in the program.

1. **Menu bar** — File, Edit, View, Tools, Refresh, Review, Run, Help.
2. **Toolbar** — icon buttons for New, Open, Save, Undo, Redo, Play/Pause, Restart, 20-year simulation, the three design layers, fit-to-window, search, the report, and the **Date** control with a **Today** button.
3. **Layers** panel (top left) — the project tree. Each layer is one drawing of the model; each design layer lists its sections and its *Virtual farm simulation*.
4. **Processing Toolbox** (bottom left) — a searchable list of ready-to-run actions.
5. **Document tabs and view bar** (above the drawing) — one tab per open layer; the view bar names the layer, gives the buttons of its sections and the **Virtual farm simulation** button.
6. **Drawing** (centre) — the active layer, with the playback bar and the legend under it.
7. **Dashboard** (right of the drawing) — four tabs: *Results, Design, Charts, Messages*. Every block can be folded and unfolded by clicking its header.
8. **Properties** (far right) — every model input, in collapsible groups.

The **status bar** shows, from left to right: the state (*Ready*), the active layer, the working date, the soil, tillage, mulch and irrigation method, *Paused* or *Running*, and the version (*SWAFL 1.1.0*).

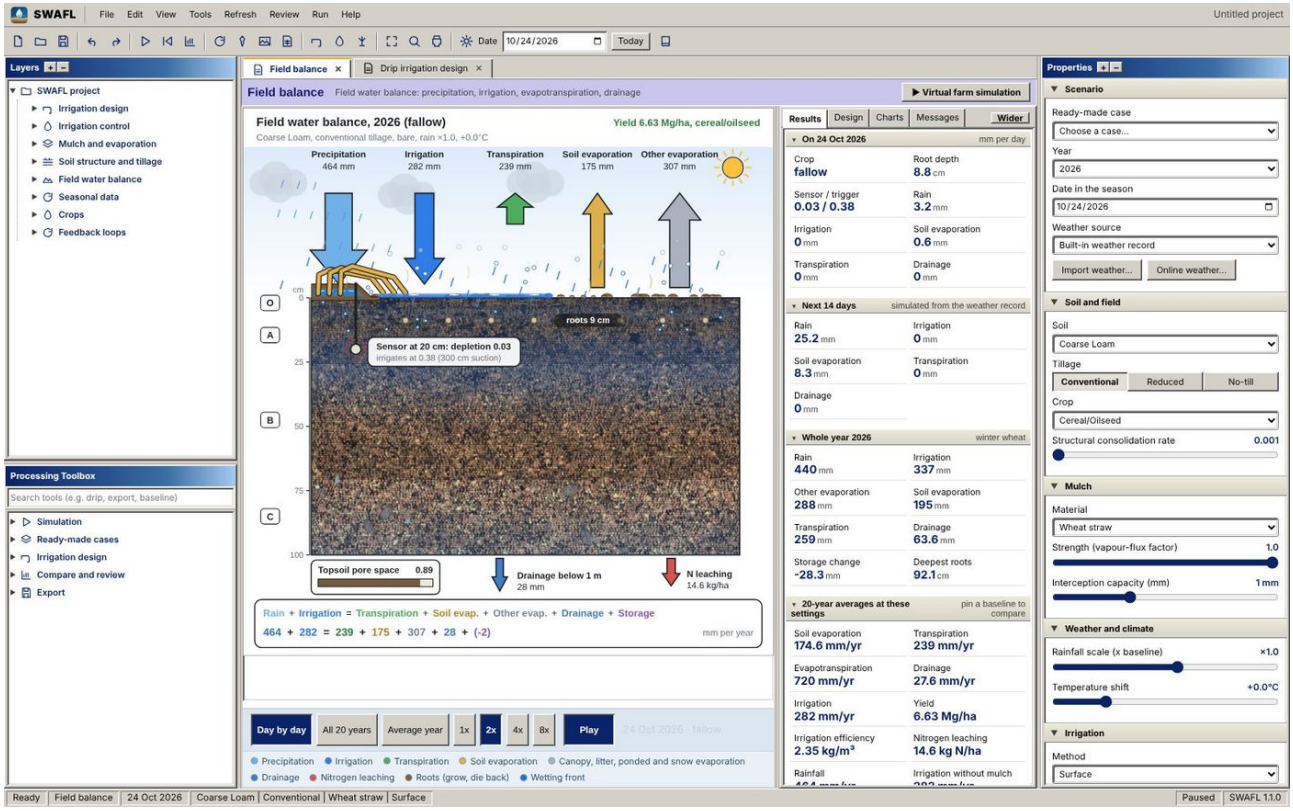


Figure 2.1: The window with the Field balance layer open.

2.2 Menu bar reference

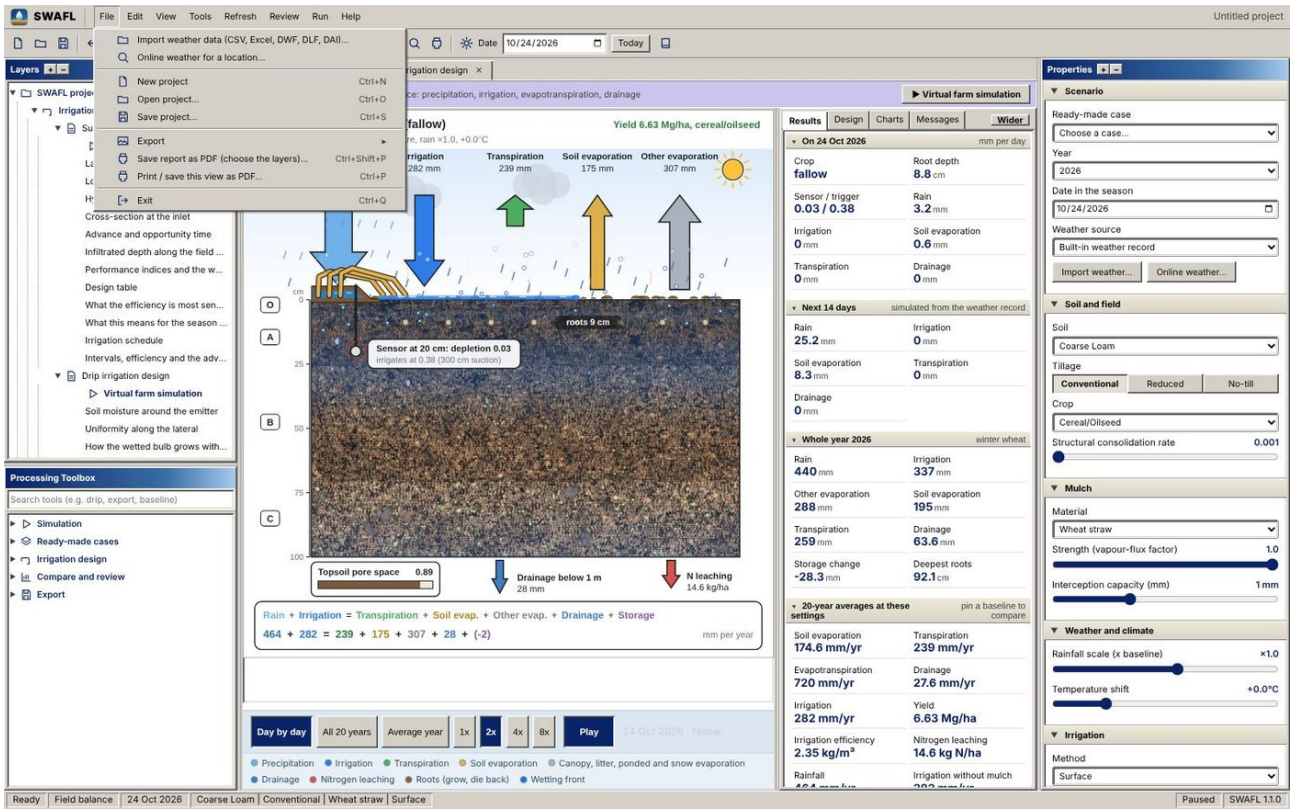


Figure 2.2: The File menu.

Menu	Item	Shortcut	Action
File	Import weather data (CSV, Excel, DWF, DLF, DAI)…		Reads a daily weather file; every layer, the dashboard and the crop water requirement then use it (Chapter 19).
	Online weather for a location…		Gets twenty years of daily weather and a 16-day forecast for a place (Chapter 19).
	New project	Ctrl+N	Resets every control to the defaults (Appendix B).
	Open project… / Save project…	Ctrl+O / Ctrl+S	Loads or saves a `.swafl` file: JSON with the version, the open layer, every control, the year and the date.
	Export ▸	Ctrl+E	Picture of the window (PNG); yearly results (CSV); design results (CSV).
	Save report as PDF (choose the layers)…	Ctrl+Shift+P	Asks which layers to include (all are ticked to start with) and writes the report (Chapter 23).
	Print / save this view as PDF…	Ctrl+P	Prints the open layer through the system print dialog.
	Exit	Ctrl+Q	Closes the program.
Edit	Undo / Redo	Ctrl+Z / Ctrl+Y	Steps through the history of control changes.
	Reset all settings; Copy settings as text		Defaults for the controls only; a text list of every control value on the clipboard.
	Pin results as baseline / Clear baseline	Ctrl+B	Freezes the 20-year averages for comparison (Chapter 14).
	Find a tool…	Ctrl+F	Moves the cursor to the Toolbox search box.
View	Design layouts ▸		Jumps to the layout drawing of the surface, drip or sprinkler design.
	Field sketch to design…; Virtual farm simulation		The field sketch (Chapter 18); the virtual farm of the design you are in (Chapter 20).
	Layers and Processing Toolbox / Properties / Dashboard		Show or hide each dock.
	Fit drawing to window / Drawing at full width (scroll) / Focus view (hide panels)	F9	Three ways of showing the drawing.
	Expand all layers / Collapse all layers; Full screen	F11	Tree display; full screen.
Tools	Field sketch to design…; Compare the irrigation systems on this field; Find the most efficient design inputs		Chapter 18.
	Import weather data…; Online weather for a location…		Chapter 19.
	Ready-made cases ▸; Irrigation design ▸; Irrigation method ▸; Mulch material ▸		Set the control directly.
	Run 20-year season simulation; Find a tool…		Chapter 5.
Refresh	Refresh display (F6); Reload simulation engine; Clear message log		

Menu	Item	Shortcut	Action
Review	Results dashboard; Design results; Yearly charts; Messages; Pin / Clear baseline; Go to top of the drawing		Switches the dashboard tab.
Run	Run / pause the season (F5); Restart from day 1 (Shift+F5); 20-year season simulation; Optimise surface layout; Ready-made cases ▸		
Help	Manual (PDF) (F1); How to use SWAFL; Troubleshooting (running the app); Keyboard shortcuts; SWAFL on GitHub; Full documentation on GitHub; Report a problem; About SWAFL		

The eight menus are pictured, opened, in Volume 2.

2.3 The command palette (Ctrl+K)

Press **Ctrl+K** anywhere to open a search box over *every* layer, section, tool, setting and date. Type part of a name (for example `drip`, `trigger`, `mulch`, `2031`) and press **Enter** to run the first match, or use the arrow keys to choose.

2.4 Layers panel

The tree has the project root *SWAFL project* and eight groups, in the order in which a field is usually designed and studied. Click a group to expand it, click a layer to open it as a tab, and click the small triangle beside a layer to list its sections; a section jumps the drawing to that part.

Group	Layers
Irrigation design	Surface irrigation design; Drip irrigation design; Sprinkler design and pipes — each with its sections and its *Virtual farm simulation*
Irrigation control	Sensor and trigger; Trigger law
Mulch and evaporation	Mulch against bare soil; Mulch control; Evaporation physics; Fate of saved water; Regime map
Soil structure and tillage	Tillage and structure; Soil hydraulic drivers; Root structure
Field water balance	Field balance; Soil, plant and air; Profile evolution
Seasonal data	Reliability over 20 seasons; Full water balance; Nitrogen budget
Crops	Crop water requirement
Feedback loops	Feedback loops

In version 1.0 every layer carried a short code (S2, V1, N4 …). The codes have gone from the program; this manual uses the layer names. Open layers stay as tabs above the drawing; closing a tab does not lose any setting, because the controls belong to the project, not to the layer.

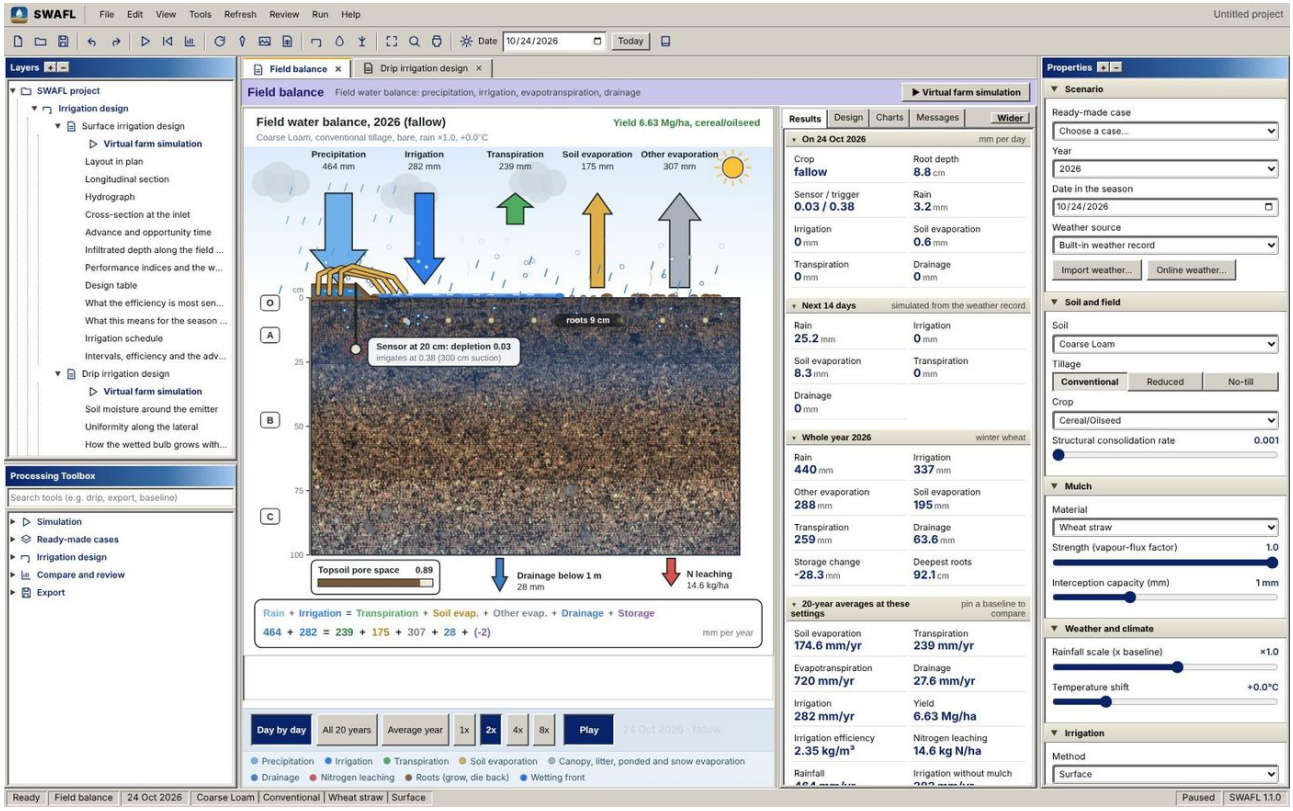


Figure 2.3: The Layers panel with every group expanded.

2.5 The view bar and the Virtual farm button

Above the drawing, the view bar names the open layer. For the design layers it carries one button per section (*Whole design, Layout design, Simulation, …*). Every layer carries the **Virtual farm simulation** button, which opens the farm of the design that belongs to the irrigation method in force; from a design layer it opens the farm of that design (Chapter 20).

2.6 Processing Toolbox

Double-click (or select and press Enter) to run an action. Typing in the search box filters all tools by name and by hidden keywords.

Group	Tools
Simulation	Play/pause the season; Restart from day 1; Run 20-year season simulation; Reset all settings
Ready-made cases	The seven cases of Section 3.8
Irrigation design	Open the surface, drip or sprinkler design; Optimise surface layout; set the irrigation method
Compare and review	Pin and clear the baseline; show the Results, Design or Charts tab
Export	PNG of the window; yearly results CSV; design results CSV; save project; print / PDF

2.7 Resizing and remembering the layout

Drag the thin splitters between the docks to resize them. The program remembers the panel widths between sessions. The *Wider* button above the dashboard widens it; **View ▸ Drawing at full width** lets the drawing keep its design width and scroll; **Focus view** (F9) hides the panels.

2.8 Reading the drawings

- **Hot spots.** Many drawings are clickable. Hovering shows a tooltip; clicking applies a setting (for example, clicking a year bar jumps to that year, clicking a soil on a chart selects that soil).
- **Value boxes.** Every input of the three design layers is a value box. Click it: a small window opens in which you type the exact value, choose its unit (for example m, km or ft; L/s, m^3/h or L/min; ha, m^2 or acre; kPa, bar, m of water or psi) or pick a value from the list of the allowed range. A value outside the range is brought back to its nearest limit.
- **Nothing runs while you type.** The design layers compute only when you press **Run simulation** (Chapter 18); the other layers follow every control at once.
- **Colours.** Water blues, soil browns, crop greens; text is dark on a white sheet; labels sit in white boxes drawn after the pictures, so no arrow covers a label.

Chapter 3

Every control, value by value

All model inputs live in the **Properties** panel (far right). Each group can be collapsed; **+** and **−** in the panel header expand or collapse all. The sliders are *indices* into lists of discrete values, because every value in the lists corresponds to a set of simulations (Chapter 5). The number shown beside a slider is the value the index stands for. The inputs of the three irrigation designs are not in Properties: they are the value boxes inside each design layer (Chapters 11–13 and 18).

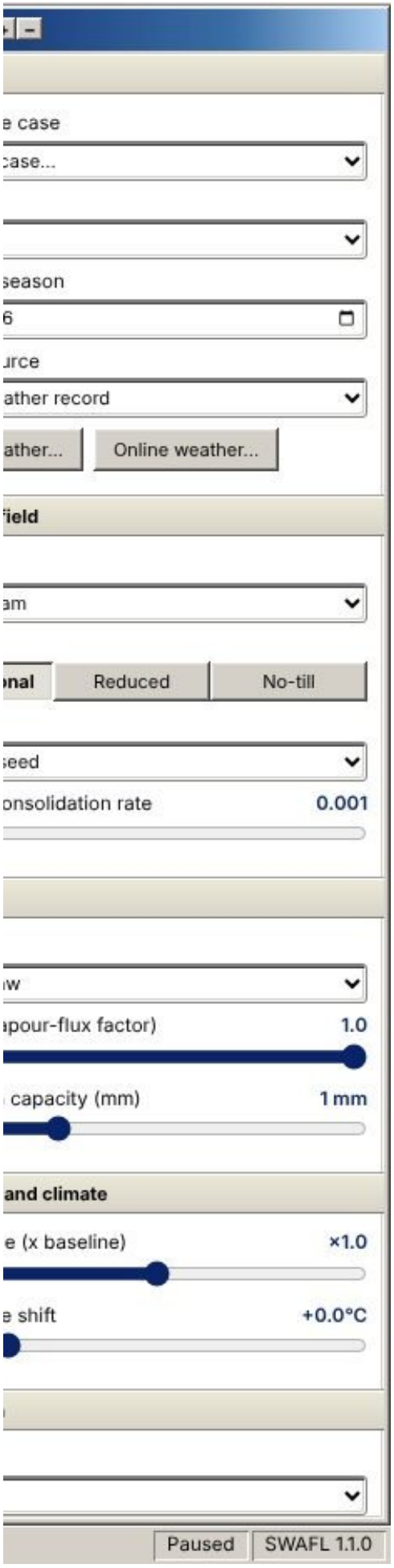


Figure 3.1: The Properties panel, with every group open (full page in Volume 2).

3.1 Scenario

Control	Values	Default	What it does
Ready-made case	seven cases (Section 3.8)	—	Replaces many controls at once, as one undoable step.
Year	1980 … 2100	the current year	The calendar year of the working date. Years 1980–1999 use the recorded weather of that year; later years use the typical year of the record, or your own weather when it covers the year (Chapter 4 and Chapter 19).
Date in the season	any calendar date	today	The working date: the day drawn, the “On …” block of the dashboard and the first day of every irrigation schedule.
Weather source	Built-in weather record; Imported weather file; Online weather of a location	built-in	Which daily weather every layer uses. The second and third choices become active only after you have imported a file or fetched the weather of a place (Chapter 19).
Import weather… / Online weather…	buttons		Open the two weather dialogs.

3.2 Soil and field

Control	Values	Default	Effect
Soil	Sand, Loamy Sand, Sandy Loam, Coarse Sand, Coarse Loam, Loam, Silt Loam, Sandy Clay Loam, Clay Loam, Silty Clay Loam	Coarse Loam	Selects the soil’s texture, its van Genuchten parameters (α, n) (Table 5.3), the baseline run, plant-available water per cm (`AWC`) and the trigger anchor. Re-computes every layer and cancels the last run of every design.
Tillage	Conventional / Reduced / No-till	Conventional	Sets the initial pore index p_0 (1.00, 0.72, 0.42) and therefore the starting structure of the topsoil (Section 8.1).
Crop	the simulated cereal/oilseed rotation and the 48 crops of the library, in the groups *Staple food crops* (9), *Cash and oil crops* (8), *Vegetables* (4) and *Trees, vines and palms* (27)	Cereal/Oilseed	Chooses the crop calendar, the crop coefficients, the root depth, the depletion fraction and the yield response (Chapter 21). The cereal rotation is the simulated crop; the library crops are transferred from it with their FAO-56 crop coefficients (Section 21.4).
Structural consolidation rate	0.001, 0.0025, 0.005, 0.01, 0.02 d^{-1}	0.001	The rate τ_c at which conventionally tilled pores close (Eq. 8.1). In the simulations it changes the results of the five finer soils (loam to silty clay loam, up to about 6 mm a year of drainage and 14 kg N/ha of leaching); in the five coarser soils every rate gives the same result.

3.3 Mulch

Control	Values	Default	Effect
Material	Rice husk, Wheat straw, Wood chips, Sawdust, Grass clippings, Black plastic film	Wheat straw	Sets both sliders below to typical values for the material and the drawn layer thickness (Table 3.1).
Strength (vapour-flux factor) φ	0.1, 0.2, … 1.0	**1.0 (no mulch)**	The fraction of the bare-soil vapour flux that passes the mulch. 1.0 means bare; 0.1 means the flux is cut by 90 %.
Interception capacity	0.5, 1, 2, 4 mm	1 mm	Water the mulch can hold and lose before it reaches the soil.

Table 3.1. Mulch materials as defined in the engine.

Material	φ	capacity (mm)	drawn thickness (relative)
Rice husk	0.5	2	1.0
Wheat straw	0.4	4	1.35
Wood chips	0.3	2	1.7
Sawdust	0.6	1	0.9
Grass clippings	0.7	4	1.2
Black plastic film	0.1	0.5	0.25

The material only *sets* the sliders. Moving a slider afterwards keeps the material for drawing but uses the slider value for the calculation. Any $\varphi \geq 0.95$ is treated as “no mulch” in the drawings.

3.4 Weather and climate

Control	Values	Default	Effect
Rainfall scale	$×$0.5, 0.6, … 1.4	$×$1.0	Multiplies the baseline annual rainfall and selects the matching runs. When weather is imported, it is set to the step nearest to the ratio of the imported to the recorded annual rain (Eq. 19.4).
Temperature shift	−1.0, −0.5, 0, +0.5, … +3.5 °C	0 °C	Selects the warming runs and changes the evaporative demand (Eq. 5.26). When weather is imported, it is set to the step whose demand is nearest the imported reference evapotranspiration (Eq. 19.5).

3.5 Irrigation

Control	Values	Default	Effect
Method	Surface; Overhead; Drip 10–20 cm; Drip 20–30 cm; Drip 40–50 cm; Drip 50–60 cm	Surface	Selects the irrigation run set and, in the day-by-day engine, where each dose is placed (Section 5.2).

Control	Values	Default	Effect
Trigger (sensor suction)	−50, −75, −100, −150, −200, −250, **−300**, −350, −400, −500, −600, −700, −800, −1000, −1200, −1500, −2000, −3000, −5000, −10000 cm	−300 cm	Irrigation starts when the soil at the sensor depth is as dry as this suction. A *less negative* number means wetter, more frequent irrigation.
Sensor depth	5, 10, 15, … 100 cm (20 values)	20 cm	Depth of the simulated tensiometer that is compared with the trigger.

Sand and loamy sand. In the simulations of these two soils the sensor-triggered runs at the reference trigger of −300 cm at 20 cm never irrigate, because the coarse soil drains past the sensor before it dries to that suction. A ratio to zero is undefined, so for these soils the irrigation of another trigger, sensor depth or method is added to the reference amount as the difference of the two simulated amounts (Eq. 5.12a). At −50 cm the sand then receives 400 mm a year instead of the reference 278 mm.

Trigger mode (Trigger law layer). The slider above applies one suction to every soil (*common* mode). The **Trigger law** layer offers an *own* mode, in which each soil gets the suction that corresponds to a soil-independent depletion fraction f^* (default 0.45): see Chapter 6.

3.6 Nutrients

Control	Values	Default	Effect
Nitrogen rate	10, 25, 40, 55, 70, 85, **100**, 115, 130, 150 % of the standard rate	100 %	Selects the nitrogen runs; changes yield and nitrogen leaching and, slightly, the maximum rooting depth (Eq. 5.18).

3.7 Ready-made cases

#	Case	Changes
0	Default: Coarse Loam, bare, normal climate	all defaults
1	Dry year on Sand, no mulch	Sand; rain $×$0.6; +1 °C
2	Wheat straw mulch on Loam, drip 10 to 20 cm	Loam; straw; drip 10–20
3	Wet year on Silty Clay Loam, no-till	Silty Clay Loam; no-till; rain $×$1.3
4	Hot year (+3 °C), tomato on Sandy Loam	Sandy Loam; tomato; +3 °C
5	Plastic film, deep drip 40 to 50 cm, Clay Loam	Clay Loam; plastic; drip 40–50
6	Low nitrogen (40 %), cowpea on Silt Loam	Silt Loam; cowpea; N 40 %

Every case first resets the controls to their defaults and then applies its changes, so cases never accumulate.

3.8 Design shortcuts

The last group of Properties opens the three designs (*Surface design, Drip design, Sprinklers and pipes*) and runs **Optimise surface**, which searches the surface design for the inflow and cut-off with the highest application efficiency at the present field.

3.9 Change-by-change: what each control moves

“Run set” means that the response is read from the simulations (Chapter 5); “engine” means that the day-by-day model of Section 5.3 responds as well.

Control changed	Annual outputs (20-year means)	Day-by-day layers and schedules	Design layers
Soil	all (baseline run for that soil)	AWC, (α, n), trigger anchor ⇒ dates, depths	intake (k, A) via Saxton–Rawls and Green–Ampt; Z_n; the sprinkler maximum rate; the soil of the virtual farm
Tillage, consolidation	drainage, evaporation, yield	pore index in the structure layers	surface storage S_d of the sprinkler design
Crop	yield, transpiration, evaporation (Section 21.4)	crop calendar, K_c, $R_{\max}$, cover	peak ET_c, Z_n, row spacing
Mulch φ, capacity	evaporation, irrigation, drainage, yield	evaporation term	interception loss of the sprinklers
Rain scale	all	rain amount per event	—
Temperature	evaporation, transpiration, irrigation, yield	E_0 and seasonal weight	air temperature in the wind drift (Eq. 13.9)
Method	irrigation, drainage, yield	dose placement	the design the virtual farm follows
Trigger	irrigation, yield, drainage	**dates of irrigation**	Z_n (as $f_T \cdot \text{AWC} \cdot R_{\max}$)
Sensor depth	irrigation, yield	**dates of irrigation**	—
Nitrogen	yield, N leaching	$R_{\max}$	—
Weather source	through the nearest rain and temperature steps	**every daily value**	the crop water requirement and the schedules
Date and year	none	**which day is shown; first date in the schedule**	the schedule table

Chapter 4

Dates, years and the clock

4.1 Why the date matters

SWAFL runs on **twenty years of daily weather** arranged on a 365-day calendar without leap days. With the built-in weather these are the twenty recorded years 1980–1999 of the study site (Chapter 5); with imported or online weather they are the last twenty complete years of your own record (Chapter 19). The program maps *any calendar date* you choose onto one of those twenty years. The working date therefore decides (i) which day is drawn, (ii) which of the twenty years acts as the *analogue* for that date, and (iii) the first line of every irrigation schedule.

4.2 The date control

The **Date** box on the toolbar and the **Year** and **Date in the season** controls in Properties ▸ Scenario all set the same working date; the year list runs from 1980 to 2100. **Today** sets the computer's date. When you change the date:

1. With the built-in weather, a year from **1980 to 1999** is its own analogue: that recorded year is drawn.
2. With imported or online weather, a year inside the twenty years of your record is its own analogue in the same way.
3. Any other year (for example 2026 or 2075) takes the **typical** year of the record as its analogue: the median of the twenty years by growing-season rain reaching the soil.
4. The engine sets the field clock to day-of-year d_{oy} of the analogue year, so that 22 October 2029 shows the crop state, soil water and irrigation events of the analogue year on 22 October.

In version 1.0 a *Weather year* control let you choose the typical, the driest or the wettest year for dates outside the record. In version 1.1 the typical year is always used for those dates; to study a dry or a wet year, choose that year inside the record (the *Full water balance* layer of the Seasonal data group shows the rain of each of the twenty years), or import weather.

With Y_0 the calendar year of the chosen date, a the analogue year index $(0, \dots, 19)$ and k the model day number $(0, \dots, 7299)$ of the twenty-year run,

$$k_{\text{start}} = 365\,a + d_{\text{oy}}, \qquad \text{date}(k) = 1\ \text{Jan of year}\ \big(Y_0 + \lfloor k/365 \rfloor - a\big) + \big(k - 365\lfloor k/365 \rfloor\big)\ \text{days}, \tag{4.1}$$

Provenance (Eq. 4.1): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

and the calendar label of the first plotted year is $\text{CAL}_0 = Y_0 - a$. The choice of analogue year is

$$a = \begin{cases} \text{argsort}_y\,(P_{\text{thr},y})_{[10]} & \text{typical (median, 11th smallest of 20): years outside the record} \\ Y_0 - Y_{\text{rec}} & Y_{\text{rec}} \le Y_0 \le Y_{\text{rec}} + 19 \end{cases} \tag{4.2}$$

Provenance (Eq. 4.2): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

where $P_{\mathrm{thr},y}$ is the sum of throughfall (rain reaching the soil) in year y (Section 5.2) and Y_{rec} is the first year of the record (1980 for the built-in weather, the first of the twenty years of your weather otherwise).

What this means for you. A date in 2029 is *not* a forecast for 2029 unless the online weather is in use and the date falls in its 16-day forecast. Otherwise it shows how this soil, crop, trigger and method would behave on that day of the year in a typical year. Chapter 17 repeats this as a validity limit.

4.3 The playback bar

Under the drawing, the **playback bar** carries the play button (F5) and, for the field layers, the *Day by day*, *All 20 years* and *Average year* views and the speed buttons. In the field layers the play button animates the field clock: each real second advances 15 model days times the speed setting, and the clock *dwells* for about $3.2/\sqrt{\text{speed}}$ s on every day on which an irrigation, a wetting front of at least 20 cm, or drainage occurs. The clock wraps at the end of the twenty years. **Restart** (Shift+F5) returns to day 1 of the working year. In the three design layers the same button is **Run simulation** (Chapter 18). The year strip that sat under the drawing in version 1.0 has gone: the year is set only in Properties and on the toolbar.

4.4 The 14-day look-ahead

The “Next 14 days” block on the Results tab sums the same simulated daily fluxes from the working date forward. It is labelled *simulated from the weather record* because the rain on those days is the analogue year’s rain. When the online weather is in use and the days fall in its 16-day forecast, the block says *from the weather forecast* instead (Chapter 19).

4.5 Detail year and sensor record

The *Sensor and trigger* and *Profile evolution* layers can show either all twenty years or one year in detail. The suction traces shown there are the study’s **daily suction at 20 cm for 1980–1999**, stored for the Coarse Loam and the Coarse Sand; for every other soil, and for every setting other than the standard one, the trace is moved by the difference between two runs of the daily model, the run at your settings and the run at the standard settings (Eq. 6.4a). The growing season of these records is the 183 days starting 1 April.

Chapter 5

The two computing engines

Every number in SWAFL comes from one of two engines, which are tied together so that they agree on the twenty-year means.

- **Engine 1, the annual meta-model.** It reads 20-year-mean results of the 10,190 Daisy simulations and scales them for the controls. It produces the Dashboard "20-year averages" and every bar, line and arrow in the annual layers. **[Data-driven]**
- **Engine 2, the daily bucket model.** A 40-layer soil-water model run for twenty years of daily weather. It produces dates, depths, schedules, root growth, water flow arrows and the animated field. It is **calibrated** so that its mean irrigation equals Engine 1's, and its fluxes are **scaled** to Engine 1's annual means (Eq. 5.24). **[Engineered, with literature components]**

The design layers (Chapters 11–13) are a third, separate set of *closed-form hydraulic models*; they take their soil, crop, trigger and depth from Engines 1 and 2 (Section 5.3).

5.1 Engine 1 — the Daisy simulations and the meta-model

5.1.1 The simulation campaign

The repository's study ran Daisy 7.1 (Abrahamsen and Hansen, 2000; University of Copenhagen), a one-dimensional soil–plant–atmosphere model that solves Richards' equation with van Genuchten–Mualem hydraulics, coupled to the WEPP dynamic structure model for the effect of tillage and consolidation on pore structure with time. Ten soils (USDA textures from sand to silty clay loam, Ap and Bt horizons) were simulated in **10,190 runs across 17 experiments (E0–E16)**. The families that the workbench reads are:

Experiment	Runs	Varied	Used for
E2	200	Trigger suction	Trigger correction, Eq. (5.11)
E3	200	Sensor depth	Sensor-depth correction, Eq. (5.11)
E4	100	Method (overhead, surface, drip at three depths)	Method correction, Eq. (5.11)
E5	300	Tillage × structural consolidation rate	Baseline **b**, Eqs. (5.4)–(5.10)
E7	200	Mulch vapour-flux factor × interception capacity	Soil evaporation $\bar{E}$, Eq. (5.4)
E11	1,000	Temperature shift × evaporation factor × rain scale	Warming ratios, Eqs. (5.4)–(5.10)
E13	1,000	Nitrogen rate × water strategy × soil	Nitrogen correction, Eq. (5.11)
E14	1,000	Mulch factor × capacity × rain scale	Mulch ratios, Eqs. (5.4)–(5.10)

(The other experiments are used by the repository figures and described in the repository README. The *Regime map* and *Fate of saved water* layers are computed from the E14 and E11

grids listed above.)

$$S_e(h) = \Big[1 + (\alpha|h|)^n\Big]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

Provenance (Eq. 5.1): **[Literature]**: van Genuchten (1980).

(van Genuchten, 1980), with relative hydraulic conductivity from the Mualem (1976) model,

$$K_r(S_e) = S_e^{1/2}\Big[1 - \big(1 - S_e^{1/m}\big)^m\Big]^2. \tag{5.2}$$

Provenance (Eq. 5.2): **[Literature]**: Mualem (1976) with van Genuchten (1980).

Equation (5.2) is used in the Soil, plant and air layer (Section 7.2); Eq. (5.1) is used for all water-retention drawings. The parameters (α, n) used by the engine are *effective* values that pass through the four anchors of the project's trigger table; for the standard textures they match the Carsel and Parrish (1988) values (Table 5.3).

5.1.2 Plant-available water and depletion

Depletion of plant-available water is defined between field capacity (suction 100 cm, pF 2.0) and wilting point (15,849 cm, pF 4.2):

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

$f = 0$ is field capacity and $f = 1$ is wilting point; $f < 0$ (wetter than field capacity) is kept as the *signed* value only in the Trigger law layer. The inverse $h(f)$ is found by bisection on $\log_{10}|h| \in [0, 4.6]$ (60 halvings).

Table 5.3. Effective van Genuchten parameters used by the engine, plant-available water, and the soil's critical suction at the soil-independent depletion $f^* = 0.45$.

Soil	α (cm^{-1})	n	AWC (mm cm^{-1})	h^* at $f^* = 0.45$ (cm)	f at 300 cm
S01 Coarse Loam	0.0961	1.353	1.5	387	0.380
S02 Coarse Sand	0.0536	1.600	0.5	262	0.496
S03 Sand	0.145	2.68	0.6	143	0.842
S04 Loamy Sand	0.124	2.28	0.9	160	0.756
S05 Sandy Loam	0.075	1.89	1.2	195	0.627
S06 Loam	0.036	1.56	1.7	284	0.468
S07 Silt Loam	0.020	1.41	2.0	401	0.367
S08 Sandy Clay Loam	0.059	1.48	1.4	311	0.439
S09 Clay Loam	0.019	1.31	1.6	503	0.314
S10 Silty Clay Loam	0.010	1.23	1.7	729	0.237

(AWC is "typical available water capacity by texture class" in mm of water per cm of soil; it is an engineered per-texture constant, not a measured value of the study soil.)

5.1.3 The meta-model

Let $\mathbf{b} = (E_b, T_b, ET_b, I_b, Y_b, D_b, N_b)$ be the 20-year-mean Daisy results for the chosen soil, tillage and consolidation rate (run set E5), and $\mathbf{b}_{\text{ref}}$ the same for conventional tillage at $\tau_c = 0.005$ d^{-1}. Let φ be the mulch strength, c the mulch capacity, ρ the rainfall scale and ΔT the warming. Define the mulch table lookups $M(\varphi, \rho)$ (E14 at capacity c) and warming table lookups $W(\Delta T, \rho)$ (E11); write

$\text{now} = M(\varphi,\rho)$, $\text{ref} = M(\varphi,1)$, $\text{nm} = M(1,\rho)$, $\text{bare} = M(1,1)$, $\text{tN} = W(\Delta T,\rho)$, $\text{tB} = W(0,\rho)$, $\text{tR} = W$

Then the engine computes (the ratio $[x/y]$ is 1 where $|y| < 10^{-6}$):

$$E = \bar{E}(\text{soil},\varphi,c)\,\left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\frac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\frac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

$\bar{E}(\text{soil},\varphi,c)$ is the linear interpolation, in φ, of the soil's simulated soil evaporation (run set E7) at capacity c, closed at $\varphi = 1$ by the bare-soil value. Fluxes are combined multiplicatively and drainage/leaching additively because ratios of small numbers (drainage near 0 mm) are unstable.

Four *set-specific corrections* then act in turn on the result. For each family $F \in \{$trigger (E2), sensor depth (E3), method (E4), nitrogen (E13, yield and N_L only)$\}$ with current key κ and baseline key κ_0 (trigger -300 cm; depth 20 cm; method Surface; nitrogen 100 %), with r_κ the soil's row at κ (nearest row if that exact run is absent):

$$Q \leftarrow Q\,\text{clip}\Big(\frac{r_\kappa.Q}{r_{\kappa_0}.Q};\,0.05,\,6\Big)\ (Q \in \{E,T,ET,I,Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_\kappa \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Families whose reference run never irrigates (release 1.1). In the sand and the loamy sand the sensor-triggered runs at the reference trigger (-300 cm at 20 cm, surface method) apply no irrigation at all: the coarse soil drains past the sensor before it dries to that suction. The ratio of Eq. (5.11) is then undefined, and version 1.0 returned 1, so that the trigger, the sensor depth and the method had no effect on the irrigation of these soils. Version 1.1 adds the simulated difference instead,

$$I \leftarrow I + \sum_{F \in \{\text{E2,E3,E4}\}} \max\big(0,\ r_\kappa.I - r_{\kappa_0}.I\big) \quad \text{when } r_{\kappa_0}.I < 1\text{ mm}, \qquad \Delta S = P + I - ET - D, \tag{5.12a}$$

Provenance (Eq. 5.12a): **[Data-driven]**: differences of Daisy run sets E2–E4; the storage term closes the balance.

so that in the sand a trigger of -50 cm (122 mm more in the runs) gives 400 mm a year instead of 278 mm, -75 cm gives 317 mm and -100 cm 285 mm; drier triggers keep the reference amount, as in the runs.

Finally the model enforces non-negativity, a soil-dependent yield ceiling $Y \le Y_{\max}(\text{soil})$ (the largest simulated yield), rain $P = P_{\text{base}}\rho$, and **water-balance closure**:

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

E_{other} is interception plus evaporation from the mulch layer, and ΔS collects the change in storage and small terms. IE converts Mg ha^{-1} per mm of irrigation to kg of grain per m^3 of irrigation water (1 mm $= 10$ m^3 ha^{-1}).

Crops. The simulated crop is the cereal/oilseed rotation; its yield is shown in Mg ha^{-1}. The 48 crops of the library are transferred from it by the ratio of two FAO-56 daily soil-water balances, one for the library crop and one for the cereal reference, run on the same soil, weather and trigger (Section 21.4, Eqs. 21.5–21.9). Their yield is shown as the relative yield, in per cent of the potential.

5.1.4 What "20-year average at these settings" means

Every Dashboard tile in the last block of the Results tab is the output of Eqs. (5.4)–(5.12) for the current controls. Every slider of Properties is an index into a list of simulated values, so no value ever lies between runs; but two controls changed together (for example, mulch and warming) use the product of ratios above, which assumes the effects are **independent**. The yield ceiling and the closure window (Eq. 5.12) stop independent effects from compounding beyond what the water and the crop allow.

5.2 Engine 2 — the daily bucket model

5.2.1 Layout and state

The soil is $N = 40$ layers of thickness $\Delta z = 2.5$ cm (100 cm total). Each layer holds water w_i (mm) between 0 and the plant-available capacity $W_{\text{max}} = \text{AWC} \cdot \Delta z$. All layers start at $0.72\, W_{\text{max}}$. Relative content $s_i = w_i / W_{\text{max}}$ and depletion $f_i = 1 - s_i$. The run is 20 years of 365 days (7,300 days). The sensor sits in layer $j_s = \text{round}(z_s / \Delta z)$ for sensor depth z_s.

The trigger suction $|h_T|$ is converted to a depletion fraction $f_T = f_{\text{used}}(|h_T|)$ by Eq. (5.3). The sensor condition "soil at suction h_T" therefore becomes the equivalent statement $f_{j_s} \geq f_T$.

5.2.2 Crop calendar and canopy cover

The cereal crop follows the four-year rotation spring barley, winter barley, winter rape, winter wheat with sowing and harvest days (relative to 1 Jan of the year) $(100, 225)$, $(-90, 205)$, $(-125, 200)$, $(-80, 230)$; winter crops are sown in the previous autumn. Legume: sown day 110, harvested 215. Vegetable: sown day 100, harvested 185. Outside the window the field is *fallow*. Canopy cover $c(d) \in [0, 1]$ is

$$c_{\text{spring}}(d) = \frac{1}{1 + \exp\!\left(-(d - d_{\text{mid}})/s\right)}, \quad s = 13 \text{ (vegetable } 8), \tag{5.13}$$

Provenance (Eq. 5.13): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

$$c_{\text{winter}}(d) = \begin{cases} 0.08 + 0.12\ \min\{1, (d - d_{\text{sow}})/60\} & d < d_{\text{pk}} - 90 \\ 0.2 + \dfrac{0.8}{1 + \exp\!\left(-(d - d_{\text{pk}} + 45)/12\right)} & \text{otherwise} \end{cases} \tag{5.14}$$

Provenance (Eq. 5.14): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

and after the peak day d_{pk} cover declines linearly to zero at harvest: $c \leftarrow c\ \max\{0,\ 1 - (d - d_{\text{pk}})/(d_{\text{harv}} - d_{\text{pk}})\}$. The mid day is $d_{\text{sow}} + 50$ (cereal), $d_{\text{sow}} + 40$ (legume), $d_{\text{sow}} + 28$ (vegetable); the peak is harvest -50 (cereal), sow $+65$ (legume), sow $+65$ (vegetable). These logistic curves are **[Engineered]**: they reproduce the shape of a canopy, not a measured leaf-area record.

5.2.3 Rainfall generator

Rain falls on a day with probability 0.28. With $\mu_0 = P/(0.28 \cdot 365)$ the mean wet-day depth of the scaled annual rain P, the depth on a wet day of simulated year y is

$$R = \min\{70,\ \mu_0\,\gamma_y\,(-\ln U)\}, \qquad U \sim \text{Uniform}(0,1), \tag{5.15}$$

Provenance (Eq. 5.15): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

an exponential distribution capped at 70 mm, with the random sequence seeded by a hash of the soil name so every run is reproducible. The year factor $\gamma_y = \text{clip}\big((m_{\text{med}}/m_y)^{0.3};\ 0.65,\ 1.45\big)$ makes years wetter or drier according to the dryness m_y of the study's 20-year rainfed suction record at 20 cm, where $m_y = 10^{\overline{\log_{10} h}}$ is the geometric-mean suction over the 183-day growing season of year y, and m_{med} is the median over the 20 years. This is how the generated weather "follows the dryness of the real record".

The canopy and litter catch the first I_c mm of every shower:

$$I_c = \mu_0 \ln\frac{1}{r}, \qquad r = \frac{\max\{40,\ P - P_{\text{other}}\}}{P}, \qquad T_f = \max\{0,\ R - I_c\}. \tag{5.16}$$

Provenance (Eq. 5.16): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

P_{other} is the interception-and-mulch loss from Eq. (5.12). Only throughfall $T_f > 0.3$ mm enters the soil. The choice of I_c makes the expected interception over the exponential rain distribution equal to the annual interception of Engine 1 (the expected value of $\min(R, I_c)$ for an exponential with mean μ_0 is $\mu_0(1 - e^{-I_c/\mu_0})$, and $e^{-I_c/\mu_0} = r$ gives $\mu_0(1 - r)$ per wet day).

5.2.4 Infiltration, redistribution and drainage

Water enters the top layer and cascades down: any layer exceeding W_{max} passes the excess to the layer below; the excess leaving layer 40 is **drainage** D_d. The depth of the deepest wetted layer is the **wetting front** (cm). Between layers a slow redistribution moves water toward the drier neighbour,

$$q_i = 0.08\,(w_{i+1} - w_i), \qquad w_i \leftarrow w_i + q_i,\ \ w_{i+1} \leftarrow w_{i+1} - q_i, \tag{5.17}$$

Provenance (Eq. 5.17): **[Engineered]**: tipping-bucket redistribution with a chosen coefficient.

shown as *capillary rise* arrows when $q_i > 0.03\,W_{\text{max}}$ upward. Drainage and the wetting front are therefore **tipping-bucket** results, not Richards solutions; that is why the engine scales them to Engine 1 (Eq. 5.24).

5.2.5 Roots

The maximum rooting depth is

$$R_{\text{max}} = \text{clip}\Big(R_c\,(0.85{+}0.3\,\text{sand})\big[1{+}0.12\log_{10}(|h_T|/300)\big]\big[1{-}0.18\,\tfrac{N-100}{100}\big];\ 25,\ 100\Big)\ \text{cm}, \tag{5.18}$$

Provenance (Eq. 5.18): **[Engineered]**: rooting rules with constants chosen for the workbench.

with $R_c = 95$ (cereal), 115 (legume), 45 (vegetable) cm, sand the sand fraction, and N the nitrogen rate in percent. (Drier triggers push roots deeper; nitrogen-rich soils root slightly shallower.) While the canopy is growing, the rooting depth R_d chases water:

$$R_d \leftarrow \max\Big\{R_d,\ \min\big\{R_{\text{max}}\,\text{clip}\big(\tfrac{c-0.04}{0.96};0,1\big)^{0.8},\ R_{\text{max}},\ R_d + 1.7\,(0.35 + \bar{s}_{\text{wet}})\,\kappa\big\}\Big\}, \tag{5.19}$$

Provenance (Eq. 5.19): **[Engineered]**: rooting rules with constants chosen for the workbench.

where $\bar{s}_{\text{wet}}$ is the mean relative water content of the four layers beneath the root front, and $\kappa = 1.3$ if the top six layers average below 0.35 (dry topsoil) and 1 otherwise. After harvest $R_d \leftarrow \max\{3, 0.965 R_d\}$; drought prunes the roots by 0.9 cm per day when the plant-available-water-weighted root fraction is below 0.12 and $R_d > 8$ cm. Root density follows

$$\rho_i \propto \exp\!\Big(-\frac{z_i}{\lambda R_d + 8}\Big)\,(0.45 + 0.9\,s_i)\Big[1 + 3\,e^{-((z_i - z_d)/12)^2}\Big]_{\text{drip only}}, \quad z_i \le R_d, \qquad \sum_i \rho_i = 1, \tag{5.20}$$

Provenance (Eq. 5.20): **[Engineered]**: rooting rules with constants chosen for the workbench.

with $\lambda = 0.32$ (cereal), 0.24 (legume), 0.40 (vegetable) and z_d the drip emitter depth. The bracket is **[Engineered]**: it raises root density up to fourfold around a drip emitter. These root functions are the *modelled* roots drawn in the R1 layer, "not a scan of real ones".

5.2.6 Transpiration, evaporation

Daily soil evaporation is the annual evaporation E_{yr} of Eq. (5.4) distributed over the year with a sinusoidal weight, reduced by canopy shade,

$$E_d = E_{\text{yr}}\,\frac{w(\text{doy})}{\sum_{k=0}^{364} w(k)}\,(1 - 0.3\,c), \qquad w(\text{doy}) = 0.35 + 0.65 \cdot \tfrac{1}{2}\Big[1 + \sin\tfrac{2\pi(\text{doy} - 100)}{365}\Big], \tag{5.21}$$

Provenance (Eq. 5.21): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

and is taken from the top four layers with weights $(0.55, 0.25, 0.12, 0.08)$, never drawing a layer below $0.05\,W_{\text{max}}$. Crop transpiration demand is the annual transpiration spread by cover, times a **calibration factor** k_t,

$$T_d = k_t\,T_{\text{yr}}\,\frac{c(d)}{\bar{C}}, \qquad \bar{C} = \tfrac{1}{20}\sum_{20\ \text{yr}} c, \tag{5.22}$$

Provenance (Eq. 5.22): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

and is divided among the layers by *availability-weighted* root density,

$$q_i^T = T_d\,\frac{\rho_i\,a_i}{\sum_j \rho_j a_j}, \qquad a_i = \text{clip}\Big(\frac{s_i - 0.15}{0.6};\,0,\,1\Big), \tag{5.23}$$

Provenance (Eq. 5.23): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

limited so that no layer falls below $0.10\,W_{\text{max}}$. The availability function a_i is a linear stress function between 15 % and 75 % of plant-available water; it is **[Engineered]** in the spirit of the FAO-56 water-stress coefficient (Allen et al., 1998, Eq. 84), which is also linear in depletion.

5.2.7 Irrigation rule and calibration

An irrigation of dose d (mm) is applied on day d if *all* of these hold: canopy cover $c > 0.2$; $95 \le$ doy ≤ 250; at least m days since the last irrigation ($m = 1$, or $m = \text{clip}(\text{round}(110\,d/\bar{I}); 2, 12)$ if $f_T < 0.1$, to space irrigations when the trigger is almost at field capacity); the yearly total would stay within the budget $6\bar{I}$ (practically no cap); the sensor layer is dry enough, $1 - s_{j_s} \ge f_T$; $f_T < 0.995$; and, for drip, the layer at the emitter depth has depletion ≥ 0.12. The dose is $d = \text{clip}(\bar{I}/10; 15, 35)$ mm, and for drip $d_{\text{eff}} = \max\{8, d/2\}$ delivered as small pulses *spread over ± 3 layers around the emitter depth*; for other methods the dose enters at the surface. $\bar{I}$ is the annual irrigation of Eq. (5.7).

Calibration of k_t. The trigger decides *when* but the root-zone water use decides *how often*. To make the day-by-day model reproduce the annual irrigation of Engine 1, $k_t \in [0.6, 8]$ is found

by eight bisection steps on the first 8 years so that the mean irrigation per year I_{sim} matches $\bar{I}$ within 6 % (if the first guess $k_t = 1$ is already within 6 %, it is kept); the result is then used for all 20 years.

5.2.8 Scaling to the annual model

The daily results are multiplied by constants so that their 20-year means equal Engine 1:

$$S_P = \frac{P}{\overline{P_{\text{sim}}}}, \quad S_{\text{thr}} = \frac{P - P_{\text{other}}}{\overline{T_{f,\text{sim}}}}, \quad S_E = \frac{E}{\overline{E_{\text{sim}}}}, \quad S_T = \frac{T}{\overline{T_{\text{sim}}}}, \quad S_D = \frac{D}{\overline{D_{\text{sim}}}}, \tag{5.24}$$

Provenance (Eq. 5.24): **[Data-driven]**: scaling of the daily series to the Engine 1 twenty-year means.

(each set to 1 if the simulated mean is below 10^{-6}), and the daily fluxes displayed on the Dashboard are $R = P_d S_P$, $T_f = \min\{R,\, T_{f,d} S_{\text{thr}}\}$, $E = E_d S_E$, $T = T_d S_T$, $D = D_d S_D$ and irrigation I unscaled. The **storage change of the day** closes the balance:

$$\Delta S_d = T_f + I - E - T - D. \tag{5.25}$$

Provenance (Eq. 5.25): **[Literature]**: water-balance identity.

5.2.9 Evaporation under warming (used by the E1 layer)

For the Evaporation physics layer a separate atmosphere factor is used (Section 9.3):

$$E_{0,\text{bare}} = 4.2\,(1{+}0.07\,\Delta T)\,\frac{0.35 + 0.65 \cdot \tfrac{1}{2}[1 + \sin(2\pi(\text{doy} - 100)/365)]}{0.675}\,\text{clip}\big(1{-}0.12(\rho{-}1);\,0.85,\,1.12\big)\sqrt{\gamma_y} \tag{5.26}$$

Provenance (Eq. 5.26): **[Engineered]**: atmosphere factor with chosen constants.

where 4.2 mm d^{-1} is the engine's season-mean bare-soil demand (the seasonal weight is divided by its mean, 0.675), the factor $1 + 0.07\Delta T$ is a 7 % per °C increase (close to the Clausius–Clapeyron rate of the saturation vapour pressure; **[Engineered]** parameter), the cloud factor reduces demand in wetter years, and $\sqrt{\gamma_y}$ links demand to the dryness of the year.

5.3 The design layers and their link to the engines

The three design layers take from the engines:

Quantity	Source	Used by
Soil texture (sand, silt, clay %)	`DATA.texture` (measured for the 10 soils)	Saxton–Rawls intake (Section 11.2)
Trigger depletion f_T	Eq. (5.3) at h_T	Required depth Z_n
Maximum rooting depth R_{max}	Eq. (5.18)	Z_n
Plant-available water per cm AWC	Table 5.3	Z_n
Peak crop use ET_c	engine's peak daily transpiration plus evaporation	Interval T (drip, sprinkler)
Date and irrigation dates	Engine 2	Irrigation schedule tables
Weather source, mulch, interception	Engine 1 and mulch capacity	sprinkler interception loss

The default required depth is therefore

$$Z_n = \text{clip}\big(f_T \cdot \text{AWC} \cdot R_{\text{max}};\,25,\,150\big)\ \text{mm} \qquad (\text{surface; drip and sprinkler use their own limits}) \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

which is the net depth that refills the root zone from the trigger depletion back to field capacity (“*from the DSS*” on the screen). Typing a value overrides it (the label then reads *tap to reset*; click the label to return to the computed value).

5.4 Reproducibility

All random numbers in the engine are generated by a deterministic generator seeded by the soil name (a 32-bit hash), so the same controls always give the same dates, depths and animations. There is no external data call; the engine file contains the simulation tables as JSON.

Chapter 6

Irrigation control layers

The two layers of this group answer one question: **when should irrigation start?** *Sensor and trigger* shows what a sensor sees through twenty years; *Trigger law* shows why the same suction means different things in different soils and how to express the trigger so that it works for every soil.

6.1 Sensor and trigger

6.1.1 What it shows

A simulated tensiometer at the **sensor depth** (default 20 cm) records soil-water suction h (cm of water, plotted on a log axis from 1 to 100,000 cm) every day. Two traces are drawn: *irrigated* (blue) and *rainfed* (orange). The red dashed line is the **trigger** you set. Pale bands are growing seasons. Under the trace, red ticks mark irrigated sensor days beyond the trigger, and the bar chart counts, for each year, the growing-season days (of 183) on which the sensor was beyond the trigger, irrigated against rainfed.

6.1.2 How to use it, step by step

1. Open *Irrigation control* ▸ *Sensor and trigger*.
2. Choose **All 20 years** to see the whole record, or **1994 detail** for the one year with full-profile output (the buttons are on the playback bar under the drawing).
3. Move **Trigger (sensor suction)** in Properties; the red line and every count update.
4. Move **Sensor depth**, the soil, the mulch, the method or any other control: both traces follow it (Eq. 6.4a), and the sentence under the drawing names the record and the settings it is drawn for.
5. Press the play button under the drawing to sweep a cursor through the record; the working year is set in Properties ▸ Scenario.
6. Read the text under the plot: *“Irrigated: 85 of 183 days beyond 300 cm, mean suction 255 cm. Rainfed: 168 of 183 days beyond, mean suction 1,945 cm. Irrigation removes 39 % of the days beyond this trigger.”*

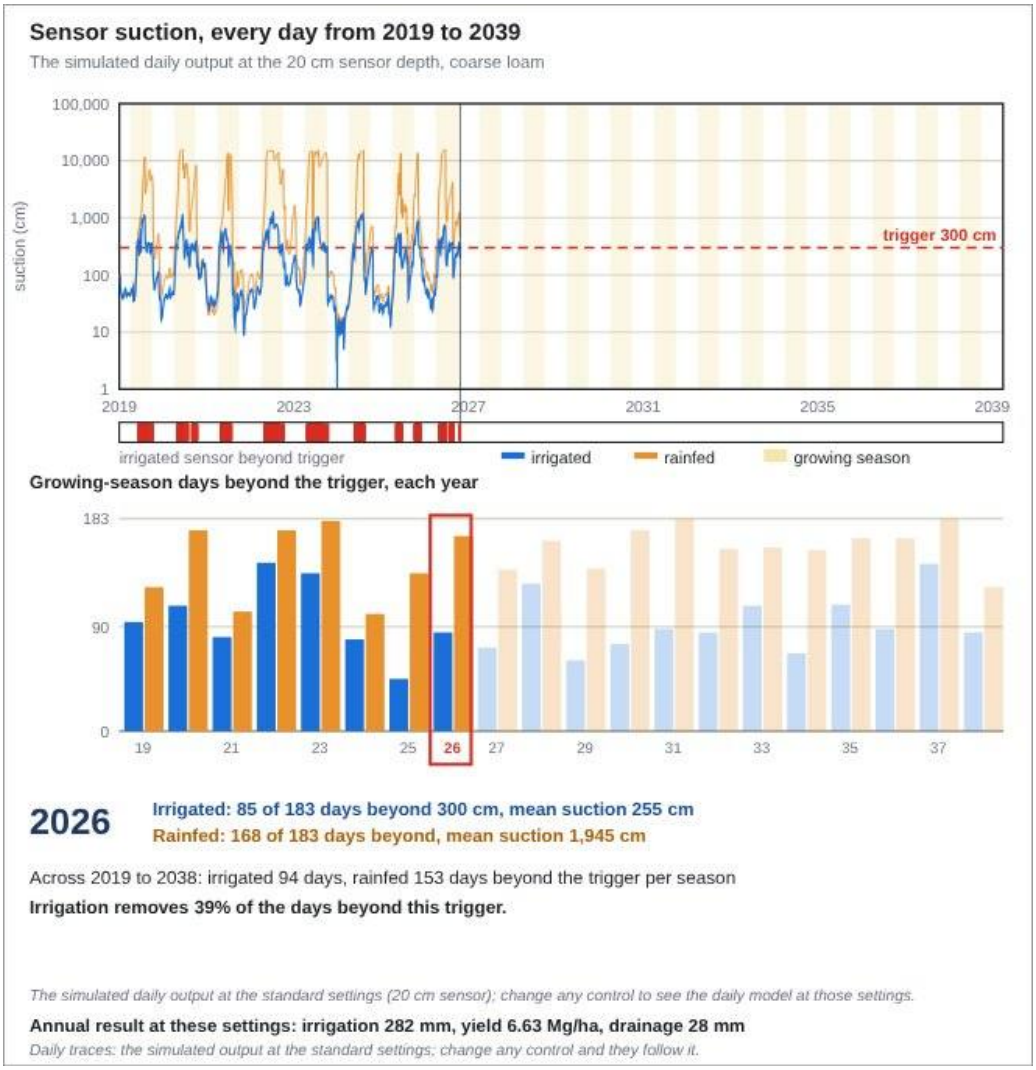


Figure 6.1: Sensor and trigger. Top: 20 years of suction at the sensor, irrigated (blue) and rainfed (orange) with the trigger (red dashed). Bottom: growing-season days beyond the trigger per year. Volume 2 shows the whole layer, part by part, with its equations.

6.1.3 How the numbers are generated

- The two suction traces start from the Daisy-simulated daily suction at 20 cm (irrigated with the study's trigger, and rainfed) for the 1980–1999 record, stored in the engine (`D20`) for the Coarse Loam and the Coarse Sand. At the standard settings of those two soils they are drawn as stored. For any other soil, sensor depth or setting the record is moved by the **delta method**: the daily model (Engine 2) is run twice, at your settings and at the standard settings of the same soil and weather, and the difference of the depletion of the sensor layer (smoothed over seven days) is added to the depletion of the stored record, which is then converted to suction in your soil:

$$h_d^{\mathrm{x}} = h_{\mathrm{soil}}\Big(f_{\mathrm{ref}}(h_d^{\mathrm{rec}}) + \tfrac{1}{7} \sum_{q=-3}^{3} [f_{d+q}^{\mathrm{set}} - f_{d+q}^{\mathrm{std}}] \Big), \tag{6.4a}$$

Provenance (Eq. 6.4a): **[Data-driven + Engineered]**: the stored simulated record carries the real weather sequence; the change made by your settings is the difference of two runs of the daily model, converted with the van Genuchten curve of your soil (the inverse of Eq. 5.3, tabulated).

with f_{ref} the depletion of the reference soil whose record is used (the nearer of the two in texture), f^{set} and f^{std} the daily depletion of the sensor layer in the two runs (irrigated or rainfed as the

trace), and $h_{\text{soil}}(f)$ the suction of your soil at depletion f. * The count of days beyond the trigger is, for year y,

$$N_y(h_T) = \#\{\, d \in \text{growing season of year } y : h_d > h_T \,\} \quad (183 \text{ days from 1 April}), \tag{6.1}$$

Provenance (Eq. 6.1): **[Data-driven]**: counts over the simulated daily series.
and the "Irrigation removes X %" statement is

$$X = 100\Big(1 - \frac{\bar{N}_{\text{irr}}}{\bar{N}_{\text{rainfed}}}\Big), \qquad \bar{N} = \tfrac{1}{20}\sum_y N_y. \tag{6.2}$$

Provenance (Eq. 6.2): **[Data-driven]**: counts over the simulated daily series.

- The annual line at the foot ("irrigation 282 mm, yield 6.63 Mg ha^{-1}, drainage 28 mm") is Engine 1 (Eqs. 5.4–5.12) at the current settings.

6.1.4 Reading it

If rainfed days beyond the trigger greatly exceed irrigated days, irrigation is doing its job; if the two bars are similar the trigger is so dry that irrigation rarely starts. A trigger of -50 cm (very wet) produces ticks on almost every day.

6.2 Trigger law

6.2.1 The idea

The same suction does not mean the same dryness in two soils. At -300 cm a sand has lost 84 % of its plant-available water while a silty clay loam has lost only 24 % (Table 5.3, last column). N1 re-expresses the trigger as the **depletion fraction** f of Eq. (5.3), the share of plant-available water already used, and shows that irrigation and yield responses from all ten soils collapse onto a common curve when plotted against f.

[Literature + study result]. The depletion form follows the management-allowed-depletion idea of FAO-56 (Allen et al., 1998, Eq. 83 and Table 22): irrigate when a fixed share p of the total available water is used. The *soil-independent* critical value $f^* = 0.45$ is the study's result (5 % yield loss), and the pooled R^2 rising from 0.48 to 0.75 (irrigation) and 0.23 to 0.70 (yield) is reported in the repository.

6.2.2 How to use it, step by step

1. Open *Irrigation control* ▸ *Trigger law*.
2. The soil column on the left is the soil by depth on the clock's day; the blue gauge beside it shows depletion from FC (0) to WP (1). **Drag the purple trigger marker** on the gauge: the soil's own trigger suction changes, and the right-hand list of ten soils moves in step.
3. Each row on the right is one soil with its trigger suction and the net irrigation depth at that depletion ("143 cm trigger, 278 mm"); the green box gives the relative yield at the soil's trigger.
4. Click a soil name to select it (it becomes the working soil).
5. **Common and own mode.** By default one suction (the Trigger slider) applies to every soil (*common* mode). **Opening the Trigger law layer switches the program to *own* mode**: each soil then uses its own critical suction for the chosen depletion, and the Trigger slider in Properties shows the nearest value on its 20-value grid (simulated runs exist only on that grid). Own mode stays on until you choose *Reset all settings* or a ready-made case, which restore the common mode.
6. Read the verdict box at the bottom: it states the soil's critical suction, the yield at the current trigger relative to the best trigger, and how many root-zone layers are beyond the trigger on this day.

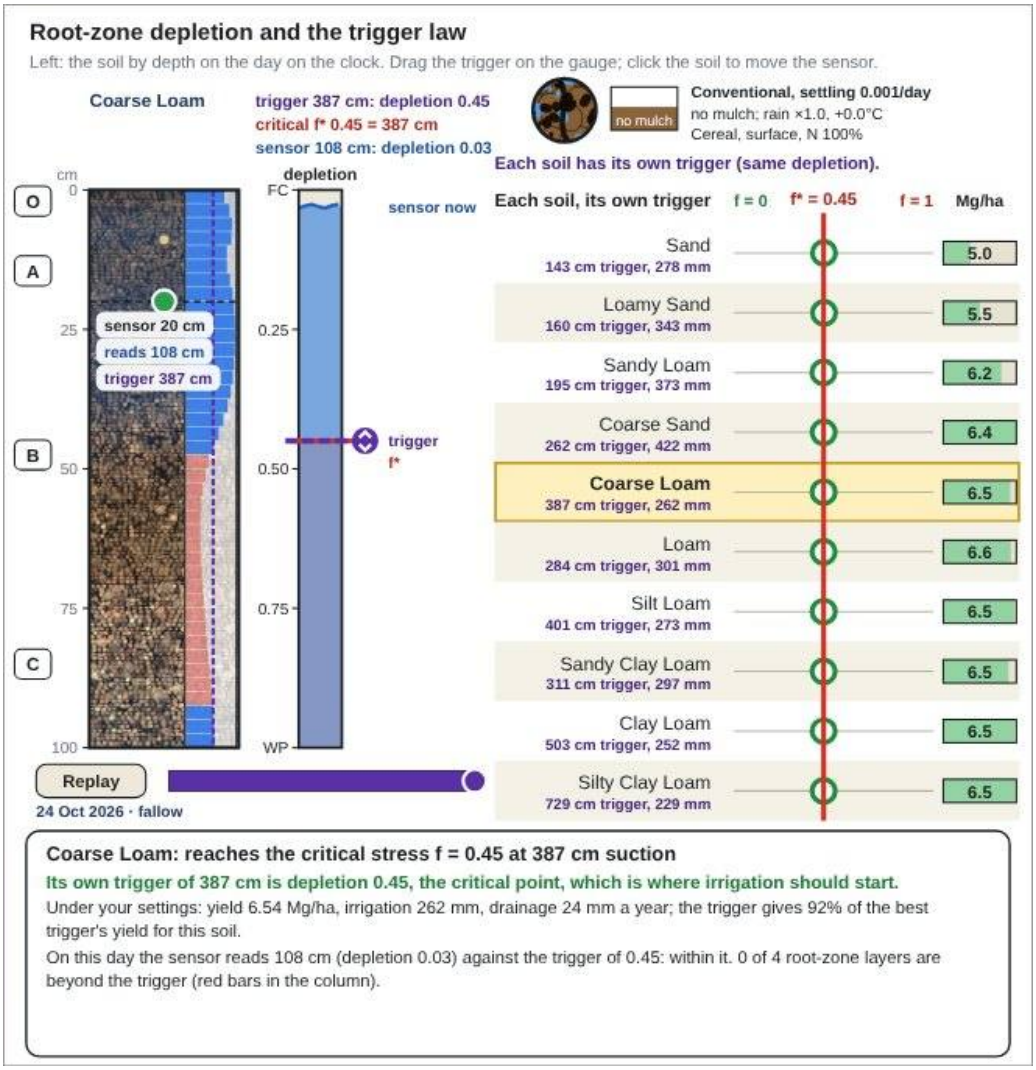


Figure 6.2: Trigger law layer: ten soils on one depletion axis, with the critical $f^* = 0.45$ (red line) and the soil's own trigger. Volume 2 shows the whole layer, part by part, with its equations.

6.2.3 Governing relations

The depletion of Eq. (5.3) is evaluated with each soil's effective van Genuchten pair (α, n) (Table 5.3). The suction at which a soil reaches a chosen depletion f is the inverse,

$$h(f) = \Big\{\, h : f(h) = f \,\Big\}, \tag{6.3}$$

Provenance (Eq. 6.3): **[Literature]**: inverse of the van Genuchten curve, solved by bisection.
solved by bisection, giving for $f^* = 0.45$ the soil-specific sensor settings of Table 5.3 (143 cm for sand to 729 cm for silty clay loam). The net irrigation per event shown beside each soil is

$$Z_n = f \cdot \mathrm{AWC} \cdot R_{\max} \quad [\mathrm{mm}] \tag{6.4}$$

Provenance (Eq. 6.4): **[Literature]**: same relation as Eq. (5.27) without the clip.
(the same relation as Eq. 5.27 without the clip). The *relative yield at the trigger* is

$$Y_{\mathrm{rel}} = \frac{Y(h_T)}{\max_h Y(h)}, \tag{6.5}$$

Provenance (Eq. 6.5): **[Data-driven]**: ratio of Daisy run-set E2 yields.
with $Y(h)$ the soil's yield at each simulated trigger (run set E2). **[Data-driven]**.

6.2.4 Reading the sensor value

The dashboard tile *Sensor / trigger* reads two numbers: the **current depletion** f at the sensor layer and the **trigger depletion** f_T. When the first reaches the second, irrigation is due. On 18 Oct 2026 the tile reads $0.02 / 0.38$ for the Coarse Loam: the soil is almost at field capacity (0.02), and the trigger is at depletion 0.38 ($= f$ at 300 cm).

Chapter 7

Field water balance layers

7.1 Field balance (the home layer)

7.1.1 What it shows

One picture of the water budget of a field for one day, one year, or the twenty-year average: rain and irrigation arriving, transpiration, soil evaporation and interception leaving upward, drainage leaving downward, and the nitrogen leached with it. The soil profile is drawn to 100 cm with three depth marks A, B and C, the sensor at the chosen depth, the roots as they grow and die back, and the wetting front.

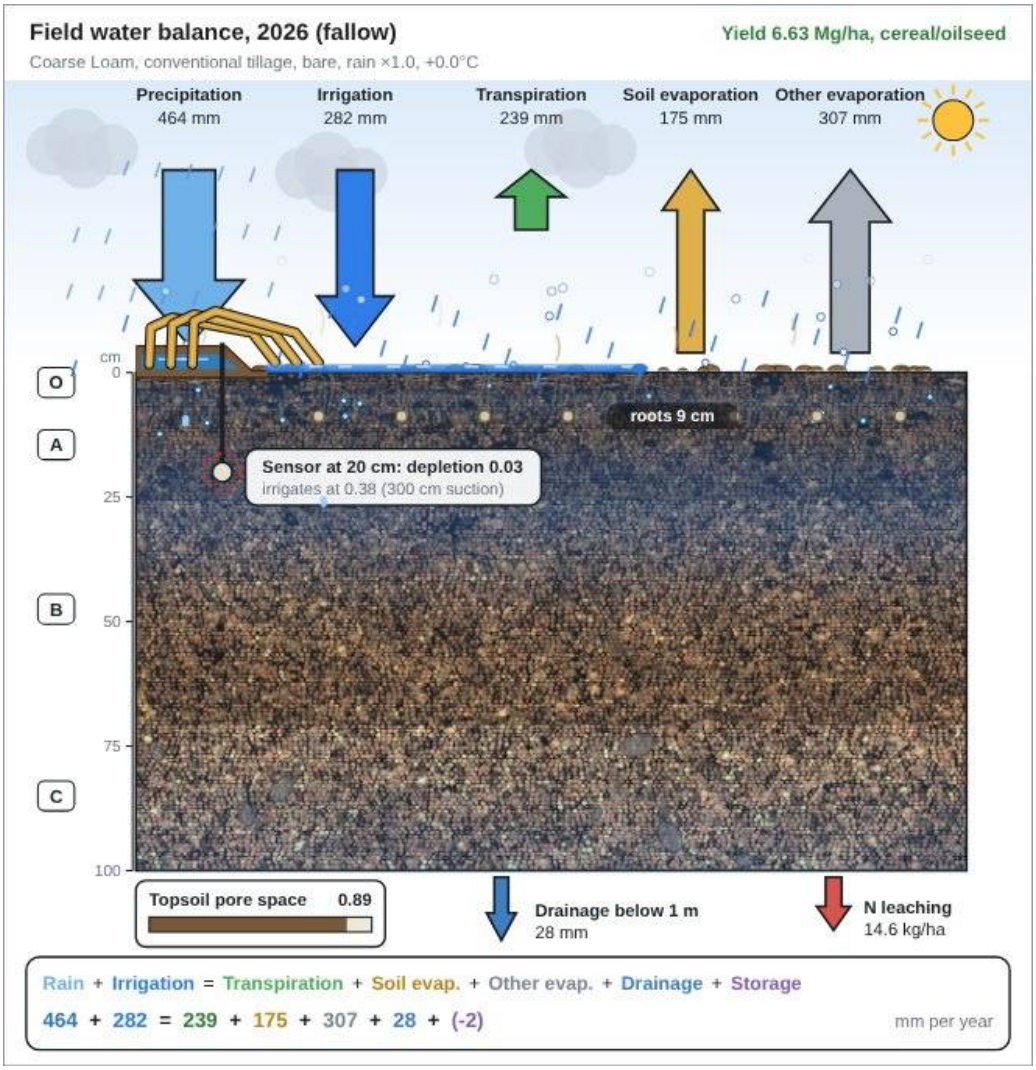


Figure 7.1: Field balance: arrows are the fluxes (mm per year in “Average year” view), the profile shows the root zone, the sensor, and drainage below 1 m. The equation strip under the picture is the closed budget. Volume 2 shows the whole layer, part by part, with its equations.

7.1.2 The control bar under the drawing

Button	Meaning
Day by day	The simulated day on the field clock; arrows show that day’s fluxes (T_f, I, T, E, interception, D).
All 20 years	Table and bars of every simulated year (rain, irrigation, interception, evaporation, transpiration, drainage, storage change, crop, deepest roots). Click a year.
Average year	The 20-year means of Engine 1 (Eqs. 5.4–5.12).
◀ year / year ▶	Previous or next simulated year.
1× 2× 4× 8× and speed slider	Play speed.
Play / Pause	Animates the clock.

7.1.3 How each output is generated

On the picture	Generated by
Precipitation, Irrigation, Transpiration, Soil evaporation, Interception, Drainage (mm) in *Average year*	Engine 1: P, I, T, E, E_{other}, D of Eqs. (5.4)–(5.12)
The same in *Day by day*	Engine 2 daily fluxes after scaling, Eqs. (5.15)–(5.25)
Equation strip “Rain + Irrigation = Transpiration + Soil evap. + Interception + Drainage + Storage”	The closure of Eq. (5.12): $P + I = T + E + E_{\text{other}} + D + \Delta S$

On the picture	Generated by
Sensor label “depletion 0.02 / irrigates at 0.38”	Engine 2 layer j_s: $f_{j_s} = 1 - s_{j_s}$ against f_T
Roots (depth, density)	Eqs. (5.19)–(5.20)
Wetting front	Depth of the deepest layer reached by the last infiltration event
Topsoil pore space bar	Pore index p of Eq. (8.1) for the chosen tillage and consolidation rate
Drainage below 1 m	D (mm), the water leaving layer 40
N leaching (kg ha^{-1})	N_L of Eq. (5.10)
Yield tag (top right)	$Y \cdot f_{\text{crop}}$ from Eq. (5.8)

7.1.4 Step-by-step: checking a budget

1. Open *Field balance* and choose **Average year**.
2. Read the five numbers over the arrows and the strip below. Check that rain plus irrigation minus the four sinks equals the storage change in brackets.
3. Change **Rainfall scale** to $×$1.3. The rain, drainage and N leaching arrows thicken.
4. Press **Pin results as baseline** (Ctrl+B) *before* each change to see the difference on the Dashboard tiles.
5. Click **Day by day**, Play, and watch irrigation events pause the clock (Section 4.3).

7.2 Soil, plant and air

7.2.1 What it shows

The soil–plant–atmosphere continuum as one water column “from the particles to the sky”: a section of the soil with roots, the plant, and the air above; five irrigation layouts (**Furrow, Basin, Border strip, Sprinkler, Drip**) that you can tap, which set the irrigation method; and, on the right, the **water potential along the path** from the air to the deep soil. Two data sources are labelled with badges: *DAISY* (taken from your simulations) and *DERIVED* (computed by the workbench along the path).

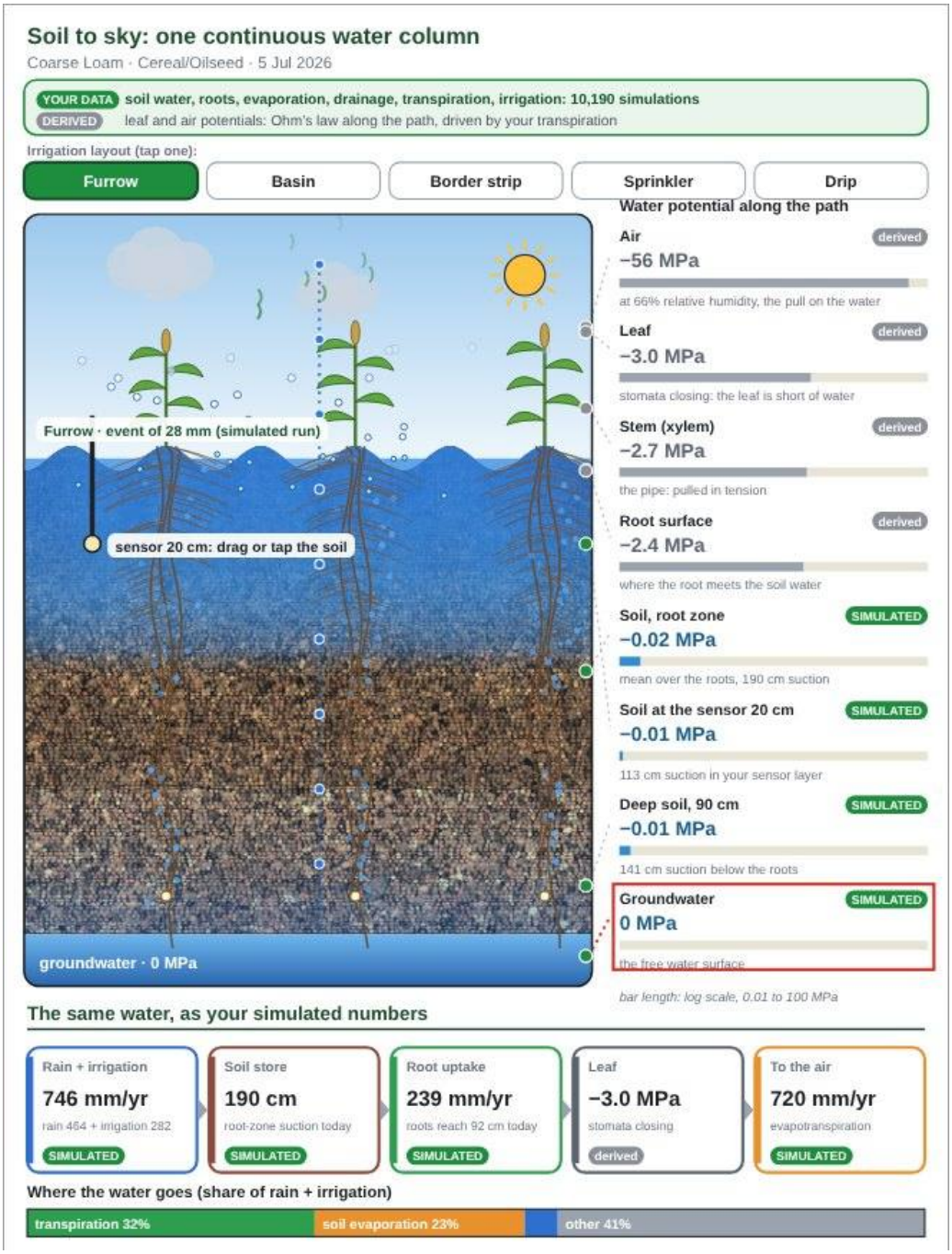


Figure 7.2: Soil, plant and air layer: potentials from the air to the deep soil. *Derived* badges mark the quantities computed along the path; *Simulated* badges mark simulation data. Volume 2 shows the whole layer, part by part, with its equations.

7.2.2 Governing equations

Soil. The suction in each layer is the suction that gives that layer's depletion, from the inverse of Eq. (5.3), floored at 30 cm:

$$h_i = \max\{30,\ h(f_i)\}, \qquad f_i = 1 - s_i, \qquad \psi_{\text{soil},i} = -h_i \cdot 9.807 \times 10^{-5}\ \text{MPa cm}^{-1}. \tag{7.1}$$

Provenance (Eq. 7.1): **[Literature]**: inverse of Eq. (5.3); the 30 cm floor is engineered.

The *root-zone* suction is the root-density-weighted mean, $h_{RZ} = \sum_i \rho_i h_i \big/ \sum_i \rho_i$, and the *sensor* and *deep-soil* values are h_{j_s} and h_{N-3}.

Soil conductivity. The relative conductivity at the root zone uses the Mualem–van Genuchten form, Eq. (5.2), with $S_e = S_e(h_{RZ})$ from Eq. (5.1).

Along the path. The plant is a chain of series resistances, in the spirit of the catenary (Ohm's-law) description of water flow through the soil–plant–atmosphere continuum (van den Honert, 1948). With E the day's transpiration (mm d^{-1}, from Eq. 5.25) and R_{soil}, R_{root}, R_{xyl}, R_{leaf} resistances in MPa per (mm d^{-1}):

$$R_{\text{soil}} = 0.03\ \max\{K_r,\ 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

where $a = 1$ when canopy cover exceeds 0.05 (a crop is present) and $a = 0$ otherwise (fallow). Stomata are drawn as *closing* when $\psi_{\text{leaf}} < -1.6$ MPa.

The air. Water potential of the air above the canopy follows the thermodynamic relation between potential and relative humidity,

$$\psi_{\text{air}} = \frac{R\,T}{V_w}\ \ln \text{RH}\ \approx\ 135.5\ \ln \text{RH}\ \ \text{MPa} \tag{7.5}$$

Provenance (Eq. 7.5): **[Literature]**: Nobel (2009).

(Nobel, 2009), where R is the gas constant, $T \approx 293$ K and $V_w = 18 \times 10^{-6}$ m^3 mol^{-1} the molar volume of water. Relative humidity is RH $= \text{clip}(0.66 - 0.03\,\Delta T;\ 0.35,\ 0.90)$ with ΔT the temperature shift. At RH = 66 % the air is at -56 MPa, as shown.

What is literature and what is engineered. Eqs. (7.1), (7.5) and (5.2) are physical relations. The resistances in (7.2)–(7.4) are **[Engineered]** values chosen so that a typical transpiring cereal reaches ψ_{leaf} of about -1 to -1.5 MPa and so that stress develops as the soil dries; they are *not* measured resistances, and the layer labels them "derived". **Caution.** Evaluated over mid-season crop days in all ten soils (Paper 3, Section 5.1), the relative conductivity at the root-zone suction is usually below the floor of 0.002 in Eq. (7.2), so R_{soil} sits at its cap (0.92 MPa per mm d^{-1}) and the mean leaf potential is -1.8 to -2.3 MPa in every soil, below the intended range. The leaf, stem and root potentials of this layer therefore illustrate the series-resistance concept and should not be read as plant water status.

7.2.3 Step-by-step

1. Open *Soil, plant and air*.
2. Click a layout. Furrow, basin and border strip set the **Surface** method; Sprinkler sets **Overhead**; Drip sets **Drip 20 to 30 cm**.
3. Drag or click in the soil to move the **sensor depth** (the control snaps to the 5-cm list).
4. Change the **date** to a growing-season day and compare the potentials: with a crop present the leaf falls below -1 MPa as the soil dries; on a fallow date the path is flat (resistance terms are zero).
5. Move the **Trigger** slider and note that the *sensor* suction tile follows the same depletion as the dashboard.

7.3 Profile evolution

7.3.1 What it shows

How the pressure head profile develops through the season in a year, side by side for **irrigated** and **rainfed** soil columns, with the sensor depth shown by a dot. In *All 20 years* mode it draws the pressure head at the sensor through every year and a heat strip of the mean suction of each growing season (darker is drier). Text states the driest and wettest rainfed season and the share of the 0–60 cm root zone beyond the trigger.

7.3.2 How it is generated

- The **1994** profile is the Daisy-simulated pressure head at every depth, irrigated and rainfed (the only year with full-depth output, as the drawing says).
- In the other years only the 20 cm sensor is available; the columns are shaded only around the sensor depth.
- *Share of the root zone beyond the trigger* is

$$\text{share} = \frac{\sum_{i:\ z_i \le 60\,\text{cm},\ h_i > h_T} \Delta z_i}{\sum_{i:\ z_i \le 60\,\text{cm}} \Delta z_i}, \tag{7.6}$$

Provenance (Eq. 7.6): **[Engineered]**: indicator defined for the workbench.

the layer-thickness-weighted fraction of the 0–60 cm profile whose suction exceeds the trigger, evaluated for the irrigated and the rainfed profile of 1994 (the year with full-depth output). * The mean suction of a season, used for the heat strip and for the “driest season” statements, is the geometric mean $m_y = 10^{\frac{1}{183}\sum_d \log_{10} h_d}$ over the growing season; the median across years m_{med} also drives the rainfall year factor γ_y (Eq. 5.15).

7.3.3 Step-by-step

1. Open *Profile evolution* and choose **All 20 years**.
2. Click a year in the heat strip to select it; click a column to put the sensor at that depth.
3. Move the trigger: the share beyond the trigger is recomputed.
4. Choose **1994 detail** to see the full profile through the season; press Play to step through it.

Chapter 8

Soil structure and tillage layers

8.1 Tillage and structure

8.1.1 What it shows

A modelled thin section (“microscope”) of the topsoil for the chosen soil and tillage, at three magnifications (10 mm, 2.5 mm, 0.6 mm), with pores filled by water or air according to a suction slider; the same place under the three tillage systems; the relative pore space against time; and the change in drainage and yield caused by tillage at your settings.

8.1.2 Pore index and consolidation

After tillage the loosened topsoil re-settles. The workbench uses a first-order relaxation of the **relative pore space** p (1 = freshly loosened) to a consolidated value $p_c = 0.30$,

$$p(t) = p_c + (p_0 - p_c)\, e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

with t in days since tillage and τ_c the **Structural consolidation rate** control (Table 3, 0.001–0.02 d^{-1}). **No-till** keeps a constant $p = 0.42$ (undisturbed pores). The half-life of loose pores is $\ln 2/\tau_c$: 693 d at 0.001 d^{-1}, 35 d at 0.02 d^{-1}. **[Engineered]**: the exponential form is the standard first-order consolidation relation also used in structure sub-models such as WEPP’s, on which Daisy’s tillage effect is based, but p_0, p_c and the no-till value are *presentation* constants for the drawing; the *hydrological effect* of tillage shown in the bars is read from the Daisy runs (E5), not from Eq. (8.1).

8.1.3 How the section is generated

The section is a random packing of mineral grains and matrix drawn on a raster, with these parameters, all **[Engineered]**:

- Grain-size classes follow the soil’s measured sand, silt and clay percentages (`DATA.texture`).
- **Saturated water content** $\theta_s = \phi$ and **residual content** θ_r are

$$\phi = \text{clip}\big(0.36 + 0.0016\,\text{clay} + 0.0009\,\text{silt};\ 0.34,\ 0.50\big), \qquad \theta_r = 0.01 + 0.0006\,\text{clay}, \tag{8.2}$$

Provenance (Eq. 8.2): **[Engineered]**: porosity relation with chosen coefficients.

with clay and silt in percent, and the retention curve is the van Genuchten function of Eq. (5.1),

$$\theta(h) = \theta_r + (\phi - \theta_r)\, S_e(h). \tag{8.3}$$

Provenance (Eq. 8.3): **[Literature]**: van Genuchten (1980).

- **Cracks and aggregates.** The number of cracks is $n_b = \text{round}(2 + 6(1 - f_{\text{till}}))$ with $f_{\text{till}} = 1,\ 0.7,\ 0.4$ for conventional, reduced, no-till; the crack width (mm) is $0.3\ \max\{0,\ p - 0.3\}^{1.3}$.

- **Which pores hold water.** A pore of radius r fills with water at suction h if $r \le r_c(h)$, where the capillary radius follows the Young–Laplace (Jurin) law,

$$r_c = \frac{2\,\sigma\cos\gamma}{\rho_w\, g\, h} \;\approx\; \frac{0.149\ \text{cm}^2}{h} \quad (\text{or } r_c[\text{mm}] \approx 1.49/h[\text{cm}]), \tag{8.4}$$

Provenance (Eq. 8.4): **[Literature]**: Young–Laplace (Jurin) capillary law.

with surface tension $\sigma = 0.0728$ N m^{-1}, contact angle $\gamma \approx 0$ and $g = 9.81$ m s^{-2} (Jurin, 1718; Hillel, 1998). In the drawing a pore is water-filled when its local half-width is at most r_c **or** when it is within a film thickness $\min\{1.6,\ 0.9\,r_c/s + 0.3\}$ pixels of a grain, which draws the water films that survive drying.

- The hover probe states, for a pore, its radius, the suction at which it empties ($h = 1.49/r$) and the present suction.

8.1.4 Controls and steps

1. Open *Soil structure and tillage* ▸ *Tillage and structure*. Press **Play** to run a year of consolidation (the day appears under the picture).
2. Drag the **suction slider** (pF 0–4.2) on the microscope: at low suction the pores are water-filled; as suction rises the large pores empty first.
3. Switch the view: *Natural* (minerals), *Particle sizes* (sand/silt/clay classes), *Water and air*.
4. Change **Tillage** or the **Structural consolidation rate** and watch the curves of $p(t)$ (Eq. 8.1) and the section change.
5. Read the bars *Change in drainage at the current settings* (conventional, reduced and no-till against no-till) and the yields. These come from the Daisy runs for the selected soil: $\Delta D_{\text{conv}} = D_{\text{conv}} - D_{\text{notill}}$, $\Delta D_{\text{red}} = D_{\text{red}} - D_{\text{notill}}$, at the current τ_c.
6. The sentence at the end states whether the consolidation rate changes anything: *"In this soil the consolidation rate changes nothing; tillage is the driver"* or *"Here the consolidation rate matters as well as tillage"*.

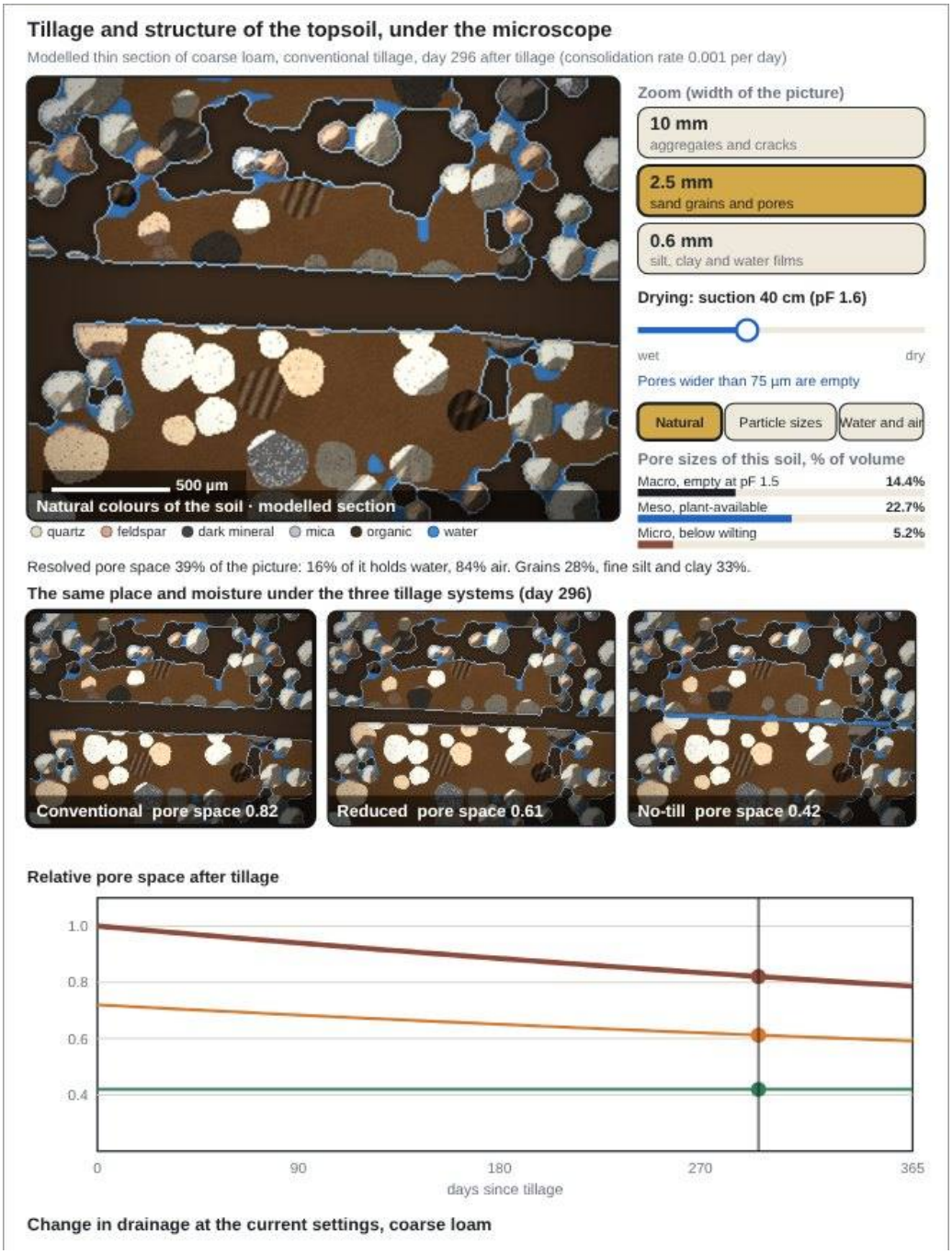


Figure 8.1: Tillage and structure layer: thin section, tillage comparison and the drainage consequences. Volume 2 shows the whole layer, part by part, with its equations.

8.2 Soil hydraulic drivers

8.2.1 What it shows

The question *“What drives change in soil hydraulic properties?”* answered for all ten soils together: for each soil sorted by clay content, a blue bar for the **tillage effect** (conventional minus no-till) and an orange bar for the **consolidation effect** (fastest minus slowest rate) on annual drainage, then scatter plots against clay content, and the 1-m soil profiles of the three tillage systems.

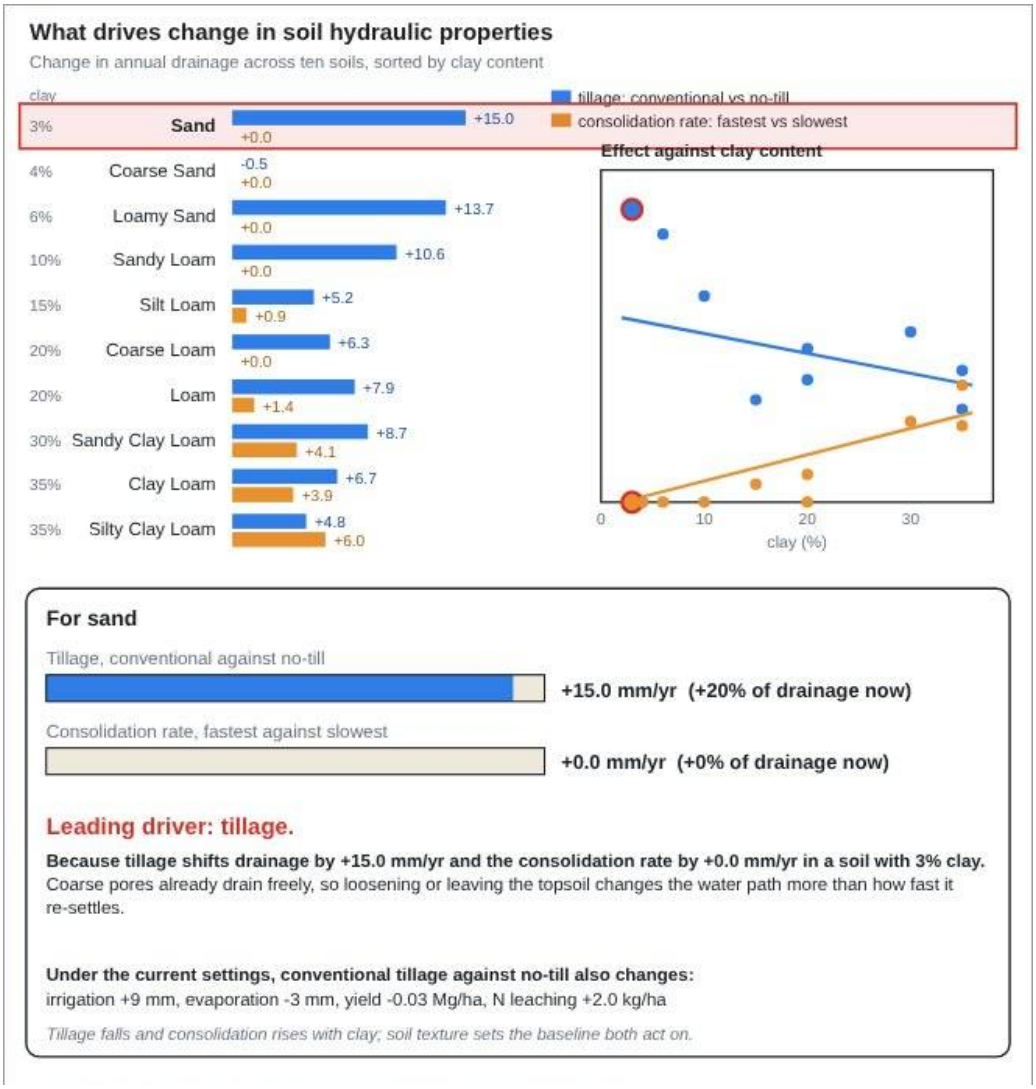


Figure 8.2: Soil hydraulic drivers. Volume 2 shows the whole layer, part by part, with its equations.

8.2.2 Generation

For each soil s the Daisy run set E5 gives annual drainage $D_s(\text{tillage}, \tau_c)$. The two bars are

$$\Delta_{\text{till}}(s) = D_s(\text{conv}, \tau^*) - D_s(\text{notill}, \tau^*), \qquad \Delta_{\text{cons}}(s) = D_s(\text{till}^*, \tau_{\text{fast}}) - D_s(\text{till}^*, \tau_{\text{slow}}), \quad (8.5)$$

Provenance (Eq. 8.5): **[Data-driven]**: differences of Daisy run-set E5 results.

with the current settings as starred values; bar length is proportional up to 16 mm yr^{-1}. The lines in the scatter plots are ordinary least-squares fits of each effect on clay content c,

$$\hat{\Delta} = a + b\,c, \qquad b = \frac{\sum_s (c_s - \bar{c})(\Delta_s - \bar{\Delta})}{\sum_s (c_s - \bar{c})^2}, \quad a = \bar{\Delta} - b\,\bar{c}. \quad (8.6)$$

Provenance (Eq. 8.6): **[Data-driven]**: ordinary least squares applied to Daisy results.

The headline “Leading driver: tillage” or “consolidation rate” compares Δ_{till} with Δ_{cons} for the selected soil, and the percentage after each bar is $100\,\Delta / D_{\text{now}}$, the share of the present drainage. The last line adds, for conventional against no-till at the current mulch and rain, the changes in irrigation, evaporation, yield and N leaching, obtained by running the meta-model (Eqs. 5.4–5.12) twice (conventional at τ index 2 and no-till) and subtracting.

The lower part of the layer draws, for the three tillage systems, a 48 cm wide, 1 m deep face of the selected soil. The upper 24 cm (the tilled layer) is generated with the pore index p of Eq.

(8.1) at day 120 (adjustable), clod and crack density from f_{till}, and the wetness factor $\text{clip}\big((p-0.30)/0.14;\,0.3,\,1.4\big)\cdot\text{clip}(0.75+0.012\,\text{clay};\,0.8,\,1.4)$; below it the subsoil is untouched by tillage. The drainage figures printed under each column are Daisy results (E5).

8.2.3 Steps

1. Open the layer. Click any soil bar or any scatter point to select that soil.
2. Click a point on the *Drainage against consolidation rate* graph to apply a rate and a tillage system at once.
3. Read the verdict, and compare with the profile drawings.

8.3 Root structure

8.3.1 What it shows

Rooting depth in all ten soils at the same day and weather: for each soil a column with the deepest roots (20-year mean, brown dashed), and ticks at the depth that holds 50 % and 90 % of the water taken up (green ticks). Below it, a time view of rooting depth through the 20 years and the daily uptake, and a set of heat maps from the repository's **E15** runs: 1,000 Daisy runs on the Coarse Loam, rooting depth 30–180 cm × ten application amounts × ten irrigation methods, mapped to yield, irrigation, drainage, transpiration, N leached and water-stress days.

8.3.2 Generation

Roots and uptake come from Engine 2 (Eqs. 5.18–5.23). For each soil the engine is run in turn with that soil's parameters (and, in own mode, its own trigger). With per-layer uptake $u_i(d)$ on day d the **depth holding a fraction p of the uptake** is

$$z_p = k_p\,\Delta z, \qquad k_p = \min\Big\{k:\ \frac{\sum_{i\le k} U_i}{\sum_i U_i} \ge p\Big\}, \qquad U_i = \sum_{d\in\text{year}} u_i(d), \quad p = 0.5,\ 0.9, \tag{8.7}$$

Provenance (Eq. 8.7): **[Engineered]**: diagnostic defined for the workbench.

with the layer index counted from 1 (z_p is the *bottom* of layer k_p). It is computed for each of the 20 years from that year's uptake, then averaged over the years; the share of uptake in the top 30 cm is $\sum_{z_i\le 30} U_i \big/ \sum_i U_i$. The model stops roots at 1 m, whereas the Daisy runs reach 120 cm, as the layer notes. The statement "*Across the soils the deepest roots differ by …*" is the range of the 20-year-mean $\max_d R_d$ across soils.

What these roots are. The drawing states it: *"These are modelled roots, not a scan of real ones"*; root density is *relative* (Eq. 5.20), not measured root length. The E15 heat maps are **[Data-driven]** and exist for the Coarse Loam only ("they do not change with the soil you picked").

8.3.3 Steps

1. Open the layer. Click a column to make that soil the working soil.
2. Click or drag on the upper graph to move the day; the lower panel shows the uptake on that day.
3. In the E15 panel choose an output (yield, irrigation, …), click a rooting depth and an application amount in the heat map; the table shows the change against the present setting and tells you the best depth, for example *"Roots at 30 cm leave the crop stressed …"*.

Chapter 9

Mulch and evaporation layers

This group follows the water that mulch keeps in the soil: how mulch lowers evaporation (V4, V6), the physics of drying (E1), where the saved water goes (N2) and when evaporation or drainage dominates (N3).

9.1 Mulch against bare soil

What it shows. Three soil blocks side by side: bare soil, the soil under *your* mulch, and under the strongest mulch (plastic film, $\varphi = 0.1$), with bars for evaporation, irrigation and drainage, the six materials as buttons (click one to apply it), and a verdict.

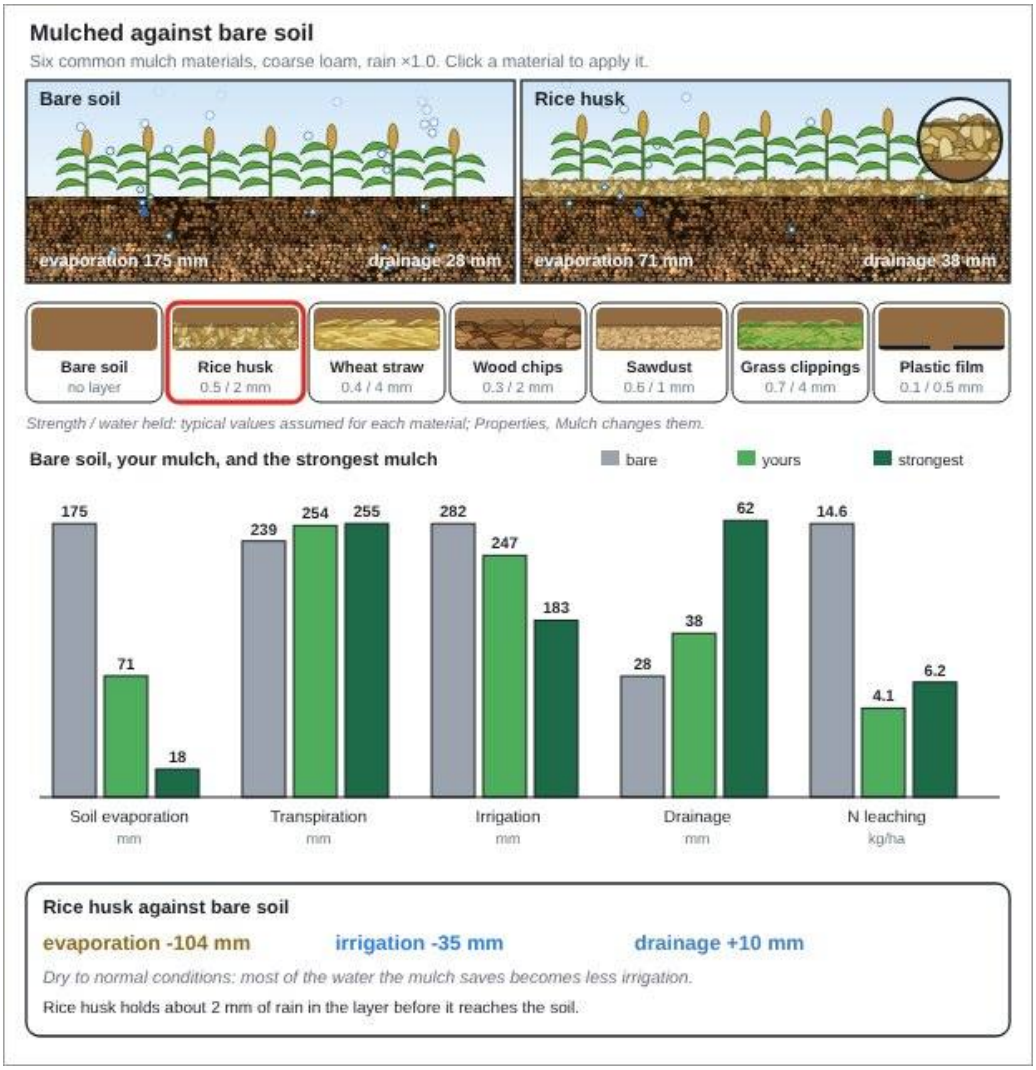


Figure 9.1: Mulch against bare soil. Volume 2 shows the whole layer, part by part, with its equations.

How the numbers are generated. All three blocks are Engine 1 evaluated at $\varphi = 1.0$ (bare), at your φ and capacity, and at $\varphi = 0.1$, at the current rainfall. With E_φ, I_φ, D_φ from Eqs. (5.4), (5.7), (5.9),

$$\Delta E = E_{1.0} - E_\varphi, \qquad \Delta I = I_{1.0} - I_\varphi, \qquad \Delta D = D_\varphi - D_{1.0}, \tag{9.1}$$

Provenance (Eq. 9.1): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

are the evaporation saved, the irrigation saved and the extra drainage, in mm yr^{-1}, printed under the blocks. The verdict compares ΔI with ΔD (“*Wet conditions: much of the water the mulch saves now leaves as drainage*” or “*Dry to normal conditions: most of the water the mulch saves becomes less irrigation*”). The block of text for plastic film states that “*plastic film stops vapour almost completely but holds no water, so rain runs to the planting holes*”, which is the capacity of 0.5 mm in Table 3.1.

Steps. (1) Open *Mulch and evaporation* ▸ *Mulch against bare soil*. (2) Click a material. (3) Raise the **Rainfall scale** to $×$1.3 and watch ΔD overtake ΔI. (4) Move the **Strength** slider to see the evaporation saving grow convexly.

9.2 Mulch control

What it shows. Two matrices. The first is **soil evaporation against mulch factor (ten rows, 0.1–1.0) by interception capacity (four columns, 0.5–4 mm)**. The second is **drainage**

against tillage system (three) by consolidation rate (five). Colour darkens with the value; a red outline marks your current setting. The text states what fraction of bare-soil evaporation your mulch removes and whether the soil responds to tillage.

Generation. Each cell of the first matrix is $E(\varphi, c)$ and each cell of the second $D(\text{till}, \tau_c)$, both from Engine 1 at the current soil, rainfall and temperature (Eqs. 5.4 and 5.9). The sentence "*No response here: this soil does not react to tillage or consolidation*" appears when the spread $\max D - \min D$ of the second matrix is below 0.5 mm yr^{-1}. The percentage in the text is $100\,(1 - E_{\varphi,c}/E_{1.0})$.

Steps. Click a cell to apply that mulch factor and capacity or that tillage and rate; the red outline moves; the Dashboard changes at once.

9.3 Evaporation physics

What it shows. A modelled thin section of the topsoil drying after a wetting, with liquid water in the pores (blue), vapour moving up (white), the **drying front** (orange) and, for comparison, the bare-soil curve (grey). To the right: the evaporation rate $e(t)$ and cumulative loss $Q(t)$ for 24 days, split into **stage 1** (the surface stays wet, loss at the atmospheric rate) and **stage 2** (a dry layer grows, loss is limited by the soil).

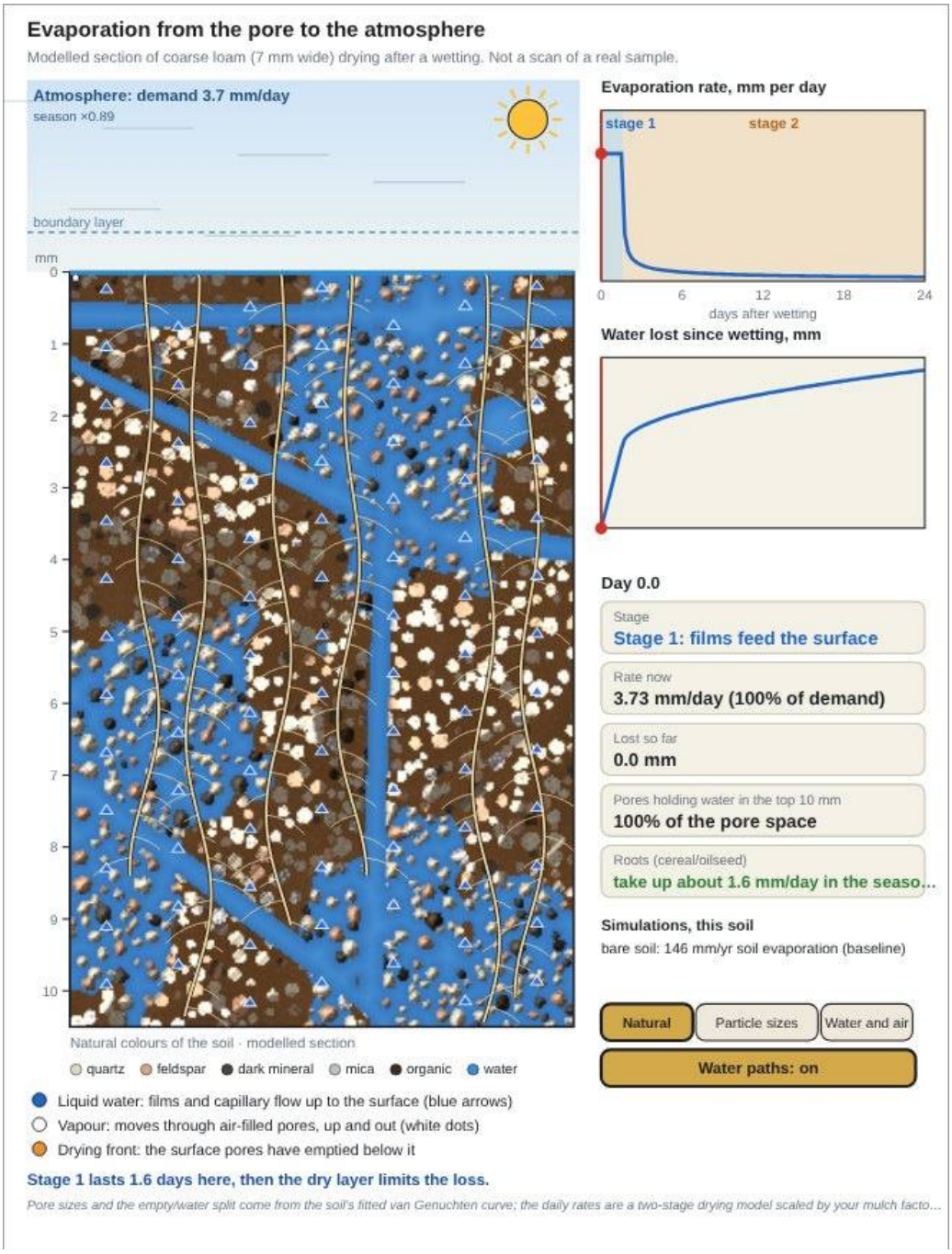


Figure 9.2: Evaporation physics layer. Volume 2 shows the whole layer, part by part, with its equations.

Governing relations. The loss is described by the classical two-stage model of soil drying (Philip, 1957; Ritchie, 1972): stage 1 at the atmospheric rate e_0, then stage 2 with a rate falling roughly as $t^{-1/2}$ (cumulative loss $\propto \sqrt{t}$). The implementation in the workbench is

$$t_1 = \frac{Q_1 - q_0}{e_0}, \qquad e(t) = \begin{cases} e_0, & t \le t_1 \\ \dfrac{e_0}{\sqrt{1 + (t - t_1)/\tau}}, & t > t_1 \end{cases} \tag{9.2}$$

Provenance (Eq. 9.2): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

$$Q(t) = q_0 + \begin{cases} e_0\, t, & t \le t_1 \\ (Q_1 - q_0) + 2\, e_0\, \tau\Big(\sqrt{1 + (t - t_1)/\tau} - 1\Big), & t > t_1 \end{cases} \tag{9.3}$$

Provenance (Eq. 9.3): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

(the rate is the time derivative of the cumulative loss, so the two are consistent), with the following parameters:

Symbol	Definition	Value or formula
$e_{0,b}$	bare-soil atmospheric demand	Eq. (5.26) (mm d^{-1})
e_0	demand under mulch and canopy	$e_0 = e_{0,b}\,\varphi\,(1 - 0.4\,c)$ with canopy cover c on the date
Q_1	stage-1 storage (mm), water held in the top 40 mm between the wet and the surface-dry state	$Q_1 = \max\{2,\ [\theta(h_i) - \theta(h_s)]\, L_1\, (0.6 + 0.4\rho)\}$, $L_1 = 40$ mm
h_i, h_s	initial and surface-dry suctions (cm)	$h_i = \dfrac{0.25}{\alpha}(1.5 - 0.5\rho)$; $h_s = 4/\alpha$
τ	stage-2 time scale (d)	$\tau = 0.006 + 0.03\,(\text{clay} + \text{silt})/100$
q_0	loss already experienced when the water was placed below the surface (drip)	$q_0 = (1 - w_f)\,0.95\,Q_1$, $w_f = \text{clip}(1 - z_d/50;\ 0.05,\ 1)$, z_d the drip depth in cm; 0 for surface methods
$\Delta\theta$	water lost across the dry layer	$\Delta\theta = \text{clip}(\theta(h_i) - \theta(10^4);\ 0.05,\ 0.4)$

$\theta(h)$ is Eq. (8.3) and α the van Genuchten parameter of Table 5.3; ρ is the rainfall scale. **[Engineered]** — every constant in the table (0.25, 1.5, 0.5, 4, 40 mm, 0.6, 0.4, 0.006, 0.03, 0.95, 50 cm, 0.05–1) is a presentation parameter chosen so that stage 1 lasts a few days in a loam and stage 2 gives the characteristic $\sqrt{t}$ fall; they are **not** calibrated to a laboratory evaporation experiment and the drawing says so ("*the daily rates are a two-stage drying model scaled by your mulch factor*"). The simulated annual evaporation of the workbench (Eq. 5.4) is *Daisy*, not this curve.

The drying front. The depth of the dry layer after stage 1 is

$$\ell(t) = \min\Big\{60,\ \frac{Q(t) - Q_1}{\Delta\theta}\Big\} \text{ mm} \quad (Q > Q_1), \tag{9.4}$$

Provenance (Eq. 9.4): **[Engineered]**: drying-front geometry chosen for drawing.

and the suction profile with depth z is an exponential approach of $\log_{10} h$ with length $\lambda = L_1/3$:

$$\log_{10} h(z) = \log_{10} h_s + \big(\log_{10} h_i - \log_{10} h_s\big)\big(1 - e^{-(z-\ell)/\lambda}\big) \quad (z \ge \ell), \qquad h = 10^5 \text{ cm for } z < \ell. \tag{9.5}$$

Provenance (Eq. 9.5): **[Engineered]**: drying-front geometry chosen for drawing.

Pore picture. Which pores are water-filled at each depth is given by the capillary rule of Eq. (8.4) with the local suction of Eq. (9.5); vapour particles are released from the drying front and rise through air-filled pores. The picture is an **illustration of the mechanism**, not a pore-network simulation.

Steps. (1) Open *Evaporation physics* and press **Play** (24 days). (2) Raise the mulch (lower φ): the grey bare-soil curve stays, your curve lengthens stage 1. (3) Change the soil: stage 1 lasts longest in the sands (largest Q_1, about 3 days for the Sand at the season-mean demand) and is shortest in the silty clay loam (about 1 day), because the window between h_i and h_s scales with $1/\alpha$; in stage 2 the fine soils lose water more slowly (larger τ). (4) Set a drip method: $q_0 > 0$ shortens stage 1 because the surface has less water to lose. (5) Hover over a pore to read its radius and the suction at which it empties.

9.4 Fate of saved water

What it shows. For the soil and mulch you set (a preview of $\varphi = 0.5$ is used if you have not set mulch), four curves against rainfall scale $\times 0.5$–$\times 1.4$: evaporation saved, irrigation saved, extra drainage, extra transpiration. A shaded band marks the rainfall above which drainage exceeds the irrigation saved (“*drainage takes over near* $\times 1.2$”). Below, a bar divides each mm of evaporation saved among less irrigation, more drainage, more transpiration and “storage and other”.

Generation. For each rainfall ρ in the ten-value list, with φ^* the mulch factor,

$$\begin{aligned} \Delta E(\rho) &= E_{1.0}(\rho) - E_{\varphi^*}(\rho), \quad \Delta I(\rho) = I_{1.0}(\rho) - I_{\varphi^*}(\rho), \\ \Delta D(\rho) &= D_{\varphi^*}(\rho) - D_{1.0}(\rho), \quad \Delta T(\rho) = T_{\varphi^*}(\rho) - T_{1.0}(\rho), \end{aligned} \tag{9.6}$$

Provenance (Eq. 9.6): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

all from Engine 1 at the current soil, tillage and temperature, and the partition at the current rainfall is

$$f_I = \frac{\max\{\Delta I, 0\}}{\max\{\Delta E, 0.1\}},\ f_D = \frac{\max\{\Delta D, 0\}}{\max\{\Delta E, 0.1\}},\ f_T = \frac{\max\{\Delta T, 0\}}{\max\{\Delta E, 0.1\}},\ f_{\text{rest}} = \max\{0,\ 1 - f_I - f_D - f_T\}. \tag{9.7}$$

Provenance (Eq. 9.7): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

The crossover rainfall is the first ρ with $\Delta D \geq \Delta I$ and $\Delta D > 1$ mm. The study reports about 60 % of the saved water going to reduced irrigation in dry years and about 63 % to extra drainage in wet years, with the crossover near $\rho \approx 1.2$ (repository README).

Steps. Choose a material; slide the **Rainfall scale**; the red dashed cursor and the bar move; read the three sentences at the bottom.

9.5 Regime map

What it shows. A map of **mulch factor (rows) against rainfall scale (columns)** coloured by which loss dominates, together with soil columns in which vapour rises (grey) or droplets fall (blue), and a verdict for your settings.

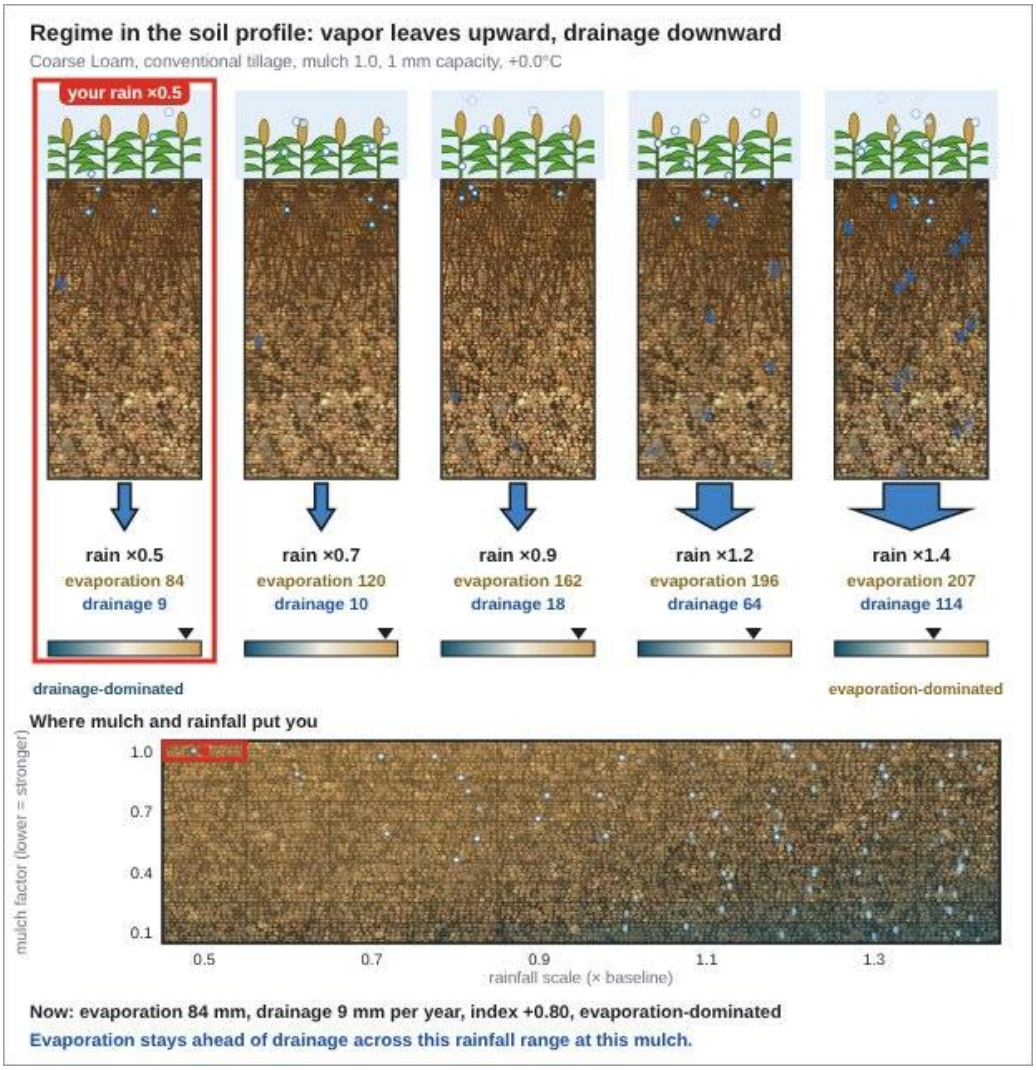


Figure 9.3: Regime map. Volume 2 shows the whole layer, part by part, with its equations.

Generation. For each cell the engine takes annual soil evaporation E and drainage D from Engine 1, and forms the **regime index**

$$\mathcal{I} = \frac{E - D}{\max\{1,\ E + D\}}, \qquad \text{regime} = \begin{cases} \text{evaporation-dominated} & \mathcal{I} > +0.05 \\ \text{drainage-dominated} & \mathcal{I} < -0.05 \\ \text{on the boundary} & |\mathcal{I}| \le 0.05 \end{cases} \tag{9.8}$$

Provenance (Eq. 9.8): **[Engineered]**: index defined for the workbench.

so $\mathcal{I} = +1$ is all evaporation and -1 all drainage. The statement "*Climate and mulch act non-additively ... reaching 15 % of irrigation at combined extremes*" is a result of the study (the interaction of mulch and climate on irrigation); it is not recomputed by the layer.

Steps. Click a cell to apply that mulch factor and rainfall; the verdict says *"the main loss is evaporation (...), so mulch will pay off here"* or *"the main loss is drainage ..., so mulch saves less"*.

Chapter 10

Feedback loops

10.1 What it shows

One large picture of a field, from sky to groundwater, on which red arrows join the parts that act on each other: temperature and the crop; rain and soil water; mulch and evaporation; the sensor trigger and irrigation; drainage and nitrogen leaching; nitrogen and the roots; tillage structure and drainage. Each arrow is numbered 1–7, and its **width grows with the size of the effect of one step** of the control behind it. Under the picture a panel lists "*What one step of each control does*"; click a row or an arrow to highlight it.

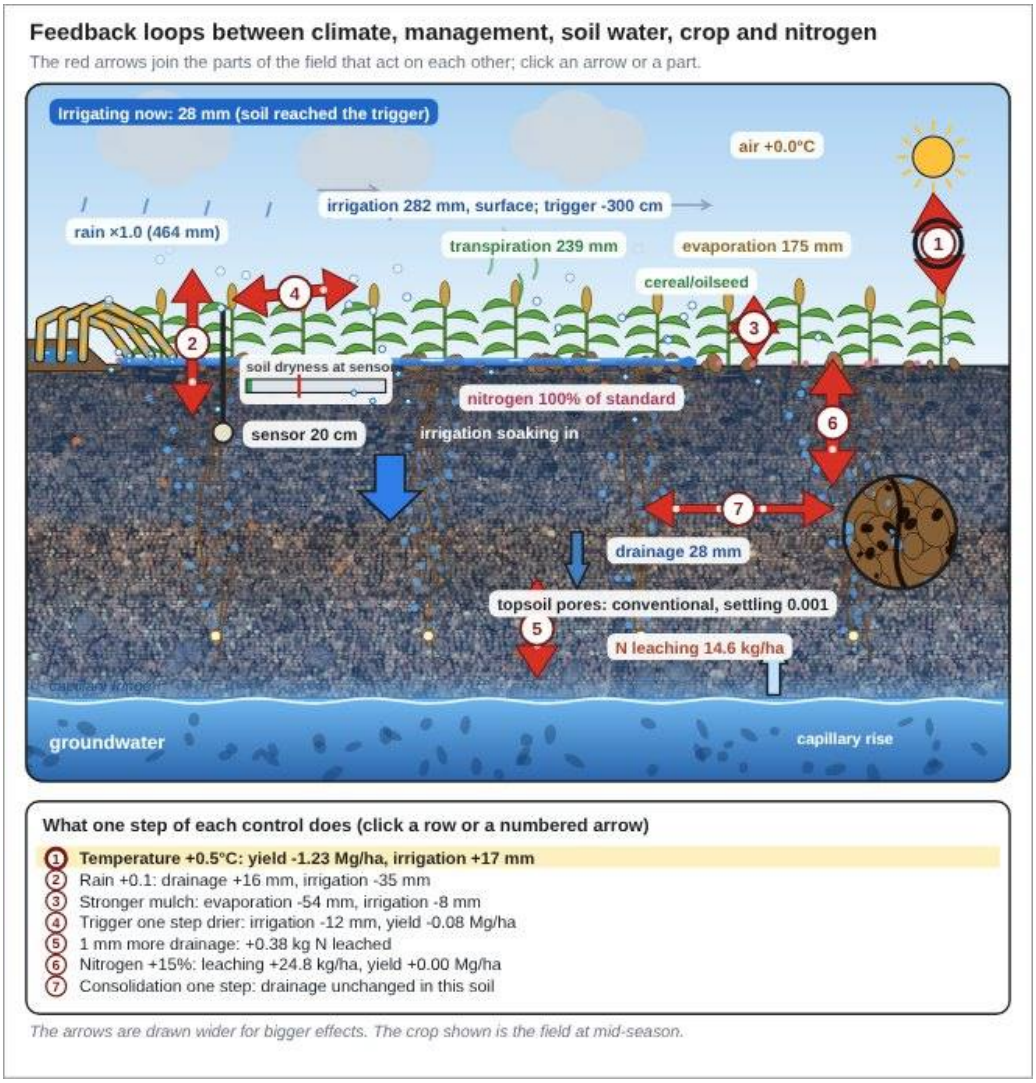


Figure 10.1: Feedback loops. Volume 2 shows the whole layer, part by part, with its equations.

The picture is also a set of controls: clicking the clouds cycles the rainfall, the sun the temperature, the mulch layer the material, the sensor the trigger (three grid steps drier each click, wrapping), the soil the tillage, the crop the crop category, the emitter or sprinkler the irrigation method, and the nitrogen band the nitrogen rate.

10.2 How the seven effects are computed

For each control the engine moves **one step** on its list (in the direction that stays inside the list: if the control is at its maximum it steps down) and re-evaluates Engine 1 (Eqs. 5.4–5.12). The change relative to the present result is printed. With m the present result and $m^{(\pm)}$ the result after one step, and f_{crop} the crop factor:

#	Control step	Printed effect	Arrow-width input μ
1	Temperature ± 0.5 °C	$\Delta Y f_{\text{crop}}$, ΔI	$8\,\lvert\Delta Y\rvert$
2	Rain ± 0.1 (scale)	ΔD, ΔI	$\lvert\Delta D\rvert/12$
3	Stronger mulch (one step lower φ)	ΔE, ΔI	$\lvert\Delta E\rvert/14$
4	Trigger one step drier or wetter	ΔI, $\Delta Y f_{\text{crop}}$	$\lvert\Delta I\rvert/8$
5	One more mm of drainage	$\Delta N_L/\Delta D$ kg N per mm (from the rain step), or "no change" if $\lvert\Delta D\rvert \le 0.5$	$\lvert\Delta D\rvert/10$
6	Nitrogen ± 15 %	ΔN_L, $\Delta Y f_{\text{crop}}$	$\lvert\Delta N_L\rvert/8$
7	Consolidation one step	ΔD (or "unchanged in this soil" if $\lvert\Delta D\rvert < 0.05$)	$\lvert\Delta D\rvert/2$

and the arrow line width in pixels is

$$w_k = 5 + \text{clip}(\mu_k;\, 0,\, 4). \tag{10.1}$$

Provenance (Eq. 10.1): **[Engineered]**: drawing rule.

The sensitivity printed in row 5, the **nitrate cost of drainage**, is the ratio

$$\kappa_N = \frac{N_L^{(\text{rain step})} - N_L}{D^{(\text{rain step})} - D} \quad [\text{kg N ha}^{-1}\ \text{mm}^{-1}], \tag{10.2}$$

Provenance (Eq. 10.2): **[Data-driven]**: one-step differences of Daisy results.

which the study reports as roughly 1.06 kg N ha^{-1} per mm of drainage (README). The numbers are **[Data-driven]** one-step differences, not partial derivatives, so a step at the end of a list can be smaller than the step in the middle.

10.3 The sentence under the picture

A status pill reports the day: *"Irrigating now: 28 mm (soil reached the trigger)"*, *"Rain day: 12 mm, no irrigation needed"* or *"Irrigation stopped: soil wet enough"*. The text for the trigger says whether *"the sensor has reached the trigger (depletion f)"*, to what depth *"the water wets the soil"* (the wetting front of the day's irrigation, Section 5.2) and, when that is shallower than the sensor, *"the sensor may stay beyond the trigger"*. If no irrigation is due, it states how many days are free in the next 14 and that rain is needed to keep it that way.

10.4 Steps

1. Open *Feedback loops*. The crop shown is the field at mid-season.
2. Click an arrow (number 4, for example). The matching row is highlighted: a line of the form *"Trigger one step drier: irrigation −X mm, yield −Y Mg/ha"* (the numbers depend on your settings).
3. Click the picture parts to change controls and watch the arrow widths change.
4. Use **Pin baseline** first to compare the 20-year averages for the whole combination.

Chapter 11

Surface irrigation design

The surface-irrigation layer designs a **border strip, furrow or basin** for *your* soil: it derives the soil's intake from texture, solves the advance of the water front, chooses a cut-off time, computes where the water goes, and then searches for a better layout. Everything on this page is a closed-form hydraulic model; no Daisy table is read, except the crop need and the trigger, which come from the DSS (Chapter 5).

The layer is a long scrolling drawing. The sub-section bar above it (*Whole design, Layout design, Simulation, Hydrograph, Advance and opportunity, Results, Design table*) jumps to a section, as do the sub-items of the layer in the Layers panel. The twelve sections are, in order:

#	Section (sub-item)	Purpose
1	Inputs and *Whole design* header	Choose the starting point, the system, intake, cut-off, and the value boxes; then Run simulation
2	Layout in plan: the advance front, shaded by the depth already infiltrated	Plan view of the field with the front
3	Longitudinal section: water running over your soil and soaking in	Side view with Z_0, Z_L, Z_n
4	Hydrograph: inflow, infiltration and runoff through time	Flow partition with time
5	Cross-section at the inlet: the wetted bulb in your soil, and the intake curve	Intake curve $Z(\tau)$ and the soil behind it
6	Advance and opportunity time	$t_x(x)$ and $\tau(x)$
7	Infiltrated depth along the field against the required depth Z_n	$Z(x)$ against Z_n
8	Performance indices and the water balance of one irrigation	E_a, E_s, DP, RO, CU, volumes
9	Design table	Iterations towards $E_a \geq 85\,\%$
10	What the efficiency is most sensitive to	Tornado chart
11	What this means for the season and the farm	Season, sets, hours
12	Irrigation schedule and *Intervals, efficiency and the advice in short*	Dates, gaps, "When to irrigate"

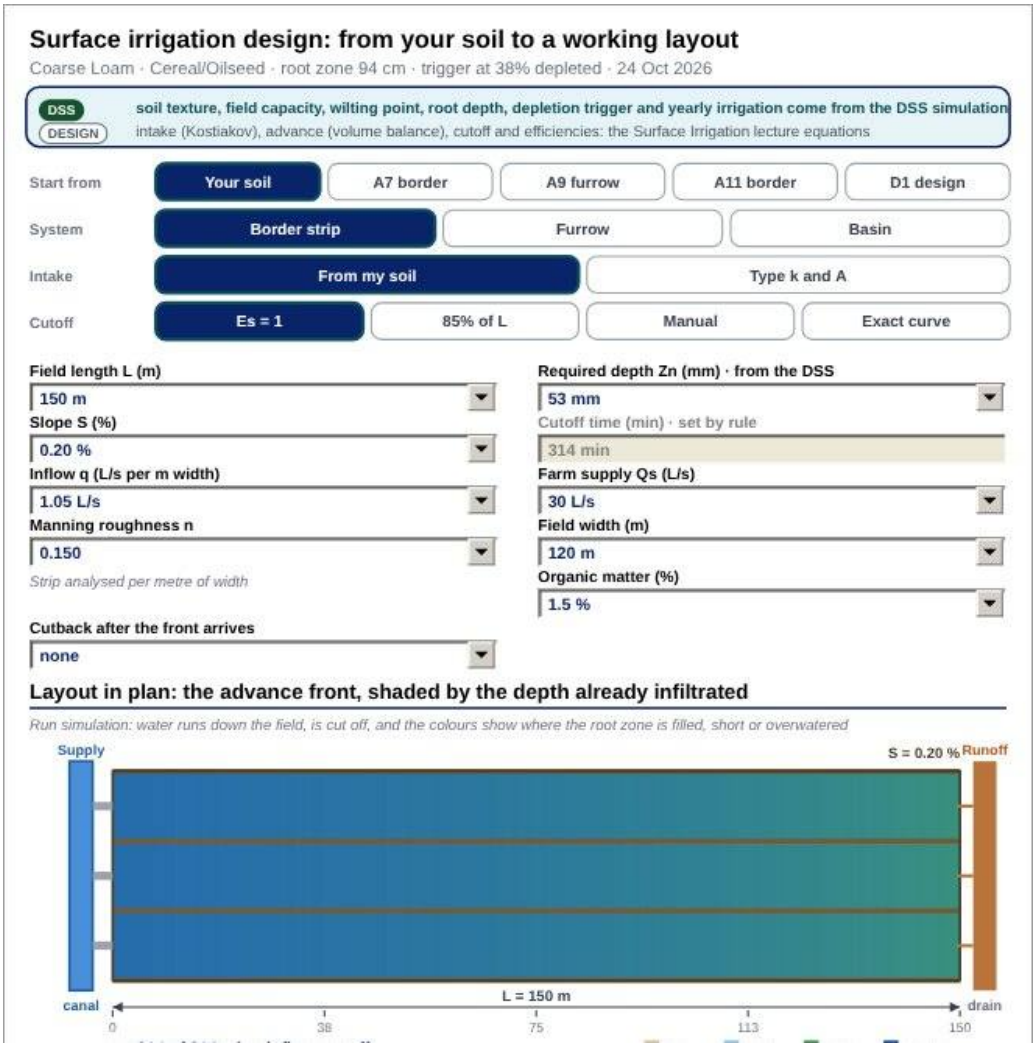


Figure 11.1: Surface irrigation design: the inputs (top) and the layout in plan. Volume 2 shows the whole layer, part by part, with its equations.

11.1 Inputs, step by step

1. Open *Irrigation design* ▸ *Surface irrigation design*. The header states the soil, crop, root-zone depth, trigger depletion and date (e.g. *"Coarse Loam · Cereal/Oilseed · root zone 94 cm · trigger at 38 % depleted · 18 Oct 2026"*). The green *DSS* badge lists what is taken from the simulation; the *DESIGN* badge what the design chooses.
2. **Start from** one of five rows:

Chip	Meaning
Your soil	Intake, required depth, cut-off and best inflow are all derived from your soil (default).
A7 border	Lecture worked example: border, $L = 200$ m, $S = 0.20$ %, $q = 2.5$ L s^{-1} m^{-1}, $n = 0.15$, $k = 2$ mm min^{-A}, $A = 0.8$, $Z_n = 90$ mm, given advance $t_L = 120$ min, cut-off 170 min.
A9 furrow	Furrow, $L = 226$ m, $S = 0.30$ %, $q = 2.0$ L s^{-1}, $n = 0.04$, $w = 0.7$ m, $k = 4$, $A = 0.7$, $Z_n = 80$ mm, $\sigma_z = 0.83$.
A11 border	Border, $L = 240$ m, $S = 0.20$ %, $q = 1.667$, $n = 0.15$, $k = 7$, $A = 0.5$, $Z_n = 80$ mm, advance from field data $t_x = x^{1.35} q^{-1.5}/300$ (a power law, Eq. 11.12), cut-off at 85 % of L.
D1 design	Furrow, $L = 240$ m, $S = 0.30$ %, $q = 1.667$, $n = 0.04$, $w = 1$ m, $k = 2$, $A = 0.7$, $Z_n = 80$ mm, approximate advance (Eq. 11.13), $\sigma_z = 0.8154$.

The four worked examples reproduce the course exercises so that the program can be checked against hand calculation; choosing one switches the intake to *Type k and A*.

3. **System**: *Border strip*, *Furrow* or *Basin*. This resets L, S, q, n to the system defaults of Table 11.1.
4. **Intake**: *From my soil* (Green–Ampt from the soil's texture, fitted to Kostiakov, Section 11.2) or *Type k and A* (your own Kostiakov constants).
5. **Cutoff**: *Es = 1* (cut off when the tail has just received Z_n), *85 % of L* (cut off when the front reaches 85 % of the length plus the storage time), *Manual* (set the time), and the toggle *Exact curve* (draw the exact Kostiakov $Z(x)$ instead of the straight-line profile of the lecture notes).
6. Set the value boxes (Table 11.1): click a box, type the exact value or pick it from the list, choose the unit. Then press **Run simulation**; the results below are computed for these inputs (Chapter 18).

Table 11.1. Surface-design inputs (value boxes): ranges (min, max, step) and system defaults.

Control	Border	Furrow	Basin	Notes
Field length L (m)	50–400, 5 (default 150)	60–400, 5 (180)	20–200, 5 (80)	
Slope S (%)	0.02–1, 0.01 (0.20)	0.05–1.5, 0.01 (0.30)	0.02–0.2, 0.01 (0.05)	
Inflow q (L s^{-1})	0.2–20 per m width, 0.05 (6)	0.2–4 per furrow, 0.05 (1.2)	5–60 per m width, 0.5 (25)	In *Your soil* mode q is set automatically (Section 11.9) until you move it
Manning n	0.02–0.25, 0.005 (0.15)	(0.04)	(0.15)	
Furrow spacing w (m)	—	0.5–1.5, 0.05 (0.75 in *Your soil*)	—	$w = 1$ m is used for border and basin (a strip 1 m wide is analysed)
Cutback after the front arrives	0.2–1.0, 0.05 (1 = none)	same	same	Fraction of q after t_L
Required depth Z_n (mm)	20–150, 1	same	same	Default *from the DSS* (Eq. 5.27); tap the label to reset
Farm supply Q_s (L s^{-1})	5–120, 1 (30)	same	same	Farm-level flow
Field width W_f (m)	30–400, 5 (120)	same	same	
Organic matter (%)	0.5–5, 0.1 (1.5)	same	same	Enters Saxton–Rawls
Kostiakov k (mm min^{-A})	0.2–10, 0.05	same	same	Visible in *Type k and A* mode
Kostiakov A	0.2–0.95, 0.01	same	same	Visible in *Type k and A* mode

11.2 Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)

Step 1: soil water properties (Saxton and Rawls, 2006). With sand fraction S, clay fraction C (both 0–1) and organic matter OM (%), the 1500-kPa and 33-kPa water contents are

$$\begin{aligned}
\theta_{1500,t} &= -0.024\,S + 0.487\,C + 0.006\,\mathrm{OM} + 0.005\,S\,\mathrm{OM} - 0.013\,C\,\mathrm{OM} + 0.068\,S\,C + 0.031,\\
\theta_{1500} &= \theta_{1500,t} + (0.14\,\theta_{1500,t} - 0.02),\\
\theta_{33,t} &= -0.251\,S + 0.195\,C + 0.011\,\mathrm{OM} + 0.006\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} + 0.452\,S\,C + 0.299,\\
\theta_{33} &= \theta_{33,t} + (1.283\,\theta_{33,t}^2 - 0.374\,\theta_{33,t} - 0.015),
\end{aligned} \tag{11.1}$$

Provenance (Eq. 11.1): **[Literature]**: Saxton and Rawls (2006), implemented as written.

and saturation and saturated conductivity (mm h^{-1}) follow from

$$\begin{aligned}
\theta_{(S-33),t} &= 0.278\,S + 0.034\,C + 0.022\,\mathrm{OM} - 0.018\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} - 0.584\,S\,C + 0.078,\\
\theta_{(S-33)} &= \theta_{(S-33),t} + \big(0.636\,\theta_{(S-33),t} - 0.107\big), \qquad \theta_s = \theta_{33} + \theta_{(S-33)} - 0.097\,S + 0.043,\\
B &= \frac{\ln 1500 - \ln 33}{\ln \theta_{33} - \ln \theta_{1500}}, \qquad \lambda = \frac{1}{B}, \qquad K_s = 1930\,(\theta_s - \theta_{33})^{\,3-\lambda}.
\end{aligned} \tag{11.2}$$

Provenance (Eq. 11.2): **[Literature]**: Saxton and Rawls (2006), implemented as written.

These are the published pedotransfer equations of Saxton and Rawls (2006), implemented as written (field capacity $\theta_{fc} = \theta_{33}$, wilting point $\theta_{wp} = \theta_{1500}$). They are regressions on a large US soil database and carry its scatter; the layer uses them for design, not for prediction at a particular site.

Step 2: wetting-front suction. The effective suction at the wetting front (mm) is taken from the van Genuchten pair (α, n) of Table 5.3, $m = 1 - 1/n$,

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

This is a rational-polynomial fit of the Rawls-type wetting-front suction as a function of m; it is an **[Engineered]** approximation (the code comment calls it the "Lisle/Rawls form") and should be read as giving the right order of magnitude (tens of mm for loams, e.g. 16 mm for the Coarse Loam), not a measured ψ_f.

Step 3: available storage in the wetting zone. The soil starts at the trigger depletion f_T and is wetted to 95 % of saturation,

$$\theta_0 = \theta_{fc} - f_T\,(\theta_{fc} - \theta_{wp}), \qquad \Delta\theta = \max\{0.05,\ 0.95\,\theta_s - \theta_0\}. \tag{11.4}$$

Provenance (Eq. 11.4): **[Engineered]**: initial and wetted water contents chosen for the workbench.

Step 4: Green–Ampt infiltration (Green and Ampt, 1911). Cumulative infiltration $F(t)$ (mm) solves

$$F - \psi_f\,\Delta\theta\,\ln\!\Big(1 + \frac{F}{\psi_f\,\Delta\theta}\Big) = K_s\,\frac{t}{60} \quad (t \text{ in min}), \tag{11.5}$$

Provenance (Eq. 11.5): **[Literature]**: Green and Ampt (1911).

by Newton iteration from the start value $\max\{K_s t/60,\ \sqrt{2\psi_f \Delta\theta K_s t/60}\}$.

Step 5: Kostiakov fit (Kostiakov, 1932). The design uses the two-parameter empirical intake law

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

and k, A are found by least squares on the log–log plot of $F(\tau)/1000$ at 25 times spaced geometrically between 10 and 600 min:

$$A = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \ln k = \bar{y} - A\,\bar{x}, \qquad x_i = \ln \tau_i,\ y_i = \ln\big(F(\tau_i)/1000\big), \tag{11.7}$$

Provenance (Eq. 11.7): **[Literature]**: ordinary least squares on the log–log plot.

with coefficient of determination $R^2 = \big(\sum (x-\bar{x})(y-\bar{y})\big)^2 \big/ \big(\sum (x-\bar{x})^2 \sum (y-\bar{y})^2\big)$, shown on the layer (e.g. *"Ks 13.1 mm/h · ψf 16 mm · Δθ 0.20 · fit R² 0.997"*). This conversion of a physically based law to a power law is what the lecture method requires; it is an approximation whose quality is reported by R^2.

Step 6: required depth. The net depth to refill the root zone is Eq. (5.27): $Z_n = \mathrm{clip}(f_T\,\mathrm{AWC}\,R_{\max};\,25,\,150)$ mm unless you type another value.

11.3 Inlet flow area (Manning)

The flow cross-section at the inlet, A_0 (m^2), is obtained from Manning's equation (Manning, 1891), in SI units,

$$Q = \frac{1}{n}\,A\,R_h^{2/3}\,S^{1/2}, \qquad R_h = \frac{A}{P}. \tag{11.8}$$

Provenance (Eq. 11.8): **[Literature]**: Manning (1891).

- **Furrow.** A V-shaped section with side slope $z = 1$ (H:V), depth limit $D = 0.25$ m: $A = zy^2$, $P = 2y\sqrt{1+z^2}$. Eq. (11.8) is solved for the flow depth y by 60 bisection steps for $Q = q$ (m^3 s^{-1}) and the slope S (with $S \geq 10^{-4}$); $A_0 = zy^2$. If $y > D$ the layer warns that the furrow overtops.
- **Border and basin.** A wide sheet: $A \approx y$ per unit width, $R_h \approx y$, so $y_0 = \left(q\,n/\sqrt{S}\right)^{0.6}$ with q in m^2 s^{-1}, and $A_0 = y_0$ (m^2 per m width).
- **D1 preset.** $A_0 = \left(q^2 n^2/792\right)^{0.37313}$ with q in m^3 min^{-1}, the empirical furrow relation of the lecture's worked example. **[Engineered, course-specific]**

11.4 Advance (volume balance)

Definition. For an inflow Q (m^3 min^{-1} per analysed width) the position $x(t)$ of the front is obtained from a volume balance of the water on the surface and in the soil (Walker and Skogerboe, 1987; Elliott and Walker, 1982):

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^A\,w\,x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^A\,w}. \tag{11.9}$$

Provenance (Eq. 11.9): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

The code also accepts an optional steady-intake term, $Z = k\tau^A + f_0\tau$ (Kostiakov–Lewis), in which Eq. (11.9) gains the term $f_0\,t\,w/(1+r)$ in the denominator, with r the iterated advance exponent (release 1.1; release 1.0 used a fixed $r = 0.55$); the interface does not expose f_0 (it is zero), and it exists so that the FAO Irrigation Paper 45 examples, which use it, can be reproduced. Here $\sigma_y = 0.8$ is the surface-storage shape factor, w the width analysed (furrow spacing, or 1 m for borders and basins), and σ_z the subsurface-storage shape factor

$$\sigma_z = \frac{A + r\,(1-A) + 1}{(1+A)(1+r)}, \tag{11.10}$$

Provenance (Eq. 11.10): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

where r is the exponent of the power-law advance $x = p\,t^r$, found by iterating (10 times, starting from $\sigma_z = 0.8$) from the times t_L and $t_{L/2}$ at which the front reaches L and $L/2$,

$$r = \frac{\ln 2}{\ln(t_L/t_{L/2})}, \tag{11.11}$$

Provenance (Eq. 11.11): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

with each time found by bisection on Eq. (11.9). Eq. (11.10) is the standard shape factor of the volume-balance method for the Kostiakov intake law (Elliott and Walker, 1982). The advance time to position x is then $t_x(x) = x^{-1}(t)$, solved on a 300-point grid.

Alternatives (set by the worked examples; the first two are Eq. 11.12):

- *Given advance:* $t_x(x) = t_L\,(x/L)^{1/r}$ with $r = 0.55$ unless set (A7).
- *Field advance* (A11): $t_x(x) = c\,q^{-q_e}\,x^{x_e}$ with $c = 1/300$, $q_e = 1.5$, $x_e = 1.35$.

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

Provenance (Eq. 11.12): **[Engineered]**: alternative advance models set by worked examples and course notes. * *Approximate* (D1, lecture formula):

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\,k\,w\,L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

Provenance (Eq. 11.13): **[Engineered]**: alternative advance models set by worked examples and course notes.

with t_V the volume-balance solution of Eq. (11.9). **[Engineered]**: this is the closed-form approximation of the course notes, not a published general law.

11.5 Cut-off, opportunity time and the infiltrated depth

The **time to infiltrate the required depth** follows from inverting Eq. (11.6),

$$t_{st} = \Big(\frac{Z_n}{k}\Big)^{1/A}. \tag{11.15}$$

Provenance (Eq. 11.15): **[Literature]**: inversion of the Kostiakov law.

The **cut-off time** t_{co} is

$$t_{co} = \begin{cases} t_L + t_{st} & \text{“}E_s = 1\text{”: the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{“85 \% of L”} \\ t_{co}^{\text{set}} & \text{“Manual”} \end{cases} \tag{11.16}$$

Provenance (Eq. 11.16): **[Engineered]**: cut-off rules chosen as design options.

The **opportunity time** at distance x is the time the water stays on the soil, $\tau(x) = t_{co} - t_x(x)$ for $t_{co} > t_x$, and the **infiltrated depth** is

$$Z(x) = k\,\big[t_{co} - t_x(x)\big]^A, \quad Z_0 = k\,t_{co}^A \text{ (inlet)}, \qquad Z_L = k\,(t_{co} - t_L)^A \text{ (tail)}. \tag{11.17}$$

Provenance (Eq. 11.17): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

The *straight-line profile* of the lecture notes replaces $Z(x)$ by the line between Z_0 and Z_L (and, for the 85 %-of-L rule, a broken line through $(0.85L,\, Z_n)$); *Exact curve* draws Eq. (11.17). If $t_{co} < t_L$ the front never reaches the end of the field and the layer shows a warning (*the stream does not reach the end*).

11.6 Volumes, efficiencies and uniformity

With w the analysed width, $N = 300$ grid intervals (90 while dragging) and the trapezoidal rule, for $Z_i = Z(x_i)$:

$$V_{\text{app}} = Q\,t_{co} \quad (\text{with cutback } c_b < 1:\ Q\,t_L + c_b\,Q\,(t_{co} - t_L)), \quad V_{\text{req}} = Z_n\,L\,w,$$
$$V_{\text{inf}} = w\int_0^L Z\,dx, \qquad V_{\text{st}} = w\int_0^L \min(Z, Z_n)\,dx,$$
$$V_{\text{dp}} = w\int_0^L \max(Z - Z_n, 0)\,dx, \qquad V_{\text{df}} = w\int_0^L \max(Z_n - Z, 0)\,dx,$$
$$V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}},\ 0). \tag{11.18}$$

Provenance (Eq. 11.18): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

The performance indices (Walker and Skogerboe, 1987; Burt et al., 1997) are

$$E_a = \frac{V_{\text{st}}}{V_{\text{app}}}, \quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}}, \quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}}, \quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}}, \quad \text{CU} = 100\Big(1 - \frac{\frac{1}{N+1}\sum_{i=0}^{N}|Z_i - \bar{Z}|}{\bar{Z}}\Big)\ \%. \tag{11.19}$$

Provenance (Eq. 11.19): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

E_a is the **application efficiency**, E_s the **storage (requirement) efficiency**, and CU the uniformity coefficient of Christiansen (1942) evaluated on the infiltrated depths. Because $V_{\text{app}} = V_{\text{inf}} + V_{\text{ro}}$ and $V_{\text{inf}} = V_{\text{st}} + V_{\text{dp}}$, the three fractions satisfy $E_a + \text{DP} + \text{RO} = 1$, which the layer prints as the closed water balance of one irrigation. The deficit V_{df} is the missing part of Z_n. The distance at which Z crosses Z_n, if it does, is $X = L\,(Z_0 - Z_n)/(Z_0 - Z_L)$.

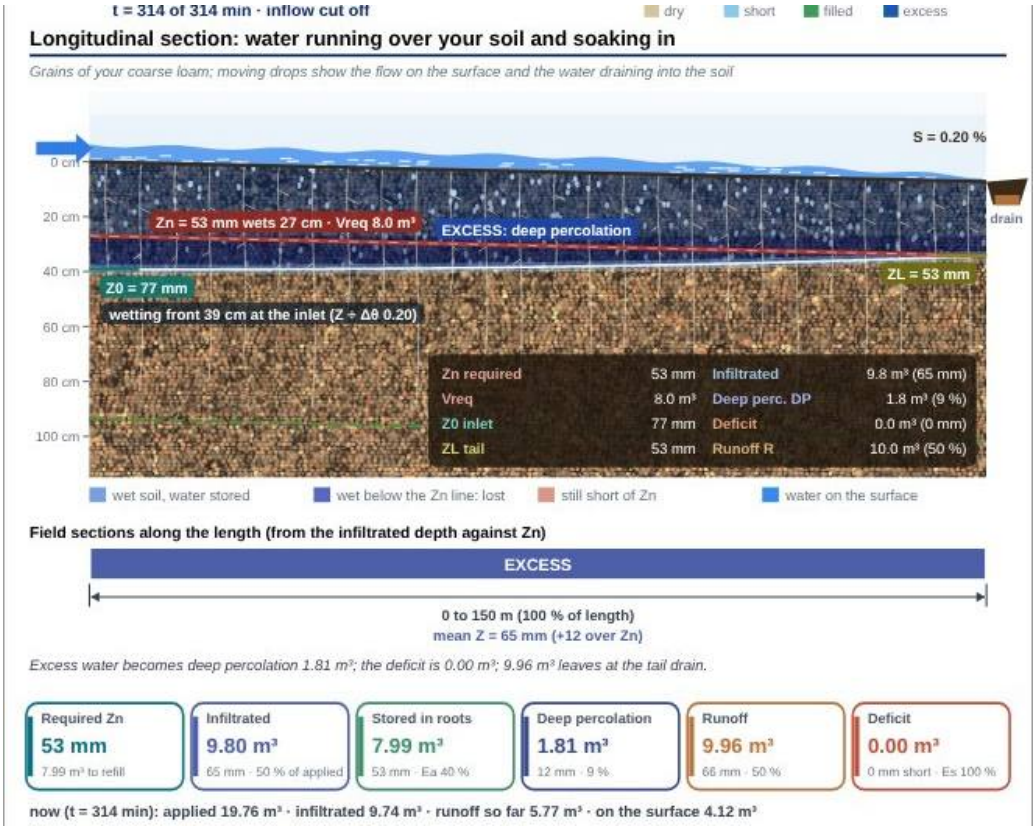


Figure 11.2: Surface design: longitudinal section with Z_0, Z_L, Z_n and the water balance of one irrigation. Volume 2 shows the whole layer, part by part, with its equations.

11.7 Hydrograph

The hydrograph (Section 4 of the layer) divides the inflow q into infiltration $I(t)$, runoff $R(t)$ and surface storage. It is built from the same advance and intake:

$$V_c(t) = w \sum_i k \left[\min(t, t_{co}) - \bar{t}_{x,i}\right]_+^A \Delta x \cdot s_c, \qquad s_c = \frac{V_{\text{inf}}}{V_c(t_{co})}, \qquad I(t) = \frac{dV_c}{dt}, \tag{11.20}$$

Provenance (Eq. 11.20): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

where $\bar{t}_{x,i}$ is the advance time at the mid-point of interval i and the factor s_c makes the cumulative infiltration at cut-off equal to V_{inf} of Eq. (11.18). While $t_L \le t \le t_{co}$ the runoff is $R(t) = \max\{0,\, q - I(t)\}$. After cut-off the surface water *recedes*: with $V_s = \max\{0,\, V_{\text{app}} - V_{\text{inf}} - \int R\,dt\}$ left on the surface, a triangular recession over $t_r = \max\{4,\, \min(0.3\,t_{co},\, 0.25\max(t_L, 10))\}$ min drains it at the tail,

$$R(t) = \frac{2V_s}{t_r}\Big(1 - \frac{t - t_{co}}{t_r}\Big), \quad t_{co} < t < t_{co} + t_r. \tag{11.21}$$

Provenance (Eq. 11.21): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

The recession shape is **[Engineered]** (the course does not specify it); it conserves the volume V_s. Surface storage at time t is $V_{\text{sto}}(t) = q\min(t, t_{co}) - V_c(t) - \int_0^t R\,dt'$.

11.8 Cross-section and intake curve

The *Cross-section at the inlet* section draws, at the inlet, the wetted shape in a soil section generated from the soil's texture, and the intake curve $Z(\tau)$ of Eq. (11.6) beside the Green–Ampt curve $F(\tau)$ of Eq. (11.5), so that the fit can be inspected by eye.

11.9 Best inflow ("Your soil" mode)

In *Your soil* mode, q is chosen automatically from 25 equally spaced values between $q_{\min}$ and $q_{\max}$ of the system as the one that maximises $E_a\,E_s$. For furrows the upper limit is the non-erosive stream

$$q_{\max} = \min\Big\{q_{\max}^{\text{sys}},\; \frac{0.04}{S}\cdot\frac{1000}{60}\Big\}\ \text{L s}^{-1} \quad (S \text{ in \%}), \tag{11.22}$$

Provenance (Eq. 11.22): **[Engineered]**: search limit; the furrow stream limit follows Criddle et al. (1956).

about $0.67/S$ L s^{-1}, in line with the classical Criddle et al. (1956) limit $Q_{\max} \approx 0.63/S$. The choice is stored in the inflow box, which becomes manual as soon as you change it.

11.10 The design table

The *Design table* automates one step at a time towards a good design. The **target** is $E_a \ge 85\,\%$ with $E_s \ge 80\,\%$ (the root zone stays at least 80 % refilled). The score of a design is

$$\text{score} = \min(E_a, 1) - 3\max(0,\, 0.8 - E_s) - 3\,[c_b < 1]\,\max(0,\, 0.05 - \text{RO}) - 5\max(0,\, E_a - 1). \tag{11.23}$$

Provenance (Eq. 11.23): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

The search tries, in this order, ten single-variable moves — **cut-off, inflow, cutback, length, cut-off, cutback, inflow, length, cut-off, cutback** — and, for each move, evaluates a candidate list and keeps the best candidate **only if the score improves by more than 0.003**:

Move	Candidates
Cut-off	$t_{co} \times \{0.5, 0.58, 0.66, 0.74, 0.8, 0.85, 0.9, 0.95, 1, 1.05, 1.1, 1.2\}$
Inflow	12 values spaced geometrically from $q_{\min}$ to $q_{\max}$
Cutback	$c_b \in \{1, 0.85, 0.7, 0.6, 0.5, 0.4, 0.3, 0.25, 0.2\}$
Length	11 values spaced geometrically from $\max(L_{\min}, 30)$ to $\min(L_{\max}, 1.5L)$, rounded to 5 m

For every candidate other than a pure cut-off move, six cut-off times are also tried (0.62 to 1.02 times the "$E_s = 1$" cut-off). The search stops when the target is met or after the ten moves. Each accepted row lists q, L, cutback, cut-off, E_a, E_s, DP and RO; **tap a row (green when it meets the target) to apply it** to the sliders. If the target cannot be reached the layer says so and suggests a shorter field, tailwater reuse, or drip or sprinklers. The same search is run by the toolbox item *Optimise surface layout (best design)*.



Design table: iterations toward an application efficiency of 85 % or more

One change per step; the root zone stays ≥ 80 % refilled. Tap a row to apply it.

#	Step	q L/s	L m	Cutback	Cutoff	Ea %	Es %	DP %	RO %	Result
0	Your current layout	1.05	150	none	314 min	**40.4**	100.0	9.2	50.4	
1	Cut off at 232 min	1.05	150	none	232 min	**47.9**	87.6	1.1	51.0	
2	Inflow 0.70 L/s	0.70	150	none	286 min	**55.6**	83.5	4.6	39.8	
3	Cut back to 20 % of q	0.70	150	20%	351 min	**76.6**	96.5	16.6	6.8	
4	Field length 70 m	0.70	70	20%	196 min	94.3	88.3	0.0	5.7	meets target

Figure 11.3: Surface design table: iterations toward $E_a \geq 85\,\%$, then the tornado chart. Volume 2 shows the whole layer, part by part, with its equations.

11.11 Sensitivity (tornado chart)

Each of seven inputs is moved up and down while the others are held, and the change in E_a is drawn as a bar (green raises E_a, red lowers it):

Input	Intake k	Intake A	Inflow q	Length L	Required depth Z_n	Roughness n	Slope S
Perturbation	± 20 %	± 10 %	± 20 %	± 20 %	± 20 %	± 30 %	± 30 %

The bars are sorted by the larger absolute change, and the printed value is $\max(|\Delta E_a^-|, |\Delta E_a^+|)$ in percentage points (*pts*). The test is local: it is the change around the *present* layout, not a global importance.

11.12 Season and farm

Using the irrigation record of Engine 2 for the year of the working date, with $\sum I$ the net irrigation of that year (mm):

$$
\begin{aligned}
n_{\text{ev}} &= \sum I \,/\, Z_n, \qquad Z_g = \frac{V_{\text{app}}}{L\,w} \times 1000 \;\; (\text{mm}), \qquad Z_{\text{season}} = n_{\text{ev}}\, Z_g, \\
N_{0M} &= \frac{Q_s}{q}, \qquad N_{\text{units}} = \frac{W_f}{w}, \quad N_{\text{sets}} = \max\!\Big(1, \frac{N_{\text{units}}}{N_{0M}}\Big), \qquad T_{\text{tot}} = \frac{N_{\text{sets}}\, t_{co}}{60} \;\; (\text{h}),
\end{aligned}
\tag{11.24}
$$

Provenance (Eq. 11.24): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

and the seasonal pumping time is $n_{\text{ev}} T_{\text{tot}}$ hours. A depth of Z_{season} mm equals $10\,Z_{\text{season}}$ m^3 per hectare. N_{0M} is the number of furrows or strips that can run at once from the farm supply Q_s.

11.13 Irrigation schedule and "When to irrigate"

The last two sections are common to all three design layers, so they are described once, here.

Schedule table. The next eight irrigation dates after the working date are listed from the DSS (Engine 2, Section 5.2): the sensor in layer j_s reaching the trigger depletion f_T. For each date the table gives

Column	Definition
#	order, soonest first
Date	calendar date of the analogue year (Chapter 4)
In	*today*, *tomorrow*, or *n days*
Net mm	the DSS irrigation depth that day, I_k
Pumped mm	gross depth I_k / E_a with the design's application efficiency (min. $E_a = 0.05$)
Run time	time of one set or one round of the design
Gap to next	days to the following irrigation

Intervals. Over the year of the working date, with irrigation days $d_1 < d_2 < \ldots$ (irrigations above 0.5 mm), the gaps $g_j = d_{j+1} - d_j$ give the mean, the shortest and the longest interval.

Advice message. The message *When to irrigate* is generated from these numbers: the *next irrigation date* and its net and pumped depth; *"Then irrigate about every $\bar{g}$ days (between $g_{\min}$ and $g_{\max}$ days, shorter in hot weather and longer after rain)"*; the efficiencies of the design (E_a, E_s) and the time of one round; and the rule *"If a rain above 10 mm is forecast, delay the irrigation by a day or two"*. If no irrigation is triggered, it says to *keep the sensor at z_s cm and irrigate when it reaches depletion f_T*.



Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	22 May 2027	210 days	28	70	22.0 h	13 days
2	4 Jun 2027	223 days	28	70	22.0 h	11 days
3	15 Jun 2027	234 days	28	70	22.0 h	9 days
4	24 Jun 2027	243 days	28	70	22.0 h	9 days
5	3 Jul 2027	252 days	28	70	22.0 h	10 days
6	13 Jul 2027	262 days	28	70	22.0 h	16 days
7	29 Jul 2027	278 days	28	70	22.0 h	251 days
8	5 Apr 2028	529 days	28	70	22.0 h	-

Figure 11.4: The irrigation schedule and the "When to irrigate" message at the end of the surface design. Volume 2 shows the whole layer, part by part, with its equations.

Reading it. The first row is today's decision. If the *Gap to next* is short (2–4 days) the trigger is wet or the soil store is small; if it is long (10+ days) the soil store is large or the weather cool. Move the **Date** to see how the schedule changes with the season, and move the **Trigger** or **Sensor depth** to see the dates shift.

Chapter 12

Drip irrigation design

The drip layer designs the **emitter, its placement, the lateral and the pipe network** for the soil, crop and trigger set in the rest of the program. It answers four questions in order: how big a wet bulb does one emitter make, how much of the water it delivers stays in the root zone, how uneven is the delivery along the lateral, and what pump and pipes does the farm need. Everything is a closed-form hydraulic or geometric model, recomputed at every slider move.

The layer's sections, in order: control bar and inputs; soil moisture around the driest, average and wettest emitter (Run simulation animates the wetting); where the water goes; uniformity along the lateral; growth of the wetted bulb with time; result tiles; layout and infrastructure; irrigation schedule; the design table; the sensitivity chart; season and farm; and the "When to irrigate" advice. The last two are shared with the other design layers and are described at the end of Chapter 11.

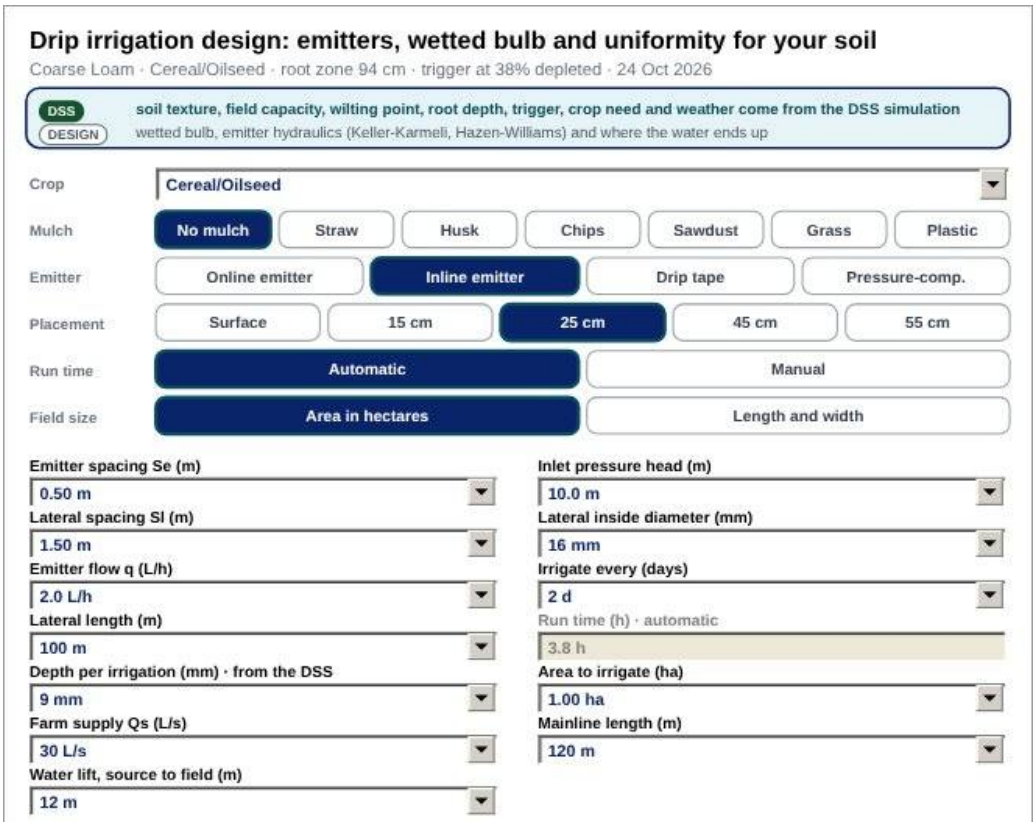


Figure 12.1: Drip design: inputs and the soil moisture around the driest, average and wettest emitter. Volume 2 shows the whole layer, part by part, with its equations.

12.1 Inputs, step by step

1. Open *Irrigation design* ▸ *Drip irrigation design*. The crop, soil, root zone, trigger and date come from the DSS (the green badge); the choices below are the design (the amber badge).
2. **Emitter** (a row of four chips). Choosing one sets the nominal flow, spacing, lateral diameter and inlet head of Table 12.1.
3. **Placement**: *Surface* (design only), or a buried line at 15, 25, 45 or 55 cm. A buried choice also sets the explorer's irrigation method to the matching Daisy run (M04, M06, M09 or M10), so that the rest of the program shows the same depth.
4. **Run time**: *Automatic* (time to deliver the depth at the driest emitters, Eq. 12.17) or *Manual*.
5. Set the value boxes of Table 12.2, then press **Run simulation**. The crop, mulch and irrigation method rows at the top are the same controls as the Properties panel.

Table 12.1. Emitter types (defaults are set when the chip is chosen; c_v manufacturer coefficient of variation, x flow exponent, n_e emitters per plant or point, $H_{\min}$ compensation threshold).

Emitter	q (L h^{-1})	S_e (m)	D (mm)	H_0 (m)	c_v	x	n_e	$H_{\min}$ (m)
Online emitter	4	0.75	16	10	0.07	0.5	1	—
Inline emitter	2	0.50	16	10	0.04	0.5	1	—
Drip tape	1	0.20	16	10	0.08	0.6	2	—
Pressure-compensating	2	0.60	16	15	0.03	0.05	1	8

Table 12.2. Drip inputs (value boxes) (range, step, default).

Control	Range, step	Default	Meaning
Emitter spacing S_e (m)	0.1–1.5, 0.05	by emitter	Along the lateral
Lateral spacing S_l (m)	0.5–3, 0.05	1.5 (1.0 for cereal, legume; 1.5 for vegetable, reset when the crop changes)	Between laterals
Emitter flow q (L h^{-1})	0.5–8, 0.1	by emitter	Nominal flow at H_0
Lateral length L_L (m)	20–300, 5	100	
Inlet pressure head H_0 (m)	4–30, 0.5	10 (15 for PC)	
Lateral inside diameter D (mm)	10–32, 1	16	
Irrigate every (days)	1–10, 1	2	Sets the default depth
Run time (h)	0.25–48, 0.25	automatic	Greyed while *Automatic*
Depth per irrigation Z_n (mm)	3–80, 1	from the DSS	Tap the label to reset
Area to irrigate A_f (ha)	0.2–10, 0.1	1	
Farm supply Q_s (L s^{-1})	2–120, 1	30	
Mainline length L_m (m)	20–600, 10	120	
Water lift, source to field (m)	0–60, 1	12	

Depth per irrigation. By default the depth is the crop need over the interval, capped by what the root zone can hold at the trigger,

$$Z_n = \min\{ Z_n^{\mathrm{DSS}},\ \max(4,\ \mathrm{ET_{pk}}\,\Delta t)\}\ \ \mathrm{mm}, \tag{12.1}$$

Provenance (Eq. 12.1): **[Engineered]**: depth and demand rules chosen for the workbench.

where Δt is the chosen interval, $Z_n^{\mathrm{DSS}} = \mathrm{clip}(f_T\,\mathrm{AWC}\,R_{\max}; 25, 150)$ is Eq. (5.27), and $\mathrm{ET_{pk}}$ is the peak five-day moving mean of simulated evaporation plus transpiration over the year, bounded to 2.5–9 mm d^{-1},

$$\mathrm{ET_{pk}} = \mathrm{clip}\Big\{ \max_d \tfrac{1}{5} \sum_{j=0}^{4} (E_{d+j} + T_{d+j});\ 2.5,\ 9 \Big\}. \tag{12.2}$$

Provenance (Eq. 12.2): **[Engineered]**: depth and demand rules chosen for the workbench.
The mean evaporative demand used for surface drying, E_0, is the mean of the simulated daily evaporation on days 141–250 of the year (when positive), clipped to 1–5 mm d^{-1}.

12.2 Lateral hydraulics: head loss and flow along the lateral

An emitter lateral of length L_L carries $N = \mathrm{round}(L_L/S_e)$ intervals and $N+1$ emitters, indexed $j = 0 \dots N$ from the inlet. The friction head loss in the segment feeding emitter j is the Hazen–Williams equation (Hazen and Williams, 1905), in SI form with a roughness coefficient $C = 140$ for plastic pipe,

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \ge j+1} q_i \;\; (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m.} \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).
The head at emitter j is $h_j = h_{j-1} - h_{f,j-1}$ with $h_0 = H_0$ (floored at 0.5 m). Emitter discharge depends on the local head through the emitter equation (Keller and Karmeli, 1974),

$$q_j = q\Big(\frac{h_j}{H_0}\Big)^{x} \quad (\text{non-compensating};\ h_j/H_0 \ge 0.01), \tag{12.4}$$

Provenance (Eq. 12.4): **[Literature]**: Keller and Karmeli (1974) emitter law.
with $x = 0.5$ for a fully turbulent orifice or long-path emitter and $x = 0.6$ for the drip tape of Table 12.1. The pressure-compensating (PC) emitter is modelled as

$$q_j = \begin{cases} q\big[1 + 0.04\,(h_j - H_0)/H_0\big], & h_j \ge H_{\min}, \\ q\,\big(\max(h_j, 0.1)/H_{\min}\big)^{0.5}, & h_j < H_{\min}, \end{cases} \tag{12.5}$$

Provenance (Eq. 12.5): **[Engineered]**: pressure-compensating emitter model with chosen constants.
i.e. a nearly flat discharge above $H_{\min} = 8$ m (a 4 % change in flow per 100 % change in head) and the orifice law below it. The flows depend on the heads and the heads on the flows, so the pair (12.3)–(12.4) is **iterated to convergence** from the uniform start $q_j = q$, $h_j = H_0$ (the first three passes are plain substitution, then the flows are averaged with the previous pass; the iteration stops when the largest relative change in any emitter flow is below 10^{-7}, at most 500 passes). Version 1 used four passes, which is sufficient only when the head loss is small; the converged solver agrees with an independent solution to 3×10^{-7} for head losses up to 60 % of the inlet head. When the friction loss exceeds 20 % of the inlet head, or the iteration fails to converge, the layer prints a red warning, because uniformity then falls and the design is outside the usual rule of thumb. The mean and minimum discharge are then

$$\bar{q} = \frac{1}{N+1}\sum_{j=0}^{N} q_j, \qquad q_{\min} = \min_j q_j, \qquad Q_{\mathrm{lat}} = \frac{N\,\bar{q}}{3600} \;\; (\mathrm{L\,s^{-1}}), \tag{12.6}$$

Provenance (Eq. 12.6): **[Literature]**: definitions of mean and minimum discharge.
and the loss along the whole lateral is $h_f = H_0 - h_N$, with $H_{\mathrm{end}} = h_N$ the head at the far end (Section “Uniformity along the lateral”).

Emission uniformity. The design emission uniformity follows Keller and Karmeli (1974),

$$E_u = \mathrm{clip}\Big\{\Big(1 - \frac{1.27\,c_v}{\sqrt{n_e}}\Big)\frac{q_{\min}}{\bar{q}};\ 0.3,\ 1\Big\}. \tag{12.7}$$

Provenance (Eq. 12.7): **[Literature]**: Keller and Karmeli (1974).

The first factor is the manufacturing variation; the second is the hydraulic variation along the lateral (here $q_{\min}$ replaces the classical quarter-lowest mean, a conservative choice). The *combined* relative standard deviation of the delivery, which drives the spread of bulbs below, is

$$\sigma_q = \sqrt{c_v^2 + \sigma_P^2}, \qquad \sigma_P^2 = \frac{1}{N+1}\sum_{j=0}^{N}\Big(\frac{q_j}{\bar{q}} - 1\Big)^2. \tag{12.8}$$

Provenance (Eq. 12.8): **[Engineered]**: combined spread chosen to drive the bulb quantiles.

Reading the lateral plot: the solid line is q_j/q in per cent against distance, the dashed line is the head as a percentage of H_0. A lateral that is too long or too narrow shows a steep fall of both. For a non-compensating emitter a 10 % loss of head costs about 5 % of flow ($x = 0.5$); the PC emitter stays flat until the head drops below $H_{\min}$.

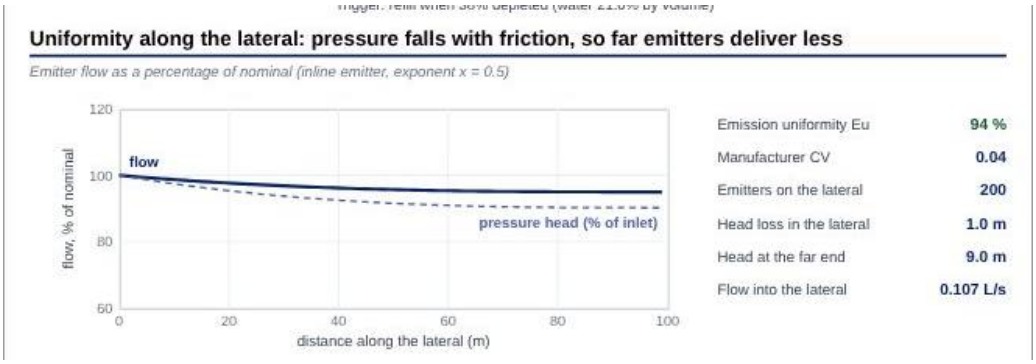


Figure 12.2: Drip design: uniformity along the lateral and the growth of the wetted bulb with run time. Volume 2 shows the whole layer, part by part, with its equations.

12.3 The wetted bulb

Water from an emitter spreads into soil that is initially at the trigger water content and is refilled up to field capacity. The layer models the wetted volume around a *buried* point source as a **half-ellipsoid** below the emitter and a short one above it (an **[Engineered]** simplification of the wetted-volume shapes used in drip design; the aspect ratio is derived from the soil, not read from a table).

Water to be stored per unit soil volume.

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

Provenance (Eq. 12.9): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Shape. With emitter depth z_d (mm), horizontal radius r (mm) at the emitter level, lower semi-axis $a\,r$ and upper semi-axis $u\,r$ with $u = 0.6$, the wetted radius at depth z is

$$\rho(z) = r\,\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \ge z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Aspect ratio. The shape factor a (depth over radius) increases when gravity is strong relative to capillarity, i.e. for a coarse soil and a low flow. It is evaluated from the ratio of the saturated conductivity to the radial capillary spreading rate $q/(2\pi\lambda^2)$ (the form is a scaling argument and the constants are **[Engineered]**),

$$a = \mathrm{clip}\Big\{1 + 0.6\,\log_{10}\Big(1 + \frac{K_s}{q\cdot 10^6/(2\pi\lambda^2)}\Big);\ 1,\ 1.9\Big\}, \qquad \lambda = \max(40, \psi_f), \tag{12.11}$$

Provenance (Eq. 12.11): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

with K_s in mm h^{-1}, q in L h^{-1} and ψ_f the wetting-front suction of Eq. (11.3) in mm. The result is a bulb that is round ($a = 1$) for a fine soil and elongated downwards (up to $a = 1.9$) for a sand.

Radius. For an emitter that delivers V litres, the wetted volume is $V_w = V \cdot 10^6/\Delta\theta$ mm^3 (1 L $= 10^6$ mm^3). The radius r is found by bisection (38 steps between 5 and 3500 mm) so that the volume of revolution of Eq. (12.10), integrated in 56 layers,

$$V_w = \int \pi\, \rho(z)^2\, dz, \tag{12.12}$$

Provenance (Eq. 12.12): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

equals V_w. When the volume is *line-like*, the cross-section instead of the volume is conserved: for drip tape ($S_e < 0.34$ m) or when neighbouring bulbs overlap ($2r > S_e$ with $S_e < 1.2$ m), the water is spread along the line, $A_w = V \cdot 10^6/(S_e \cdot 1000)/\Delta\theta$ mm^2 per unit length of lateral and the profile area is $2\int \rho\, dz$. The panel is then labelled "line (bulbs merge)" rather than "point". The wetted share of the soil surface is

$$P_w = \mathrm{clip}\Big\{\frac{2r}{1000\, S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

Provenance (Eq. 12.13): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Where the water goes. Of the V litres, the part that falls inside the bulb below the root-zone depth $R_{\max}$ is lost to deep percolation, a fraction B_{below}/V_w of V:

$$V_{\text{dp}}^{(1)} = V\, \frac{\int_{z>R_{\max}} \pi\rho^2\, dz}{V_w}, \tag{12.14}$$

Provenance (Eq. 12.14): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

and the part evaporated from the wet surface patch is limited to half the delivery,

$$V_{\text{ev}} = \min\{0.5\, V,\ \pi\rho(0)^2 \cdot 10^{-6} \cdot 2E_0\} \quad (\text{L},\ \text{area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

(for a line source the area is $2\rho(0)S_e \cdot 10^{-6}$ m^2 per emitter). The factor 2 corresponds to about two days of evaporation from a wet surface patch of mean demand E_0; this is an **[Engineered]** simplification of the two-stage evaporation of Chapter 9. For a surface line ($z_d = 0$) the bulb is half-spherical and the evaporating patch is largest, which is why surface drip stores less than buried drip in the table. The remainder is available to the roots, $V_{\text{root}} = V - V_{\text{dp}}^{(1)} - V_{\text{ev}}$.

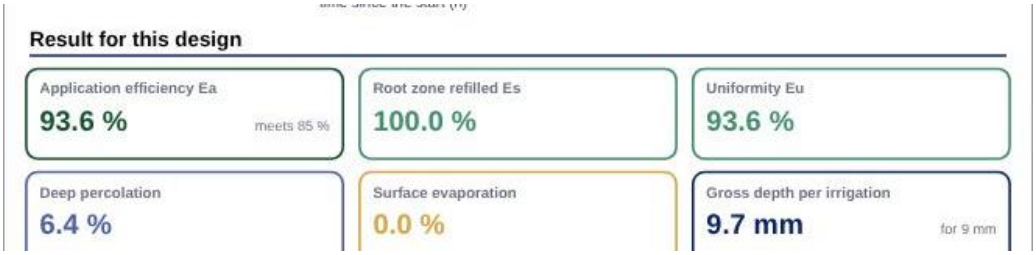


Figure 12.3: Drip design: result tiles and the water partition of one irrigation. Volume 2 shows the whole layer, part by part, with its equations.

12.4 Spread between emitters and the efficiencies

Emitters do not all deliver the same volume. The layer evaluates five representative emitters, at standard-normal quantiles $z \in \{-1.28, -0.52, 0, 0.52, 1.28\}$ (the 10th, 30th, 50th, 70th and 90th percentiles, equal weights), each delivering

$$V_i = \max\{0.02\,\bar{V},\ \bar{V}\,(1 + \sigma_q z_i)\}, \qquad \bar{V} = \bar{q}\,t, \tag{12.16}$$

Provenance (Eq. 12.16): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

where t is the run time (h). For each, the bulb, deep percolation and evaporation are computed with Eqs. (12.10)–(12.15). The water that the root zone can accept is limited by the need,

$$V_{\text{req}} = Z_n\, S_e\, S_l\ \ (\text{L}), \qquad V_{\text{st},i} = \min\{V_{\text{root},i},\ V_{\text{req}}\}, \qquad V_{\text{ex},i} = V_{\text{root},i} - V_{\text{st},i}. \tag{12.17}$$

Provenance (Eq. 12.17): **[Literature]**: definition of the requirement over the area served by one emitter.

Because 1 mm over 1 m^2 is 1 L, V_{req} is the depth needed over the area $S_e S_l$ served by one emitter. **Automatic run time** is the time at which the *driest* emitters deliver the need,

$$t = \text{clip}\Big\{\frac{V_{\text{req}}}{\bar{q}\,E_u};\ 0.05,\ 96\Big\}\ \text{h}. \tag{12.18}$$

Provenance (Eq. 12.18): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

The efficiencies are means over the five quantiles ($\langle\cdot\rangle$):

$$E_a = \frac{\langle V_{\text{st}}\rangle}{\langle V\rangle}, \quad E_s = \frac{\langle V_{\text{st}}\rangle}{V_{\text{req}}}, \quad \text{DP} = \frac{\langle V_{\text{dp}}^{(1)} + V_{\text{ex}}\rangle}{\langle V\rangle}, \quad \text{EV} = \frac{\langle V_{\text{ev}}\rangle}{\langle V\rangle}, \quad Z_{\text{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

Provenance (Eq. 12.19): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

so that $E_a + \text{DP} + \text{EV} = 1$. Reading them: E_a is how much of the pumped water ends up stored in the root zone; E_s is how well the root zone is refilled (a value below 85 % turns the "Root zone refilled" line red and says *short*); E_u is the lateral uniformity; Z_{gross} is the depth to pump per irrigation. The layer colours $E_a \geq 85$ % green, 65–85 % amber and below 65 % red; $E_u \geq 90$ % is green and 80–90 % amber.

12.5 Layout and infrastructure

The farm side of the design uses the area A_f and the supply Q_s.

Laterals and zones.

$$W_f = \frac{10^4 A_f}{L_L}, \quad N_L = \Big\lfloor \frac{W_f}{S_l} \Big\rfloor, \quad n_{\text{zone}} = \Big\lceil \frac{N_L\, Q_{\text{lat}}}{Q_s} \Big\rceil, \quad n_l = \Big\lceil \frac{N_L}{n_{\text{zone}}} \Big\rceil, \quad Q_z = n_l\, Q_{\text{lat}}, \tag{12.20}$$

Provenance (Eq. 12.20): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

where W_f is the width of the field perpendicular to the laterals, n_{zone} the number of zones that must run in turn so that Q_s is not exceeded, and Q_z (L s^{-1}) the flow of one zone. The total number of emitters is $N_L(\text{round}(L_L/S_e) + 1)$.

Pipe sizing. Submain length $L_{sm} = n_l S_l$, mainline length L_m (value box). The pipe diameter is the smallest of the commercial series

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\}\ \text{mm} \tag{12.21}$$

Provenance (Eq. 12.21): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

for which the mean velocity $v = 4Q_z/(\pi D^2) \le 1.5$ m s^{-1} (a common design limit that keeps friction and water-hammer moderate; see Keller and Bliesner, 1990, for pipe design practice). The submain loss is 0.4 of the full-length Hazen–Williams loss, Eq. (12.3) with $Q_j = Q_z$ (the Christiansen (1942) reduction factor F for a pipe with many equally spaced outlets is $F \to 1/2.852 = 0.35$ for $m = 1.852$, so 0.4 is slightly conservative), and the mainline loss is the full value:

$$h_{f,sm} = 0.4\,\frac{10.67\,L_{sm}\,Q_z^{1.852}}{C^{1.852}D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\,L_m\,Q_z^{1.852}}{C^{1.852}D_m^{4.871}}. \tag{12.22}$$

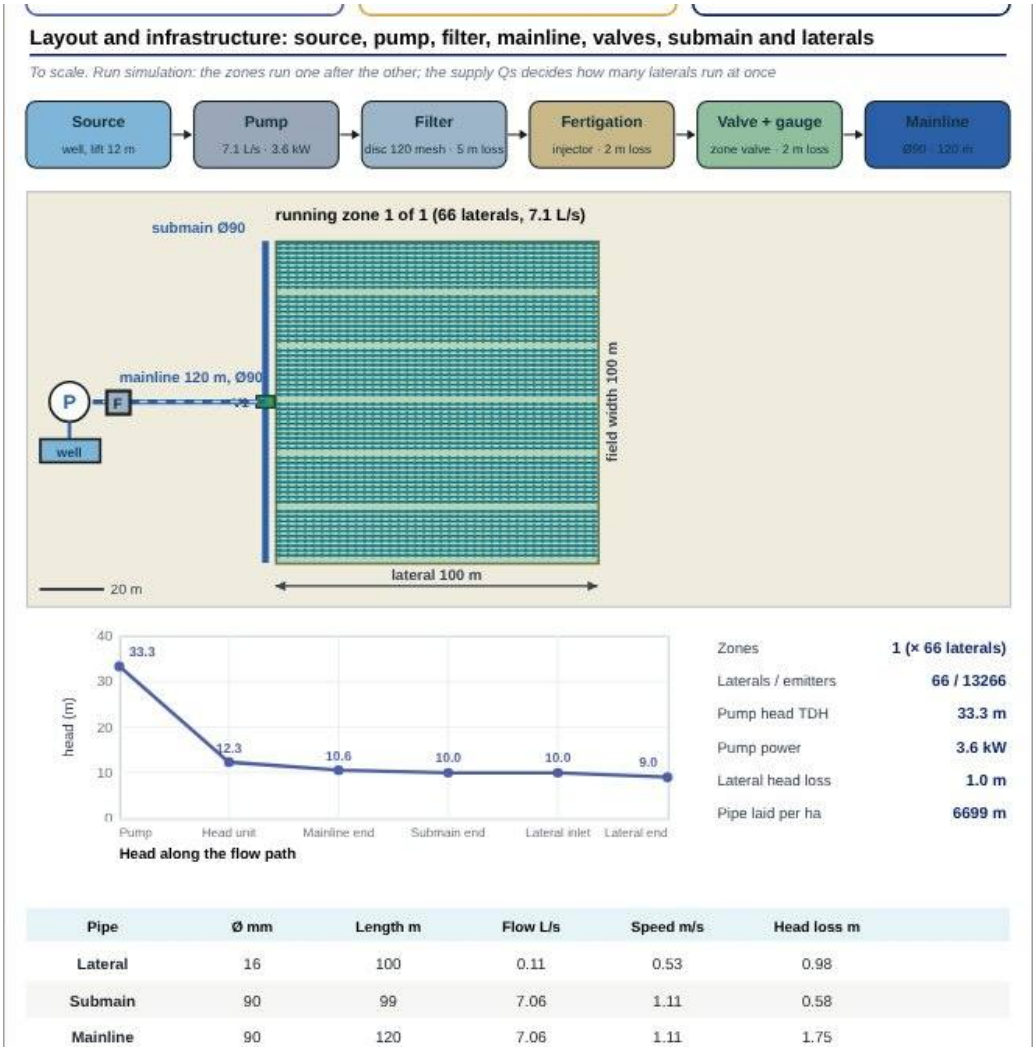
Provenance (Eq. 12.22): **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

Pump. The total dynamic head and the shaft power are

$$\mathrm{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9\ \text{m}, \qquad P = \frac{\rho g\,Q_z\,\mathrm{TDH}}{\eta} = \frac{9.81\,(Q_z/1000)\,\mathrm{TDH}}{0.65}\ \text{kW}. \tag{12.23}$$

Provenance (Eq. 12.23): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

The constant 9 m covers the filter, valves and fittings, and $\eta = 0.65$ is the overall pump-and-motor efficiency; both are **[Engineered]** design allowances, not measured values. The laterals' own loss is already inside H_0.



Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	16	100	0.11	0.53	0.98
Submain	90	99	7.06	1.11	0.58
Mainline	90	120	7.06	1.11	1.75

Figure 12.4: Drip design: layout and infrastructure, to scale, with the zones running in turn. Volume 2 shows the whole layer, part by part, with its equations.

The drawing shows the source, pump, filter, mainline, valves, submain and laterals to scale; during the run the zones run one after the other, and the head along the flow path is drawn underneath. The table beside it gives zones (× laterals per zone), laterals and emitters, TDH, pump power, lateral head loss and pipe laid per hectare, $(N_L L_L + L_{sm})/A_f$.

12.6 Season and farm

With y_I the net seasonal irrigation of the DSS for the year of the working date, E_u and E_a as above,

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\, Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\text{L s}^{-1}\,\text{ha}^{-1}), \tag{12.24}$$

Provenance (Eq. 12.24): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\, t, \qquad H_{\text{pump}} = n_{\text{ev}}\, t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\, P. \tag{12.25}$$

Provenance (Eq. 12.25): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

The layer prints these as a table (net irrigation need and number of events, gross water per irrigation and per season in mm and $\text{m}^3\ \text{ha}^{-1}$, peak crop need, flow per hectare, sets for A_f

with Q_s, hours of pumping and energy). Then the *Irrigation schedule* and the *When to irrigate* advice, described at the end of Chapter 11, close the layer.

12.7 The design table

The optimiser takes the current design and tries one change at a time, in this order: **placement depth, run time, emitter flow, emitter spacing, placement depth, run time, emitter kind, lateral spacing, emitter flow, run time**. For each move the candidates are:

Move	Candidates
Placement depth	Surface, 15, 25, 45, 55 cm
Run time	$t_0 \times \{0.55, 0.65, 0.75, 0.85, 0.95, 1, 1.1, 1.25\}$, with t_0 the automatic run time of the current design
Emitter flow q	0.6, 1, 1.5, 2, 3, 4, 6, 8 L h^{-1} (run time reset to automatic)
Emitter spacing S_e	0.2, 0.3, 0.4, 0.5, 0.6, 0.75, 1, 1.25 m
Emitter kind	the four emitters of Table 12.1, each with its own default q, S_e
Lateral spacing S_l	0.75, 1, 1.25, 1.5, 1.8, 2.2, 2.6 m

The score is

$$\text{score} = \min(E_a, 1) - 2.5\ \max(0,\ 0.9 - E_s), \tag{12.26}$$

Provenance (Eq. 12.26): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

so a design is penalised when the root zone is not refilled to 90 %. A move is accepted when its best candidate improves the score by more than 0.002; the search stops when $E_a \geq 85$ % **and** $E_s \geq 85$ %, or after ten moves. One move is performed per screen update, and the search pauses while a slider is being dragged. Each accepted row shows the step, q, S_e, depth, run time, E_a, E_s and DP; a green row meets the target, and **tapping any row applies that design** to the sliders (including the placement depth and the explorer's irrigation method). If the target is not met, the layer reports the best found and suggests a shorter interval or a different emitter.



Figure 12.5: Drip design table and the sensitivity chart. Volume 2 shows the whole layer, part by part, with its equations.

12.8 Sensitivity

As for the surface layer, each input is moved up and down and the change of E_a is shown as a bar: emitter flow q (± 25 %), emitter spacing (± 25 %), lateral spacing (± 25 %), placement depth (± 30 %, at least 10 cm), depth per irrigation (± 25 %) and run time (± 20 %). The result is a local sensitivity at the present design.

12.9 How to use the layer

- **To choose an emitter for a soil.** Select the soil in the Properties panel, open the layer and compare the four chips with the same run time: the tile of E_a and the line chart of the bulb show which emitter keeps the bulb inside the root zone.
- **To check a lateral length.** Increase L_L until E_u turns amber; the lateral plot shows where the head falls below the pressure at which the emitter holds its flow.

- **To size a pump.** Fix A_f, Q_s, L_m and the lift; read TDH and kW from the layout table.
- **To see the effect of the trigger.** Move *Trigger* in the Properties panel: f_T changes $\Delta\theta$ (Eq. 12.9) and Z_n (Eq. 12.1), and the bulb and the run time adjust.

Chapter 13

Sprinkler irrigation design

The sprinkler layer designs the **nozzle, pressure, spacing and layout** of a sprinkler system for the soil, crop, slope and wind of the case, and then checks the result against the standard course rules (spacing, application rate, uniformity, pipes and pump). It has two parts that are built on the same run: a *simulation* part (the overlapped spray pattern, the wind drift, the soil intake that causes runoff, and the efficiencies) and a *textbook* part (the ten design checks, the pipe and pump sizing, the kinds of sprinkler system).

The layer reads top to bottom: **A** site, **B** sprinkler and layout, **C** operation and **D** farm and pipes (the inputs); the layout in plan; the profile along the lateral; the intake against the application rate; result tiles and the water partition; the textbook checks and system plans; layout and infrastructure; the irrigation schedule; the design table with the sensitivity chart; the season and the advice.

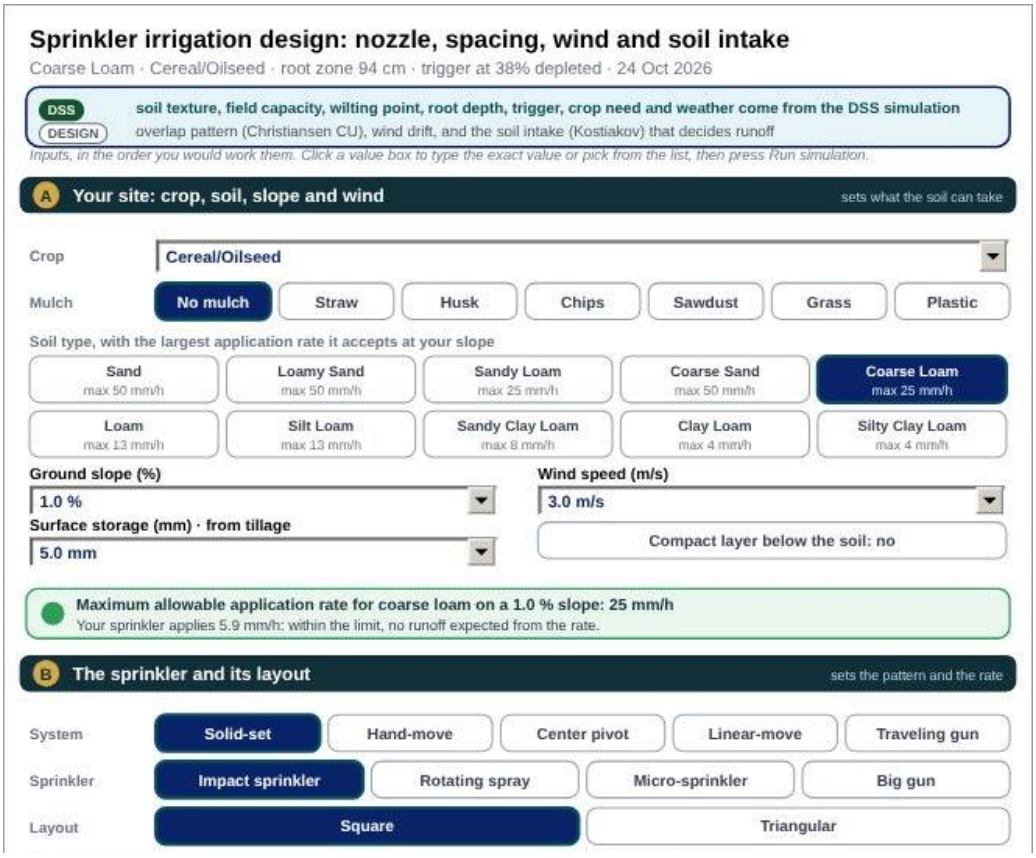


Figure 13.1: Sprinkler design: inputs A to D and the layout in plan with the overlapping spray patterns. Volume 2 shows the whole layer, part by part, with its equations.

13.1 Inputs, step by step

A. Your site. Crop, mulch and irrigation method are the same controls as in the Properties panel. The soil row shows each soil with the largest application rate it accepts at the present slope (Table 13.3). Sliders: ground slope S (0–16 %, step 0.5, default 1), wind speed u (0–12 m s^{-1}, step 0.5, default 3), surface storage S_d (0–20 mm, step 0.5; default from tillage, below) and a toggle *Compact layer below the soil* (which lowers the allowed rate, Table 13.3). A banner states the maximum allowable application rate for the soil and slope and whether the present sprinkler is within it.

B. The sprinkler and its layout. *System* (five chips, see *Kinds of sprinkler system* below), *Sprinkler* (four types, Table 13.1), *Layout* (square or triangular), then value boxes for the nozzle diameter d, operating pressure P, trajectory angle α (8–45°, default 25°), spacing along the lateral S_s and between laterals S_l. Choosing a type resets d, P, S_s, S_l to its defaults and run time to automatic.

Table 13.1. Sprinkler types: ranges (min–max, step) and defaults; c_R range coefficient, m profile exponent, F drift factor. The impact value $c_R = 1.35$ is the Keller and Bliesner (1990) coefficient for impact sprinklers (changed from 1.15 in version 1; the earlier value under-predicted the wetted diameter of their worked example by 14 %). The values for the other three types are **[Engineered]** and have not been verified against published data.

Type	d (mm)	P (kPa)	S_s (m)	S_l (m)	c_R	Profile	F
Impact	2.4–8, 0.1 (4.4)	200–500, 10 (300)	6–30, 1 (12)	6–36, 1 (18)	1.35	$1 - x^{1.3}$	1.0
Rotating spray	1.5–5, 0.1 (3.2)	70–250, 5 (170)	3–15, 0.5 (6)	3–15, 0.5 (6)	0.80	ring	1.6
Micro-sprinkler	0.8–2.5, 0.05 (1.3)	100–250, 5 (180)	1.5–8, 0.25 (3)	1.5–8, 0.25 (3)	0.70	ring-2	0.7
Big gun	10–25, 0.5 (16)	350–700, 10 (500)	20–80, 1 (30)	20–90, 1 (36)	0.95	$1 - x$	1.3

C. How you run it. *Run time* automatic or manual; depth per irrigation Z_n (5–100 mm; default Eq. 5.27, tap to reset); run time T (10–1200 min, step 5); operating hours per day (4–24, default 16).

D. Farm, water supply and pipes. Area A_f (0.5–40 ha, step 0.5, default 5), farm supply Q_s (5–200 L s^{-1}, default 40), lateral length L_L (40–500 m, step 10, default 150), mainline length L_m (20–600 m, default 120) and lift (0–60 m, default 12).

Tillage and surface storage. Unless set by hand, the depression storage S_d follows the tillage of the Properties panel: 5 mm conventional, 3.5 mm reduced, 2 mm none. On a slope the effective storage is reduced,

$$S_d^{\text{eff}} = \frac{\max(0, S_d)}{1 + 0.25\,S} \qquad (S \text{ in \%}). \tag{13.1}$$

Provenance (Eq. 13.1): **[Engineered]**: depression storage reduction with slope.

This is an **[Engineered]** relation: storage in depressions decreases as the slope rises (compare the slope dependence of depression storage in Hillel, 1998).

13.2 Nozzle discharge and wetted radius

For a nozzle of diameter d (mm) at operating pressure P (kPa) the pressure head is $h = P/9.81$ m. The discharge follows the orifice equation (Torricelli) with a discharge coefficient of 0.95,

$$q = 0.95\,A_n\sqrt{2g\,h} \times 6 \times 10^4 \;\; (\text{L min}^{-1}), \qquad A_n = \pi\Big(\frac{d}{2000}\Big)^2 \text{ m}^2, \tag{13.2}$$

Provenance (Eq. 13.2): **[Literature]**: Torricelli orifice equation.

and the wetted radius is (an empirical range law of the form used in sprinkler manuals, Keller and Bliesner, 1990, with the coefficient c_R of Table 13.1 chosen per type)

$$R = c_R \sqrt{d\,h}\; f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \Big[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\Big]^{0.45}, \tag{13.3}$$

Provenance (Eq. 13.3): **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

with d in mm, h in m and α in degrees (3.1α as an angle in degrees). The angle function peaks near $\alpha = 29^\circ$ and equals 1 at the default 25° ($\sin 77.5^\circ = 0.9763$). **[Engineered]**: the exponent 0.45 and the 3.1α term are chosen to give a mild optimum, not fitted to a data set. The reach R is printed in the plan as a percentage of the spacing along the lateral.

13.3 The spray pattern and the overlap

Radial profile. A single sprinkler deposits water at radius r with a shape $\hat{p}(x)$, $x = r/R_e$, taken from Table 13.1:

$$\hat{p}(x) = \begin{cases} 1 - x^m, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^6), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^5), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \hat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

These are **[Engineered]** shapes that reproduce the qualitative patterns published for each class: a single sprinkler with a rapidly falling profile (impact, gun), and a pattern that peaks away from the centre (spray and micro types).

Wind. Wind elongates the pattern downwind and shifts it:

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

where θ is the angle from the wind direction and u is the wind speed in m s^{-1}. The pattern is computed on a 45×45 grid over $\pm 1.4R$ and normalised so that its volume equals the discharge, $\iint p\,dA = q$, with p in mm min^{-1} (1 mm over 1 m^2 is 1 L).

Overlap. The application rate at a point of the repeating cell is the sum over neighbouring sprinklers (square grid $S_s \times S_l$; triangular layout: every second lateral shifted by $S_s/2$, repeating cell $S_s \times 2S_l$),

$$a(x, y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\text{mm min}^{-1}), \qquad \bar{a} = \frac{1}{n}\sum_k a_k, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

evaluated on a 14×14 grid of the cell (14×24 for the triangular layout). The mean *gross* application rate is AR $= 60\bar{a}$ mm h^{-1} and the **Christiansen coefficient of uniformity** (Christiansen, 1942) is

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

A CU $\geq$ 85 % is the usual requirement for a sprinkler system (Burt et al., 1997; Keller and Bliesner, 1990); the tile is green from 84 %. The *Layout in plan* shows the circles of reach, the wind arrow and the depth infiltrated so far (Play animates the time): green is the depth needed, red-brown is short, blue is more than needed, and brown hatching is water running off.

13.4 Wind drift and evaporation loss

The fraction of the water lost between nozzle and soil is

$$w_d = \frac{1}{100}\,\mathrm{clip}\Big\{(1.2{+}0.5u{+}0.04u^2)\,F\Big(\frac{P}{250}\Big)^{0.7}\big[1{+}0.025\,(T_a{-}20)\big]\Big[0.6{+}0.4\,\frac{\sin\alpha}{0.4226}\Big];\,0.5,\,45\Big\}. \tag{13.8}$$

Provenance (Eq. 13.8): **[Engineered]**: loss and ponding rules chosen for the workbench.

where F is the type factor of Table 13.1, P is in kPa, u in m s^{-1} and the air temperature $T_a = 24\,^{\circ}\mathrm{C} + \Delta T$, with ΔT the warming of the Properties panel. The form follows the known trends — loss rises with wind, with pressure (finer droplets), with temperature, and falls with a flatter trajectory — in line with the ranges of 5–15 % usually quoted in textbooks (Keller and Bliesner, 1990); the numerical coefficients are **[Engineered]**. w_d is clipped to 0.5–45 %.

13.5 The soil intake, ponding and runoff

At each cell the water balance is computed with the Kostiakov intake of the soil (Eq. 11.6, from the texture; $K = k \times 1000$ mm min^{-A}). The applied rate after drift is $a_k(1 - w_d)$ and for a run of T min the supply is $\mathrm{sup} = a\,T$. Ponding begins when the application rate exceeds the intake rate, which the layer takes as $a = Kt^{A-1}$, i.e. at

$$t_p = \Big(\frac{a}{K}\Big)^{1/(A-1)}, \qquad F = \begin{cases} \mathrm{sup}, & T \le t_p \\ K\,T^A, & T > t_p \end{cases} \qquad R = \mathrm{sup}{-}F, \quad \mathrm{RO} = \max(0, R{-}S_d^{\mathrm{eff}}), \quad I = \mathrm{sup}{-}\mathrm{RO}. \tag{13.9}$$

Provenance (Eq. 13.9): **[Engineered]**: loss and ponding rules chosen for the workbench.

This is a **simplified point balance** (it uses the cumulative law $F = K\,T^A$ after ponding, and it ignores re-infiltration of water that has run to a lower cell, which makes the layer slightly pessimistic about runoff in a field with surface storage and slightly optimistic on a steep slope). A portion of the infiltrated depth is retained on the crop canopy and mulch and evaporates,

$$I_c = \min\{I,\ I_{\mathrm{int}}\}, \qquad I_{\mathrm{int}} = 0.6\,\mathrm{CAP}\,[\mathrm{VFF} < 0.95] + 1.2\,\mathrm{LAI}_{\mathrm{live}}, \tag{13.10}$$

Provenance (Eq. 13.10): **[Engineered]**: loss and ponding rules chosen for the workbench.

where CAP is the mulch storage capacity (mm, Table 3.1), the indicator is 1 when a mulch is present, and $\mathrm{LAI}_{\mathrm{live}}$ is the live leaf-area index of the simulated crop on the working date. The depth that reaches the soil at the cell is $Z_k = I - I_c$.

The intake figure shows the rate the soil accepts, $\dot{I}(t) = 60\,K\,A\,t^{A-1}$ mm h^{-1}, the mean applied rate $\overline{AR}(1 - w_d)$, the shaded area where the applied rate exceeds the intake (runoff), and the time at which ponding starts. (The plotted curve is the derivative of Kt^A; the point balance uses the ponding condition $a = Kt^{A-1}$, so the two differ by the constant factor A in the intake rate, a documented simplification.)

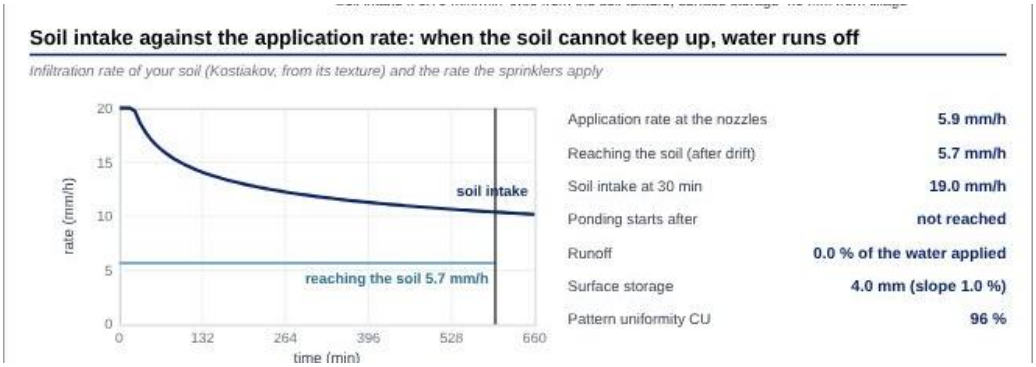


Figure 13.2: Sprinkler design: depth applied and infiltrated along the lateral, and the soil intake against the application rate. Volume 2 shows the whole layer, part by part, with its equations.

13.6 Efficiencies and run time

With Z_k the infiltrated depth at cell k and gross $= \bar{a}\,T$ the mean depth discharged:

$$E_a = \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, \qquad E_s = \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, \quad \text{DU} = \frac{\overline{Z}_{25}}{\overline{Z}},$$
$$\text{DP} = \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, \quad \text{RO} = \frac{\overline{\text{RO}_k}}{\text{gross}}, \qquad \text{WD} = w_d + \frac{\overline{I_c}}{\text{gross}}, \tag{13.11}$$

Provenance (Eq. 13.11): **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

where $\overline{Z}_{25}$ is the mean of the lowest quarter of the Z_k (the distribution uniformity of Merriam and Keller, 1978). The partition is closed, $E_a + \text{DP} + \text{RO} + \text{WD} = 1$. The tiles show E_a (green from 85 %), E_s, CU, runoff, deep percolation and the combined drift, evaporation and interception loss, with a stacked bar of the partition.

Automatic run time. The time T is the smallest value for which the driest quarter of the field receives the required depth,

$$T = \min\{T : \ \overline{Z}_{25}(T) \geq Z_n\}, \qquad T \in [1,\, 2400] \text{ min}, \tag{13.12}$$

Provenance (Eq. 13.12): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

found by 18 bisection steps. If the soil cannot take the water as fast as it is applied, longer runs only raise runoff and the layer shows the run time and low E_a together.

13.7 Maximum allowable application rate

The allowable rate by soil and slope follows the standard table of sprinkler manuals (Keller and Bliesner, 1990, Table of maximum application rates; reproduced here for the textbook check).

Table 13.3. Maximum application rate (mm h^{-1}) by soil type and slope.

Soil group	0–5 %	5–8 %	8–12 %	12–16 %
Deep coarse sandy soils	50	38	25	13
Coarse sandy, compact layer below	38	25	19	10
Deep sandy loam	25	20	15	10
Sandy loam, compact layer below	19	13	10	8
Deep silt loam	13	10	8	5
Silt loam, compact layer below	8	6	4	2.5
Clay or clay loam, strong structure	4	2.5	2	1.5

The ten soils of the program are assigned to rows: Coarse Sand, Sand, Loamy Sand → 1; Sandy Loam, Coarse Loam → 3; Loam, Silt Loam → 5; Sandy Clay Loam → 6; Clay Loam and Silty Clay

Loam → 7. The toggle *Compact layer below the soil* moves an odd row to the next (compact) row. The slope selects the column.

13.8 The ten textbook checks

The "Does your design pass?" panel runs the design through the course rules. Let $h_s = P/9.81$ (m), $q_h = 0.06\,q$ (m^3 h^{-1}), $R_w = R$, the wind in km h^{-1} $W = 3.6\,u$, and the wind factor

$$k = \begin{cases} 1, & W \le 5.5 \\ 0.9, & 5.5 < W \le 13 \\ 0.8, & 13 < W \le 22 \\ 0.7, & W > 22. \end{cases} \tag{13.13}$$

Provenance (Eq. 13.13): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

The spacing limits, the field layout and the time budget are

$$S_s \le k\,R_w, \qquad S_l \le 1.5\,R_w, \qquad L_y = \frac{10^4 A_f}{L_x}, \quad N = \lfloor L_x/S_s \rfloor, \quad N_m = \lfloor L_y/S_l \rfloor, \tag{13.14}$$

Provenance (Eq. 13.14): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

$$t_{\text{set}} = \frac{Z_{\text{gross}}}{\text{AR}} \ \text{(h)}, \qquad n_{\text{sets/day}} = \left\lfloor \frac{h_{\text{op}}}{t_{\text{set}}} \right\rfloor, \qquad T_{\text{int}} = \max\!\left(1, \left\lfloor \frac{Z_n}{\text{ET}_{\text{pk}}} \right\rfloor\right) \text{ days}, \tag{13.15}$$

Provenance (Eq. 13.15): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

where $L_x = L_L$ is the lateral length, h_{op} the operating hours per day and Z_{gross} the gross depth per irrigation. For $n = 1 \dots 4$ laterals running at once, the days needed and the inflow are

$$N_d(n) = \left\lceil \frac{N_m}{n\, n_{\text{sets/day}}} \right\rceil, \qquad Q_{\text{in}}(n) = n\,N\,q_h, \tag{13.16}$$

Provenance (Eq. 13.16): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

and the program picks the smallest n with $N_d \le T_{\text{int}}$ and $Q_{\text{in}} \le 3.6\,Q_s$ (m^3 h^{-1}); if none satisfies both it picks the smallest n meeting the time, else $n = 4$.

Pipes. Friction losses use the Blasius form (Blasius, 1913) for smooth plastic pipe,

$$J = \frac{0.472\,Q^{1.75}}{D^{4.75}} \ \ (\text{m m}^{-1};\ Q \text{ in L h}^{-1},\ D \text{ in mm}), \tag{13.17}$$

Provenance (Eq. 13.17): **[Literature]**: Blasius friction law for smooth pipe.

with the Christiansen reduction factor $F = 1/2.75 + 1/(2N)$ for N equally spaced outlets and a 10 % allowance for local losses. The lateral pipe is the smallest of the standard internal diameters $\{46.4,\ 58,\ 63,\ 65.8,\ 75,\ 84.9,\ 100\}$ mm for which

$$h_{f,\text{lat}} = 1.1\,F\,J(Q_L, D)\,(L_x - S_s/2) \le 0.2\,h_s, \qquad Q_L = 1000\,N\,q_h, \tag{13.18}$$

Provenance (Eq. 13.18): **[Literature]**: 20 % rule of Keller and Bliesner (1990).

the classical 20 %-of-operating-head rule that keeps discharge variation along the lateral under about 10 % (Keller and Bliesner, 1990). The mainline (flow $Q_m = n\,Q_L$, length $L_y - S_l/2$) is the smallest diameter not below the lateral's with $J < 0.05$ m m^{-1}.

Manometric head and pump.

$$H_{\text{man}} = z_{\text{lift}} + J\,L_{\text{conv}} + J\,L_{\text{main}} + h_s + 0.75\,h_{f,\text{lat}}, \tag{13.19}$$

Provenance (Eq. 13.19): **[Engineered]**: head and sets bookkeeping.

$$P_{\text{CV}} = \frac{1000\,(Q_m/3.6\times10^{6})\,H_{\text{man}}}{75\,\eta}, \quad \eta = 0.75, \qquad P_{\text{kW}} = 0.7355\,P_{\text{CV}}, \tag{13.20}$$

Provenance (Eq. 13.20): **[Literature]**: hydraulic power definition.

with L_{conv} the conveyance length (the mainline length box) and $0.75\,h_{f,\text{lat}}$ the head at the lateral inlet above h_s (the textbook inlet-head rule). The ten checks (spacing along the lateral, spacing between laterals, wind below 16 km h^{-1}, rate below the allowed value, rate below K_s, CU ≥ 85 %, lateral loss ≤ 20 % of h_s, mainline gradient, the whole field within the interval, flow within the farm supply) are shown with a tick or an exclamation point and the numbers; "N of 10 checks pass" closes the panel. The textbook sequence in the layer also plots the class of sprinkler (low, medium or high pressure) from h_s: < 20 m, 20–45 m, > 45 m.



Figure 13.3: Sprinkler design: the ten textbook checks and the kinds of sprinkler system. Volume 2 shows the whole layer, part by part, with its equations.

13.9 Kinds of sprinkler system

Five systems are offered, each with its efficiency range (Keller and Bliesner, 1990; Burt et al., 1997 report the range of field values):

System	Principle	Typical E_a
Solid-set	Fixed buried laterals; valves open some at a time	70–75 %
Hand-move	The lateral is moved by hand after every set (1–4 sets a day)	70–75 %
Center pivot	A rotating lateral on towers about 30 m apart; the corners stay dry	65–70 %
Linear-move	A lateral travelling in a straight line	65–70 %
Traveling gun	One gun on a hose reel covers a 220° sector, strip by strip	65–70 %

The plan of the selected system is drawn to scale for the present field; the run moves it (the active lateral, the pivot arm, the travelling lateral or the gun strip). A bar shows where the design's E_a lies against the range for the system.

13.10 Layout and infrastructure

The *Layout and infrastructure* block of this layer uses the same hydraulic tools as the drip layer, with these differences. With $N_s = \lfloor L_L/S_s \rfloor$ sprinklers per lateral and $Q_{\text{lat}} = N_s q/60$ (L s^{-1}),

the number of laterals that can run together is $\lfloor Q_s/Q_{\text{lat}} \rfloor$ and the sets

$$N_L = \Big\lfloor \frac{10^4 A_f / L_L}{S_l} \Big\rfloor, \quad n_l = \min\Big(N_L,\ \Big\lfloor \frac{Q_s}{Q_{\text{lat}}} \Big\rfloor\Big), \quad n_{\text{set}} = \Big\lceil \frac{N_L}{n_l} \Big\rceil, \quad Q_z = n_l Q_{\text{lat}}. \tag{13.21}$$

Provenance (Eq. 13.21): **[Engineered]**: head and sets bookkeeping.

The lateral loss uses the Hazen–Williams equation (Eq. 12.3, $C = 140$) with the Christiansen factor for N_s outlets and the exponent $m = 1.852$,

$$F_m = \frac{1}{2.852} + \frac{1}{2N_s} + \frac{\sqrt{0.852}}{6N_s^2}, \qquad h_{f,\text{lat}} = F_m \, h_{HW}(Q_{\text{lat}}, D, L_L), \tag{13.22}$$

Provenance (Eq. 13.22): **[Literature]**: Christiansen reduction factor with Hazen–Williams exponent.

and the lateral pipe is the smallest of the series (12.21) with $h_{f,\text{lat}} \le 0.2\,h$ and $v \le 2.0$ m s^{-1}. The lateral inlet head is $H_0 = h + 1.5 + h_{f,\text{lat}}$ m (1.5 m of riser). The submain, mainline and pump then follow Eqs. (12.22)–(12.23) exactly as for drip, with H_0 from this paragraph. The block's table gives the sets ($\times$ laterals), laterals and sprinklers, TDH, power, lateral loss (≤ 20 %) and pipe laid per hectare.

(Two parallel calculations therefore appear in this layer: the **simulation block** sizes the pipe by Hazen–Williams with $C = 140$, while the **textbook block** uses the Blasius equation and the commercial plastic-pipe sizes of the course. They are separate presentations of the same design question; the numbers will differ by a few per cent, and the textbook block is the one to compare with hand calculations from the course.)

13.11 The design table and sensitivity

The optimiser works like the drip one (Eq. 12.26): score $\min(E_a, 1) - 2.5\max(0, 0.9 - E_s)$, target $E_a \ge 85$ % and $E_s \ge 85$ %, accept a move only if the score rises by more than 0.002. The ten moves are **pressure, spacing along the lateral, run time, nozzle, spacing between laterals, run time, pressure, triangular layout, run time, spacing along the lateral**. For pressure, nozzle and the two spacings the candidates are nine equally spaced values over the range of Table 13.1 (rounded to the step); for the layout the two options; for the run time $T_0 \times \{0.6, 0.7, 0.8, 0.9, 1, 1.1, 1.25\}$ with T_0 the automatic time. Tapping a row applies it. If the target is not met, the layer suggests a lower application rate (a smaller nozzle or wider spacing) or another sprinkler type.

The sensitivity chart moves eight inputs and shows the change of E_a: soil intake k (± 20 %), wind (± 50 %), pressure (± 20 %), nozzle (± 15 %), the two spacings (± 20 %), the depth per irrigation (± 25 %) and the ground slope (± 50 %).

13.12 Season and farm

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \qquad n_{\text{sp}} = \text{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\text{tot}} = \frac{n_{\text{sp}}\, q}{60}, \qquad n_{\text{set}} = \Big\lceil \frac{Q_{\text{tot}}}{Q_s} \Big\rceil, \qquad t_{\text{round}} = \frac{n_{\text{set}} T}{60} \text{ h}. \tag{13.23}$$

Provenance (Eq. 13.23): **[Engineered]**: head and sets bookkeeping.

The gross water per irrigation is gross $= \bar{a}\,T$ mm, per season n_{ev} gross mm ($= 10\,n_{\text{ev}}$ gross m^3 ha^{-1}). The table also gives one sprinkler's discharge and reach, the sprinklers for A_f, the sets with Q_s, and the pumping hours and energy of the season. The layer closes with the *Irrigation schedule* and the *When to irrigate* message (end of Chapter 11).

13.13 How to use the layer

- **To pick a sprinkler for a soil.** Choose the soil, set the slope and the wind, then compare the types: the banner tells you the largest allowed rate, the intake plot shows where runoff

starts, and the checks name the failing rule.

- **To reduce runoff.** Lower the application rate: a smaller nozzle or lower pressure reduces q; a wider spacing spreads it over a larger area; a triangular layout improves CU at the same rate.
- **To reduce drift.** Reduce the pressure, use a flatter angle, or use a micro-sprinkler ($F = 0.7$); the wind arrow and the w_d share of the partition show the effect.
- **To size the system for the farm.** Set the area, farm supply and operating hours; read the sets, the pipe diameters and the pump power from the layout block, and the textbook block for the hand-calculation version.

Chapter 14

The Dashboard: how every number is produced

The Dashboard is the panel to the right of the drawing. It has four tabs (*Results*, *Design*, *Charts*, *Messages*) and the buttons **Wider** (enlarges or shrinks the panel), **Pin** (stores the present 20-year averages as a baseline; Ctrl+B) and **Clear** (removes the baseline). It is laid out like the Properties panel: a white sheet with grey group bars. **Every block can be folded and unfolded on its own** by clicking its bar; the folding is remembered while the program runs, also when the numbers refresh. The dashboard re-reads the engine each time the state, the date or the weather changes, so it always agrees with the drawing.

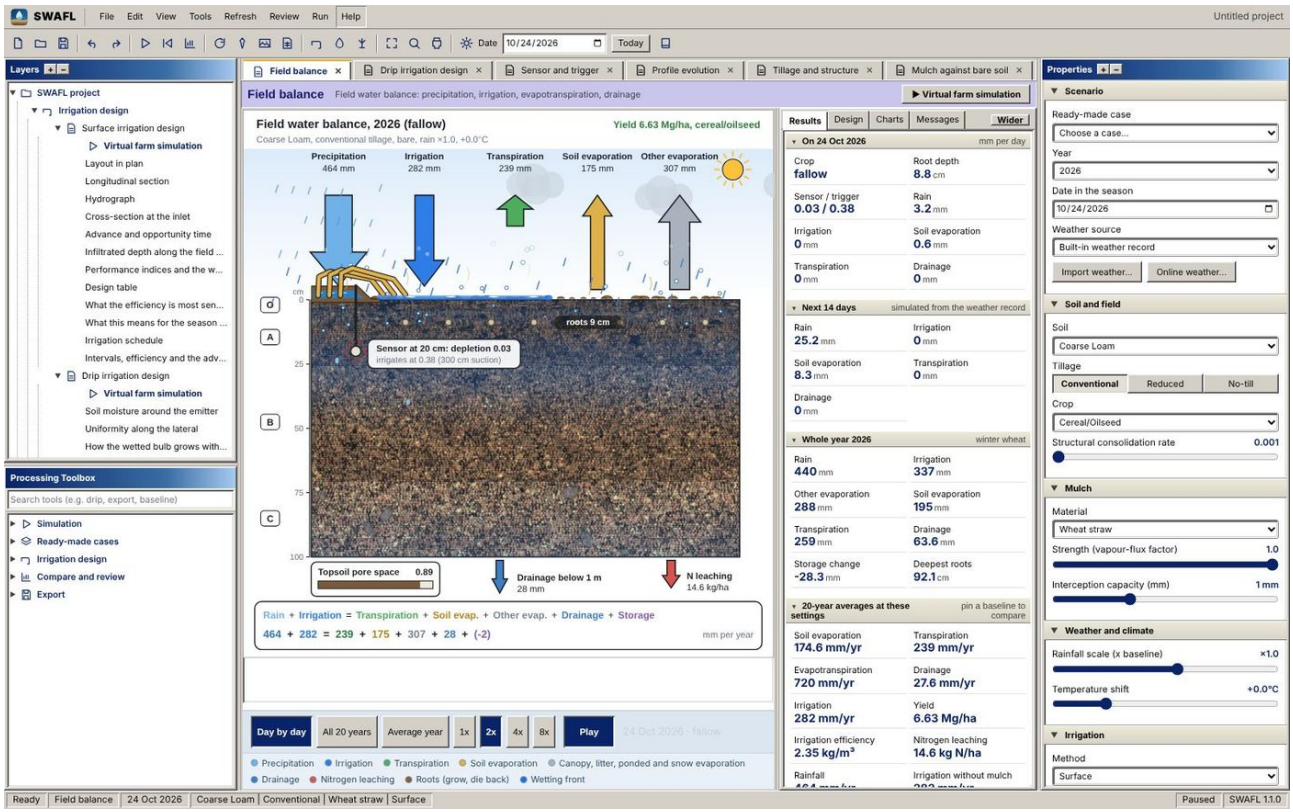


Figure 14.1: The Dashboard, Results tab, with the Field balance layer open.

14.1 Results

The tab has four blocks.

Block 1: On the working date (mm per day unless stated). Crop, root depth (cm), *Sensor / trigger* (the simulated relative depletion f of the sensor layer and the trigger f_T; irrigation fires

when $f \geq f_T$, Eq. 6.1), rain, irrigation, soil evaporation, transpiration and drainage on that day. They are the Engine 2 daily series of Chapter 5, read at the day index k of the working date:

$$\text{Rain} = P_k, \quad \text{Irr} = I_k, \quad \text{Evap} = E_k, \quad \text{Transp} = T_k, \quad \text{Drain} = D_k \ \ (\text{mm d}^{-1}). \tag{14.1}$$

Provenance (Eq. 14.1): **[Engineered]**: bookkeeping of Engine 2 series.

Block 2: Next 14 days. Sums of the same series over the following $N = 14$ days,

$$\Sigma X = \sum_{j=1}^{N} X_{k+j}, \qquad X \in \{P, I, E, T, D\}. \tag{14.2}$$

Provenance (Eq. 14.2): **[Engineered]**: bookkeeping of Engine 2 series.

The bar of the block says *simulated from the weather record* when the days come from the analogue year, and *from the weather forecast* when the online weather is in use and the days fall in its 16-day forecast (Chapter 19).

Block 3: Whole year. For the calendar year of the working date: rain, irrigation, other evaporation (interception, litter, ponded water and snow), soil evaporation, transpiration, drainage, the change in storage and the deepest roots (cm). The water balance of the year closes as

$$\Delta S = P + I - E_{\text{other}} - E - T - D, \tag{14.3}$$

Provenance (Eq. 14.3): **[Literature]**: water-balance identity.

and the tile *Storage change* is ΔS.

Block 4: 20-year averages at these settings. The annual-model outputs of Engine 1 (Chapter 5): soil evaporation, transpiration, evapotranspiration ET $= E + T$, drainage, irrigation, yield (Mg/ha for the simulated cereal, per cent of the potential for a library crop), irrigation efficiency, nitrogen leaching, rainfall, and *Irrigation without mulch* (the same case with no mulch, for the saving).

14.1.1 The pinned baseline

Press **Pin** (or Ctrl+B): the tiles of Block 4 are stored. After any change the tiles show the difference from the pinned values,

$$\Delta_i = x_i - x_i^{\text{pin}}, \tag{14.4}$$

Provenance (Eq. 14.4): **[Engineered]**: bookkeeping.

with ▲ (increase) or ▼ (decrease), green when the change is in the better direction and red when it is worse, and "= no change" when $|\Delta_i| < 10^{-9}$. Lower is better for soil evaporation, drainage, irrigation and nitrogen leaching; higher is better for yield and irrigation efficiency; transpiration, evapotranspiration, rainfall and *Irrigation without mulch* are neutral.

14.2 Design

With a design layer open, the tab shows the results of its **last run**. Before the first run, or after an input changed, it says that the design waits for a run and carries the two buttons **Run simulation** and **Find the most efficient inputs** (Chapter 18). After a run it shows, in folding blocks:

Block	Contents
Key results	Surface: E_a, E_s, deep percolation, runoff, CU, cut-off, fill and advance times, the volumes. Drip: E_a, E_s, E_u, deep percolation, wetted fraction P_w, gross application, irrigation time, volumes per emitter. Sprinkler: E_a, E_s, DU, CU, deep percolation, runoff, gross and net application, set time, discharge, application rate.
Warning	For a surface design whose stream does not reach the end of the field in a practical time.

Block	Contents
Most efficient inputs	The best combination found by the search of Section 18.4, with its E_a, E_s, deep percolation and runoff, and **Apply these inputs and run**; or *Already the most efficient* when no combination tried beats your inputs.
Virtual farm	The button that opens the farm of this design (Chapter 20).

The definitions are Eqs. (11.18)–(11.19), (12.19) and (13.11). The Design tab and the *Design results (CSV)* export read the same object, so the numbers agree digit for digit.

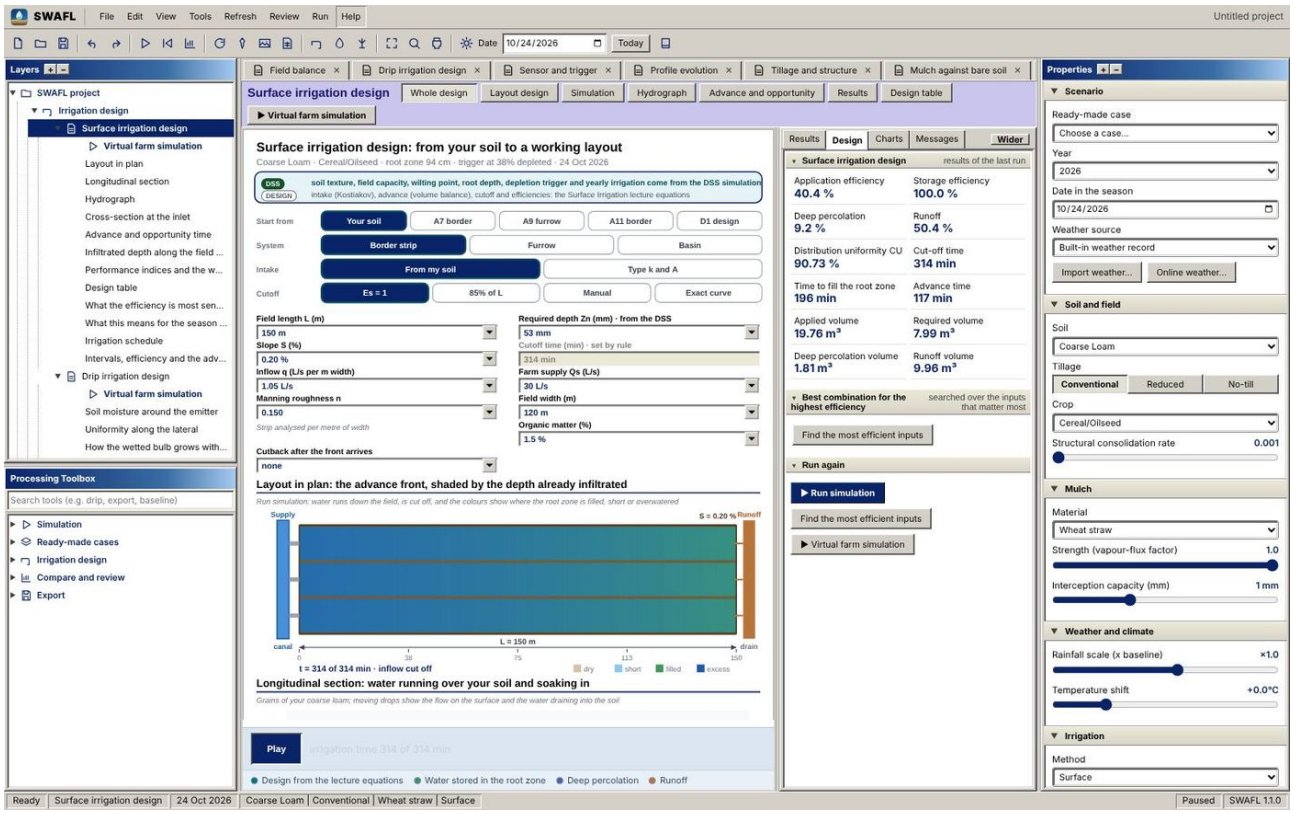


Figure 14.2: The Dashboard, Design tab, after a run of the surface design.

14.3 Charts

With a design layer open and run, the tab draws the charts of the run:

Design	Charts
Surface	Hydrograph: inflow, infiltration and runoff against time, with the advance time t_L and the cut-off t_{co} marked; advance, cut-off and recession along the field; infiltrated depth along the field against the required depth; the times of the irrigation (t_L, t_s to fill the root zone, t_{co}, the recession time t_r and the total); where the water goes.
Drip	Hydrograph at the pump (the zones run one after the other at the zone flow); the times of the irrigation (run time of a zone, the whole field, the interval); pressure head along the lateral; emitter flow along the lateral; where the water goes.
Sprinkler	Hydrograph at the pump (the sets run one after the other); the times of the irrigation; soil intake rate against the application rate; depth caught at each point, sorted (uniformity); where the water goes.

With any other layer open, three charts of the twenty years around the working date (calendar

years on the horizontal axis, mm per year): *Rainfall and irrigation*, *Drainage below the root zone* and *Evaporation and transpiration*. They are the annual sums of the Engine 2 daily series,

$$Y^{(y)} = \sum_{d \in y} X_d, \qquad X \in \{P, I, D, E, T\}. \tag{14.5}$$

Provenance (Eq. 14.5): **[Engineered]**: bookkeeping of Engine 2 series.

Every chart sits in its own folding block; a chart that is unfolded is drawn again at once. The pump hydrograph of drip and sprinklers is the step series

$$Q_{\text{pump}}(t) = Q_z \quad \text{for } t \in [\,0,\ n_z\, t_{\text{run}}\,), \qquad 0 \text{ afterwards}, \tag{14.6}$$

Provenance (Eq. 14.6): **[Engineered]**: zones of the layout (Eqs. 12.20–12.23 and the sprinkler analogue) run in turn.

with Q_z the flow of one zone or set, n_z the number of zones or sets and t_{run} the run time of one zone (drip) or the set time (sprinklers).

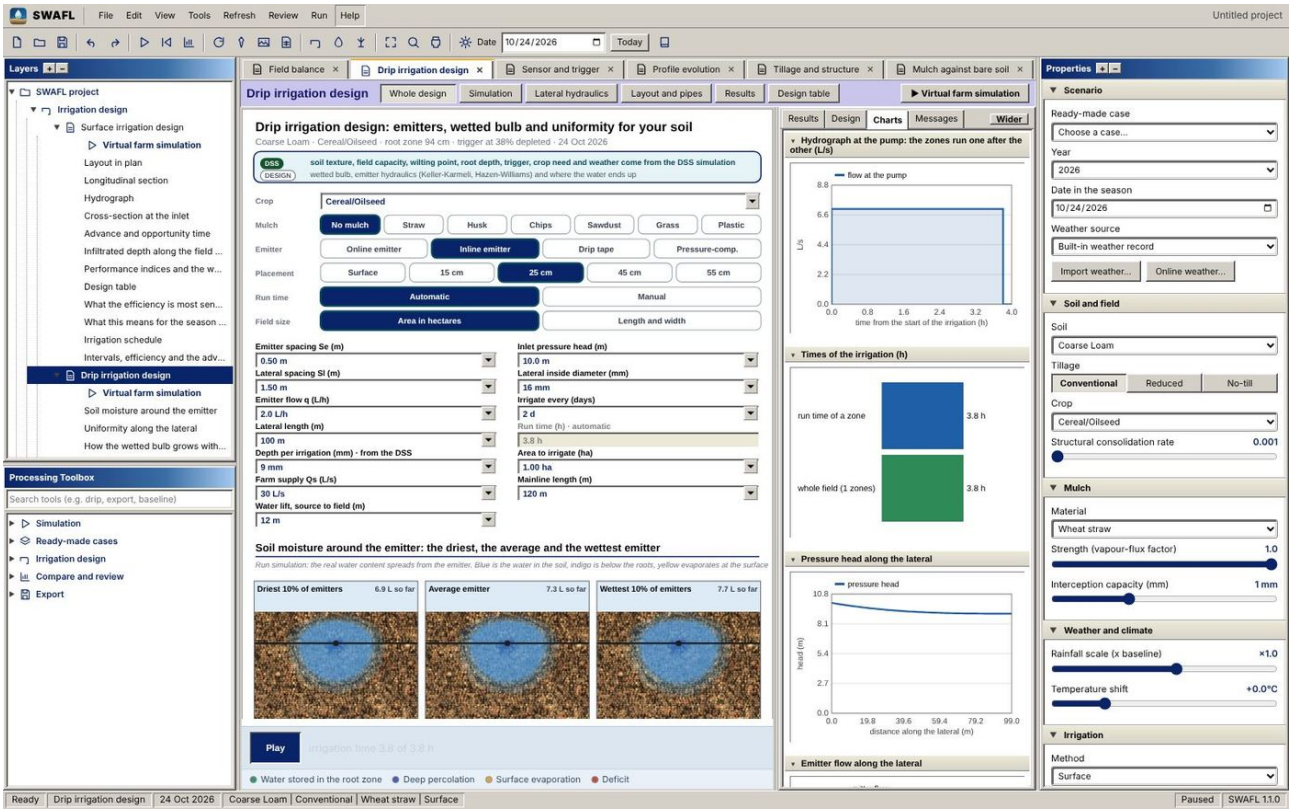


Figure 14.3: The Dashboard, Charts tab, after a run of the drip design.

14.4 Messages

A time-stamped log of what the program did (weather imported, design run, report saved, baseline pinned), newest first, in two folding blocks: *Warnings* (only when there are any) and *All messages*. Use it to find the path of a saved file.

Chapter 15

Worked workflows

Each workflow is a complete sequence of clicks, what to expect, and the equations behind it.

15.1 Workflow 1: How much does straw mulch save?

1. Properties ▸ Scenario ▸ *Ready-made case* ▸ *Default* (Coarse Loam, bare, normal climate).
2. Press **Pin** above the dashboard (Ctrl+B). The tiles of the 20-year block are stored.
3. In Properties ▸ Mulch choose **Wheat straw**; the strength becomes 0.4 and the capacity 4 mm. Read the arrows: soil evaporation falls (green), irrigation falls (green).
4. Open *Mulch and evaporation* ▸ *Fate of saved water* to see where the saved water goes (less irrigation, more drainage, more transpiration).

Why: the strength φ scales the vapour flux through the mulch, and the capacity the water the mulch holds (Chapter 9); *Irrigation without mulch* minus *Irrigation* is the saving on the whole-year balance.

15.2 Workflow 2: When should I irrigate?

1. Set the working date to today and choose the soil and crop.
2. Open *Irrigation design* ▸ *Surface irrigation design* (or drip, or sprinklers), set the inputs and press **Run simulation**. Scroll to the *Irrigation schedule*: the first row is the next irrigation date; the *When to irrigate* message at the end repeats it with the mean interval.
3. Change the **Trigger** in Properties to a drier value (for example $-300 \rightarrow -1000$ cm): the run is cancelled; run again: the interval lengthens and the number of irrigations falls (Eq. 6.3 shifts f_T).
4. Change the **Sensor depth**: a shallow sensor dries first and triggers earlier.

15.3 Workflow 3: Design a border strip for a loam

1. Set the soil to Loam; open *Surface irrigation design*; *Start from: Your soil*; *System: Border*.
2. Type the run length of your field in the *Field length* box (for example 200 m), the width and the slope.
3. Press **Run simulation**. Read E_a and E_s in *Result*, and the hydrograph and the times on the Charts tab.
4. Press **Find the most efficient inputs**. The search tries inflows, cut-off rules, cutbacks and shorter runs (Section 18.4); if it finds a better combination, the Design tab offers **Apply these inputs and run**.
5. Read the sensitivity chart: if the intake constants dominate, the soil intake is the main uncertainty, so measure it in the field.

15.4 Workflow 4: Choose and size a drip system

1. Choose the crop, the soil and the emitter (inline).

2. *Field size*: **Length and width**; type the length and width of the field. The area and the lateral length follow (Eq. 18.1).
3. *Placement*: compare *Surface*, *25 cm* and *45 cm*: each change cancels the run; run each and compare E_a and the losses.
4. Set the farm supply Q_s; read zones, pump head and power in the layout block (Eqs. 12.20–12.23), and the pump hydrograph on the Charts tab.
5. Open the *Virtual farm simulation* of the drip design to watch one irrigation in the field and its 3D water fate (Chapter 20).

15.5 Workflow 5: Check a sprinkler against the course rules

1. Choose the soil and the slope (for example Sandy Loam, 3 %); the banner states the allowed rate (Table 13.3).
2. Choose *Impact sprinkler*; set the nozzle, the pressure and the spacing in their value boxes; press **Run simulation**; read CU and the application rate.
3. Open *Does your design pass?*; fix the red checks in the order the text names; run again.
4. Compare the system chips (solid-set, hand-move, center pivot, linear-move, travelling gun) and their efficiency ranges.

15.6 Workflow 6: My own weather, a warmer future

1. *File* ▸ *Import weather data* and choose your daily weather file, or *File* ▸ *Online weather for a location* (Chapter 19). Every layer now runs on it; Properties shows *Imported weather file* (or *Online weather of a location*) as the weather source.
2. Pin the baseline.
3. Set the temperature shift to +2 °C and the rainfall scale to ×0.8. The arrows show the effect of the change on your own climate.
4. Switch back to *Built-in weather record* in Properties ▸ Scenario at any time: your earlier rainfall and temperature settings return.

15.7 Workflow 7: A report for a client

1. *File* ▸ *Save report as PDF (choose the layers)…*.
2. Untick the groups you do not need (for example *Seasonal data* and *Feedback loops*), keep the irrigation design and the virtual farm.
3. **Make the PDF**, choose where to save it. Each layer gets your inputs, its key numbers and its full drawing; long drawings continue on the next pages, cut at their section headings (Chapter 23).

15.8 Workflow 8: Save, share and reproduce

Save the project (*File* ▸ *Save project*) after each case, export the yearly CSV and the design CSV, and note the version shown in the status bar (*SWAFL 1.1.0*). Reopen the `.swafl` file to restore the state exactly.

Chapter 16

Files, exports and troubleshooting

16.1 What can be saved

Item	Menu / toolbox	File	Contents
Report	*File* ▸ *Save report as PDF (choose the layers)…*, Ctrl+Shift+P, toolbar *Report*	`SWAFL_full_report_<soil>_<date>.pdf`	The layers you ticked, each with your inputs, its key numbers and its full drawing (Chapter 23)
Picture of the window	*File* ▸ *Export* ▸ *Picture of the window (PNG)*, Ctrl+E	`SWAFL_<layer>.png`	The whole drawing of the open layer, top to bottom
Yearly results	*File* ▸ *Export* ▸ *Yearly results (CSV)*	`SWAFL_yearly_results.csv`	Two comment lines with soil, tillage, crop, mulch and method; then `Year, Rain_mm, Irrigation_mm, Drainage_mm, Evaporation_mm, Transpiration_mm`
Design results	*File* ▸ *Export* ▸ *Design results (CSV)*	`SWAFL_design_<id>.csv`	Two columns `Quantity,Value`: every number the open design layer computed
Project	*File* ▸ *Save project*	`SWAFL_project.swafl`	JSON: app name and version, the open layer, the state, the year and the date
Virtual farm video	*Record video (20 s)* on the farm	`SWAFL_virtual_farm_<date>.webm`	Twenty seconds of the farm, from two days before the next irrigation
Print	*File* ▸ *Print / save this view as PDF*, Ctrl+P	PDF	The open layer through the system print dialog

In the yearly CSV, *Year* is the index 1–20 of the 20-year run starting at the calendar year shown on the Charts tab; the values are the annual sums of Eq. (14.5), rounded to 0.01 mm. In the design CSV the quantity names are the internal names of the model (for example `Ea`, `Es`, `dp`, `ro` for the surface layer; `Eu`, `gross`, `t` for drip; `CU`, `DU`, `AR`, `T` for sprinkler); the efficiencies are fractions (0–1).

The imported weather is not stored in the project file; import it again after opening a project, or keep the file beside the project.

16.2 Troubleshooting

Problem	Fix
“Running scripts is disabled” in PowerShell	Use a Command Prompt terminal, or run `Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned`, or type `cmd /c npm install`.
`npm` is not recognised	Install Node.js, then close and reopen the terminal.

Problem	Fix
`ENOENT … package.json not found`	You are in the wrong folder; the zip makes a folder inside a folder. `cd` into the one that contains `package.json`.
Install or build is slow or fails inside OneDrive	Copy the project to `C:\SWAFL_Desktop` and run it there.
“Windows protected your PC”	Click *More info* ▸ *Run anyway* (the app is not code-signed).
The window still shows version 1.0	An old `SWAFL.exe` is still on the Desktop or pinned. Delete it and run `BUILD_AND_INSTALL_SWAFL.bat` again: the build removes the old `dist` folder first and the program clears its cache on start.
White or empty window	Wait 10 seconds (the engine loads), then *Refresh* ▸ *Reload simulation engine*. Close any second SWAFL window.
A design shows *Run the design* instead of results	The inputs or a setting changed since the last run. Press **Run simulation**.
The virtual farm asks to run a design	The farm follows the last run of its design (Chapter 20). Press **Run the … design now**.
A design layer shows “the stream does not reach the end”	The cut-off time is shorter than the advance time: increase the cut-off or the inflow (Eq. 11.16), or shorten the run.
E_a is low and runoff is high (sprinklers)	The application rate is above the soil intake: reduce the nozzle or the pressure, widen the spacing (Chapter 13).
A weather file is refused	Read the message: the file needs a date column (or year, month and day) and a rain column with their names in a header row. A Daisy setup file (`.dai`) holds no weather: the message names the weather file it uses; import that file (Chapter 19).
The online weather is not reachable	The computer is offline or the service does not answer. The current weather stays in use; try again later or import a file.

If a problem remains, open an issue on the project page, and attach the `.swafl` file and the text of the *Messages* tab.

Chapter 17

Validity, limits and honest use

The program is a **teaching and design-exploration tool** built on a research data set. This chapter says what each part can and cannot support.

Engine 1 (the annual meta-model) reproduces, by interpolation, the Daisy 7.1 runs it was fitted to (run sets E2, E3, E4, E5, E7, E11, E13 and E14; the study as a whole has 10,190 runs). Within the ranges of the controls it is an *emulator* of those runs: it is not a new physical model, and it does not extrapolate outside the soils, mulches, climates and irrigation methods that were simulated. It was calibrated to one location's weather record (1980–1999) and its projections to 2100 are **scenarios built from analogue years**, not forecasts.

Engine 2 (the daily bucket model) is a 40-layer, 1-D, daily water balance. It uses closed-form root, evaporation and transpiration functions with a single calibrated coefficient k_t so that its seasonal transpiration matches Engine 1. It does not solve the Richards equation, has no lateral flow, no runoff on slopes and no water table (upward flow is a redistribution between layers only). Its day-to-day irrigation dates are therefore indicative: use them to compare scenarios, not to schedule a real irrigation to the day.

Design layers (surface, drip, sprinkler) are textbook models (Chapters 11–13). They are suitable for teaching, for comparing layouts, and for a first sizing. They are **not a substitute for field evaluation** of intake, advance, pressure and flow, or for a hydraulic design by a qualified engineer. In particular:

- the intake of the soil is derived from texture by pedotransfer functions with their own scatter; measure it in the field for real design;
- the sprinkler pattern is a parametric shape, not a measured catch-can test; the wind-drift loss is a parametric estimate;
- the drip bulb is a geometric simplification of two-dimensional soil water redistribution;
- pump power and pipe sizes use stated allowances (9 m of fittings, 0.65 pump efficiency, $C = 140$) that you should replace with your own;
- the "Engineered" items of the label scheme (Section "About this manual") are modelling choices, not published laws.

Data and reproducibility. The same inputs and version give the same outputs (no random numbers in the design layers; the rainfall generator and all animations of the engine use a deterministic generator seeded by the soil name). Save the `.swafl` file and the version number with any result you report.

Your weather (version 1.1). Imported or online weather drives the daily model and the crop water requirement day by day. The annual model, however, can only move along the simulated steps of rainfall ($×0.5–1.4) and warming (-1$ to +3.5 °C) of the study site (Eqs. 19.4–19.5): a climate far outside those steps (for example a humid tropical climate with 2,000 mm of rain) is represented by the nearest step, and the annual results should then be read as the direction of a change, not as its size. The online forecast covers 16 days; dates after it are the typical year of your record.

The crop library (version 1.1). The 48 library crops are not simulated: they are transferred from the simulated cereal rotation by the ratio of two FAO-56 balances (Section 21.4). The transfer is reliable for the water need and the irrigation (which follow from K_c and ET_0), less so for drainage and nitrogen, and the yield is a relative yield (FAO-33), not a harvest in tonnes.

The virtual farm (version 1.1). The farm is a picture of the design result and of the daily

model: the partition of the water is that of the design run, the wetted depth follows Eq. (20.2), and the plants are drawings, not a crop model.

Not included. The package does not contain a soil-erosion model, a crop-pest model, an economic optimiser, or a groundwater model.

When you cite results from the program, state the version, the case (soil, crop, mulch, method, trigger, date and weather source) and whether the numbers come from Engine 1, Engine 2 or a design layer.

Chapter 18

Running a design and finding the most efficient inputs

Version 1.1 changes how the three irrigation designs are used. Their inputs are set first, and nothing is computed while you type; the results, charts, design table, schedule and virtual farm appear only when you press **Run simulation**. This chapter describes the run, the value boxes, the field size, the search for the most efficient inputs, the field sketch and the comparison of the three systems.

18.1 The run

When a design layer opens, its inputs are drawn as usual and, below them, the results are replaced by the **Run the design** box:

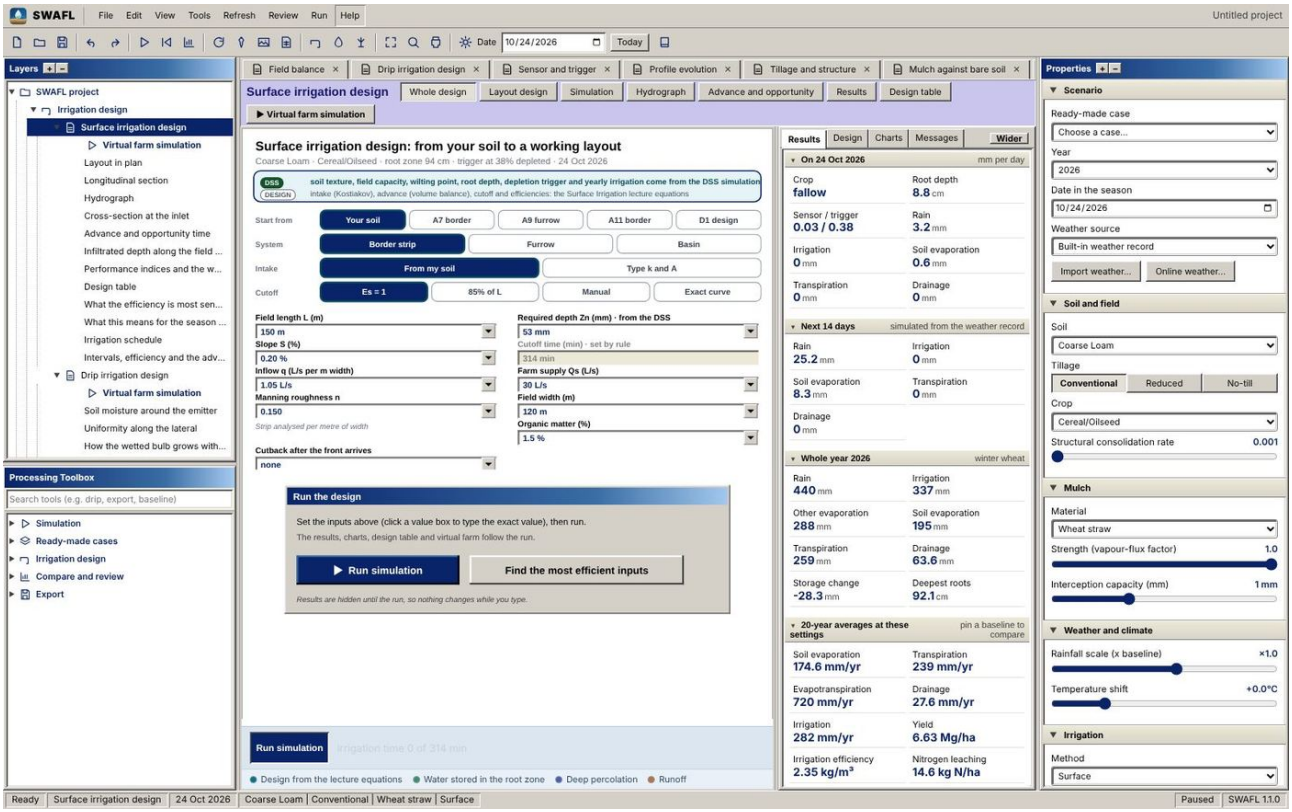


Figure 18.1: The surface design before a run: the inputs are set, the results wait for the run.

- **Run simulation** computes the design with the inputs above it. The run animates the irrigation from its start to its end (the time is shown on the playback bar, for example *irrigation time 0 of 314 min*), and then holds the finished state.

- **Find the most efficient inputs** searches the inputs for the highest application efficiency (Section 18.4), puts the best combination into the value boxes and runs it.
- The play button under the drawing is labelled **Run simulation** in the design layers and does the same.

The last run stays valid until something that changes the result changes: a value box, a chip of the design (system, emitter, placement, sprinkler type, layout, run-time mode, mulch, crop, field size), or a setting of Properties other than the irrigation method (soil, tillage, consolidation, crop, mulch, rain, temperature, trigger, sensor depth, nitrogen, year or date). The box then returns with the reason, for example *The settings in Properties changed: the results wait for a new run*. The Design and Charts tabs of the dashboard follow the same rule.

Formally, a run stores the key κ_{run} of the state with which it was made; the results are shown only while

$$\kappa(\text{state now}) = \kappa_{\text{run}}, \qquad \kappa = \big(\text{every Properties control except the irrigation method, weather mode}\big), \tag{18.0}$$

Provenance (Eq. 18.0): **[Engineered]**: bookkeeping of the run.

and no input of the design has been changed since.

18.2 Value boxes and units

Every input of a design is a value box. Clicking it opens a small window:

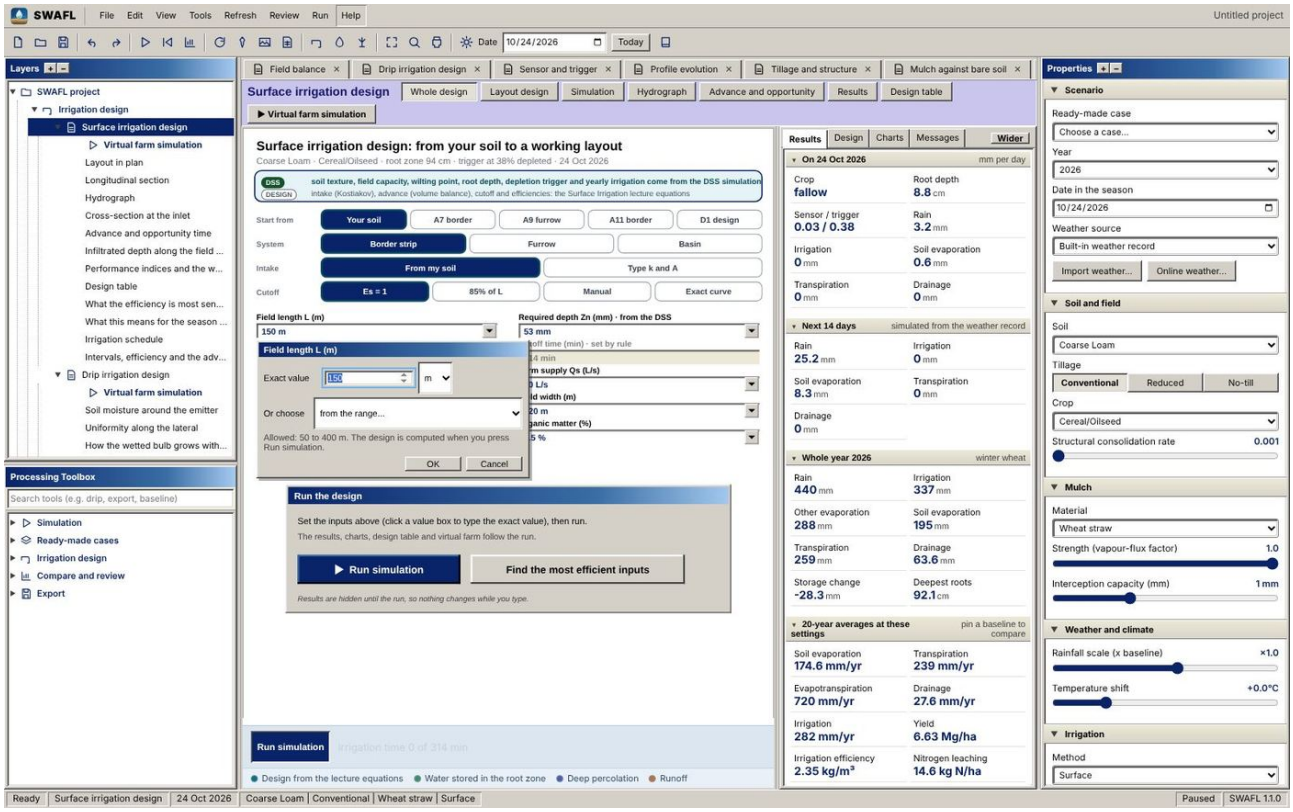


Figure 18.2: The value window of a surface-design input: exact value, unit, list of the allowed range.

- type the exact value, in the unit chosen beside it;
- or pick a value from the list, which runs across the allowed range;
- lengths can be given in m, km or ft; flows in L/s, m³/h or L/min (emitter flow in L/h or L/s); areas in ha, m² or acre; pressures in kPa, bar, m of water or psi; pressure heads in m, kPa, bar or ft; times in min or h; spacings in m or cm.

A value outside the allowed range is set to the nearest limit, and the window says so. The value

is converted to the unit of the model, $x_{\text{model}} = x_{\text{typed}}\, c_u$, with c_u the factor of the chosen unit (for example 1000 for km, 1/3.6 for m³/h to L/s, 9.80665 for m of water to kPa).

18.3 Field size: an area, or a length and a width

The drip and sprinkler designs need the area to irrigate and the length of the laterals. The row **Field size** offers two ways of giving them:

- **Area in hectares** — type the area; the lateral length is its own value box.
- **Length and width** — type the length and the width of the field in metres. The area and the lateral length then follow, and the lateral-length box is greyed out with the note *from the field length*:

$$A_f = \frac{L\,W}{10^4}\ \text{[ha]}, \qquad L_{\text{lat}} = \text{clip}\Big(\frac{L}{\lceil L/L_{\max}\rceil};\ L_{\min},\ L_{\max}\Big), \tag{18.1}$$

Provenance (Eq. 18.1): **[Engineered]**: geometry; laterals longer than $L_{\max}$ are split into equal runs from a central submain.

with $L_{\min} = 20$ m and $L_{\max} = 300$ m for drip laterals and $L_{\min} = 40$ m and $L_{\max} = 500$ m for sprinkler laterals. The number of laterals follows from the width, $n_l = \lfloor W_f / S_l \rfloor$ with $W_f = A_f \cdot 10^4 / L_{\text{lat}}$ (Eq. 12.20 and its sprinkler analogue in Chapter 13), and with it the zones, the pipes and the pump.

The surface design is given by its run length and width directly (*Field length L*, *Field width*), and its area is their product.

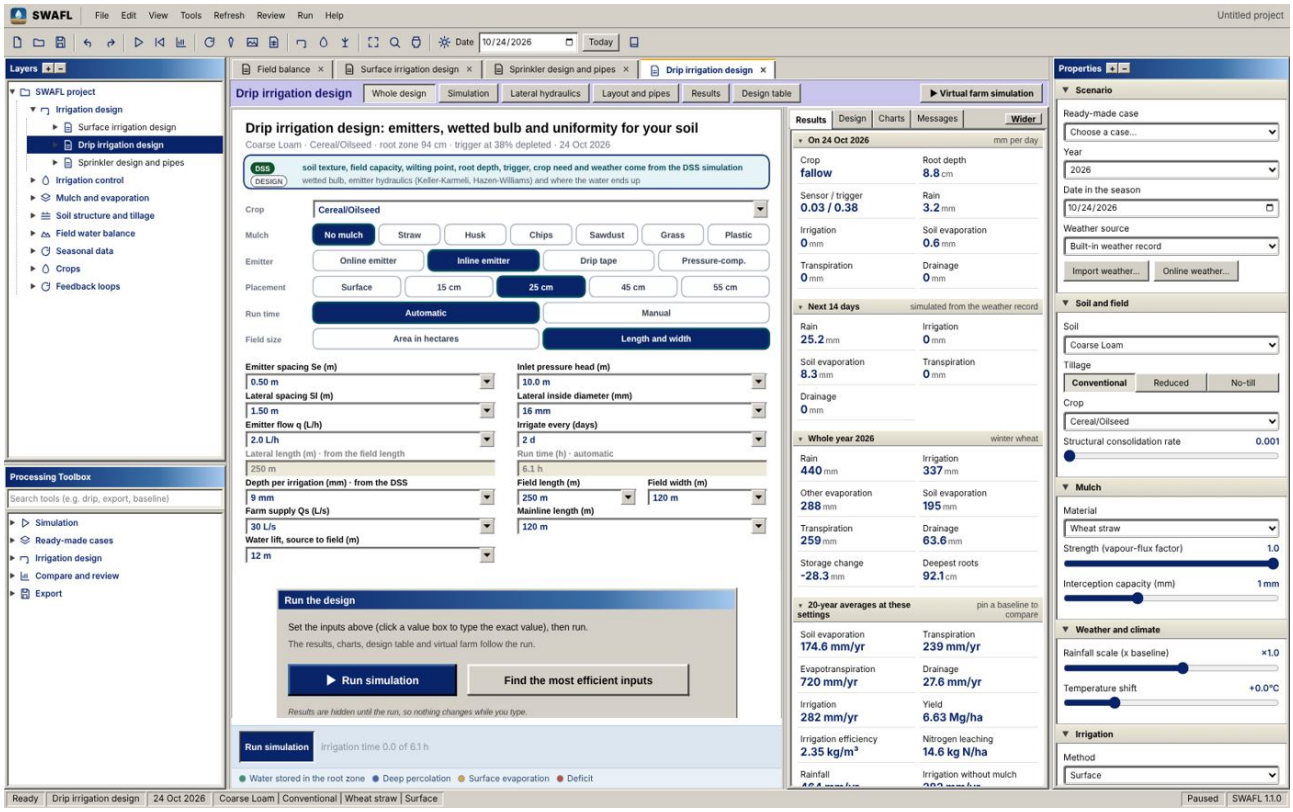


Figure 18.3: The drip design with the field given by its length and width.

18.4 The search for the most efficient inputs

Find the most efficient inputs (in the Run box, on the Design tab, or Tools ▸ *Find the most efficient design inputs*) tries a grid of the inputs that matter most and keeps the one with the highest score

$$s = E_a - 3\,\max\!\left(0,\; 0.95 - E_s\right) + \begin{cases} 0 & \text{surface} \\ 0.05\,E_u & \text{drip} \\ 0.05\,\mathrm{CU} - 2\,R_o & \text{sprinkler} \end{cases} \tag{18.2}$$

Provenance (Eq. 18.2): **[Engineered]**: the application efficiency, with a penalty when the root zone is refilled to less than 95 % (an efficient design that leaves the crop short is not efficient) and a small reward for uniformity.

Design	Grid tried
Surface	11 inflows spaced geometrically across the allowed range, $q_i = q_{\min}(q_{\max}/q_{\min})^{i/10}$; the cut-off rules *when the tail is filled* and *at 85 % of the length*; cutback to 100, 75 or 50 % of the inflow; and the run length L/n for $n = 1$ to 4 (the field split into n runs, each at least the shortest length of the system).
Drip	emitter flow 1, 1.6, 2, 3, 4 or 8 L/h; emitter spacing 0.3, 0.5, 0.75 or 1 m; interval 1 to 4 days.
Sprinkler	five nozzle diameters and four pressures across the range of the sprinkler type; for each, a square spacing of 0.4 to 0.65 times the wetted diameter.

Your present inputs take part in the search. A combination is recommended only when its score is higher than the score of your present inputs; otherwise the Design tab says *Already the most efficient: your present inputs* and offers nothing to apply. The search takes about 0.1–3.5 s and stops after that time.

Applying a combination writes its values into the value boxes, so you can see and change each of them, and runs the design.

18.5 The field sketch

View or *Tools* ▸ *Field sketch to design…* opens a grid on which you click the corners of your field and the water source. The sketch gives

$$A = \tfrac{1}{2}\left|\sum_i (x_i y_{i+1} - x_{i+1} y_i)\right|, \qquad L_{\text{run}} = \max_{\text{along the longest side}} - \min, \qquad W = \text{extent across it}, \tag{18.3}$$

Provenance (Eq. 18.3): **[Literature]**: shoelace formula and projection on the direction of the longest side.

and the mainline length from the source to the nearest corner. **Turn the sketch into the design** fills the run length, width, area, lateral and mainline lengths and the slope of the chosen design (border strips, furrows, basins, drip or sprinklers), opens it and waits for the run.

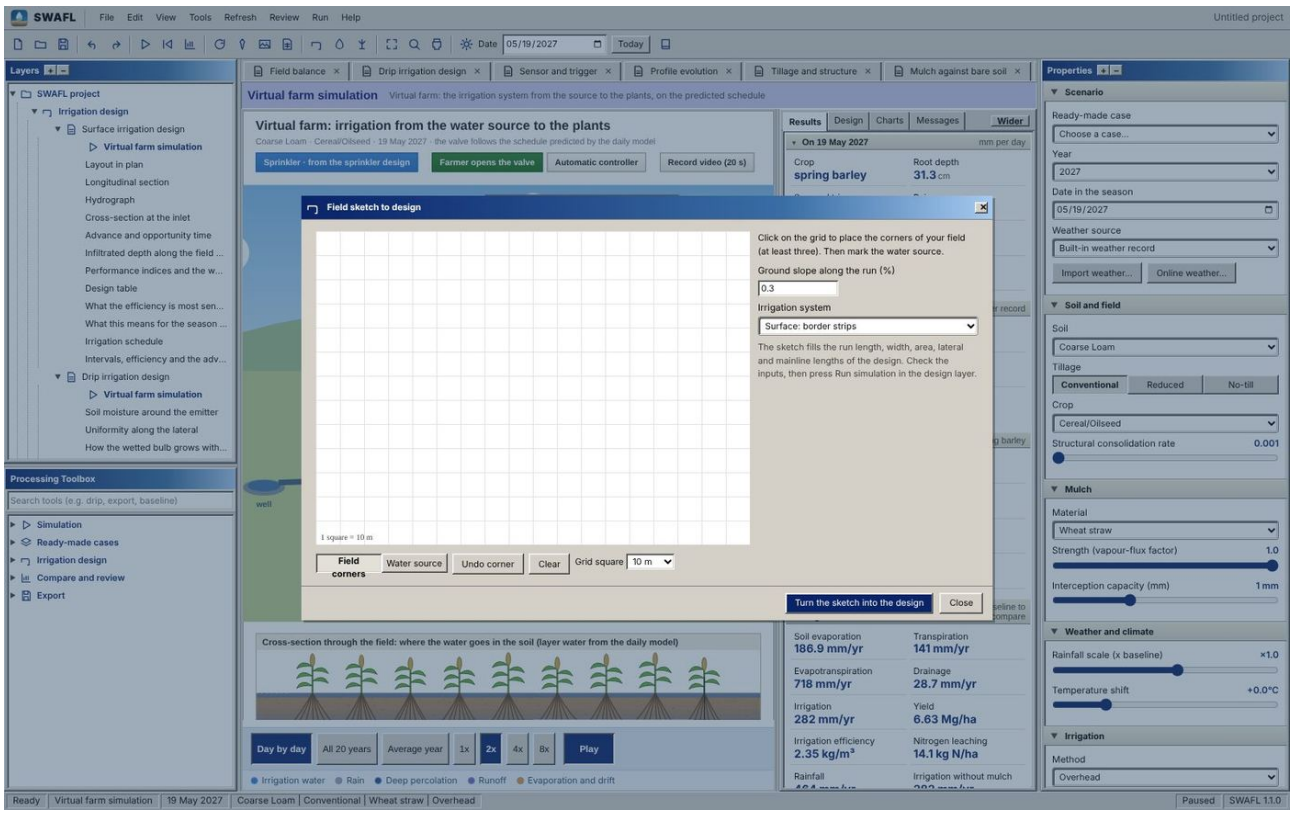


Figure 18.4: The field sketch.

18.6 Comparing the three systems

Tools ▸ *Compare the irrigation systems on this field* computes the surface, drip and sprinkler designs with the inputs they have now, on the same soil, crop and trigger, and lists their application efficiency, root zone filled, deep percolation, runoff or evaporation and the time per irrigation side by side.

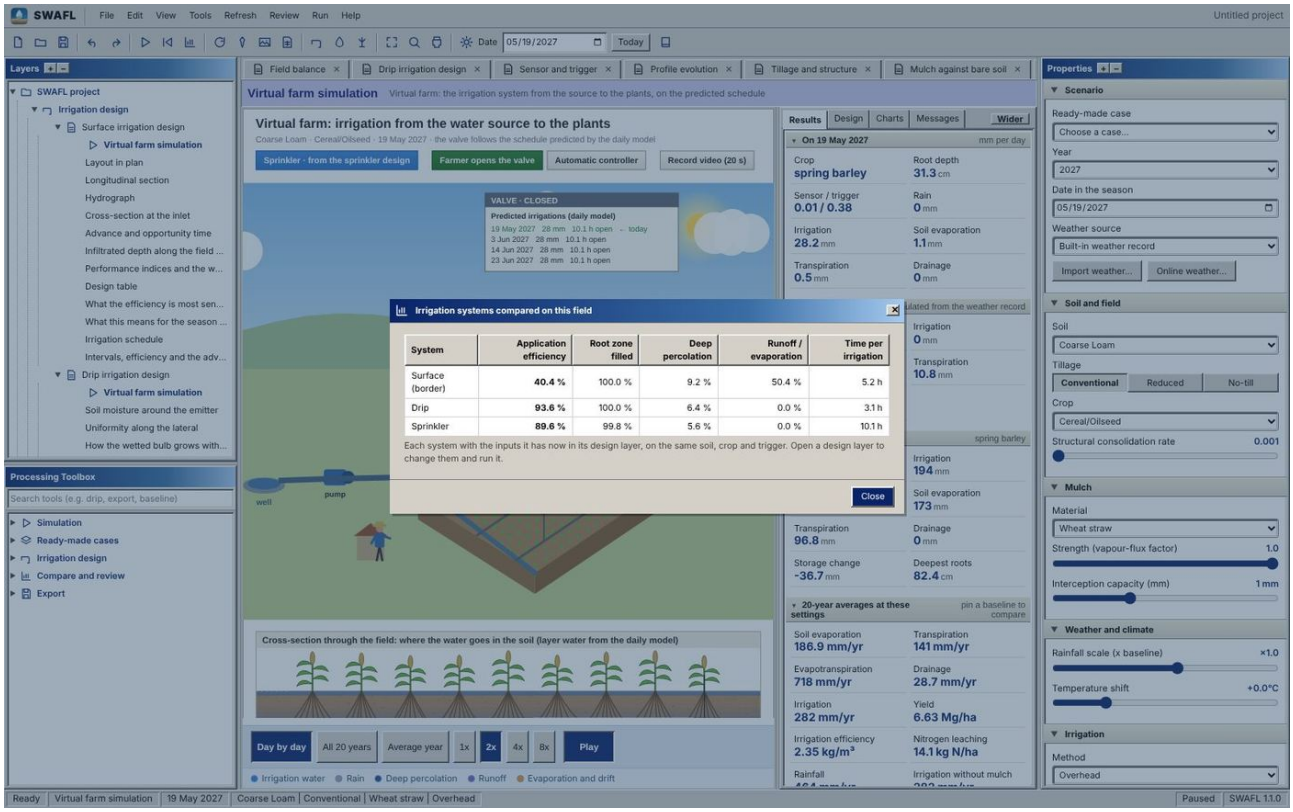


Figure 18.5: The comparison of the three systems.

Chapter 19

Weather data: your own file, or the weather of a place

Without weather data of your own, SWAFL runs on the twenty recorded years of its study site. Version 1.1 lets every layer run on **your weather**: a daily weather file, or the measured daily weather of any place on Earth with its 16-day forecast. The change is seen at once in every layer, the dashboard, the schedules and the crop water requirement.

19.1 Where to start

- **File ▸ Import weather data (CSV, Excel, DWF, DLF, DAI)…**, Tools ▸ *Import weather data…*, or the **Import weather…** button in Properties ▸ Scenario.
- **File ▸ Online weather for a location…**, Tools ▸ *Online weather for a location…*, or the **Online weather…** button.
- Properties ▸ Scenario ▸ **Weather source** switches between *Built-in weather record*, *Imported weather file* and *Online weather of a location*. The last two can be chosen only after a file has been imported or the weather of a place fetched.

19.2 Importing a file

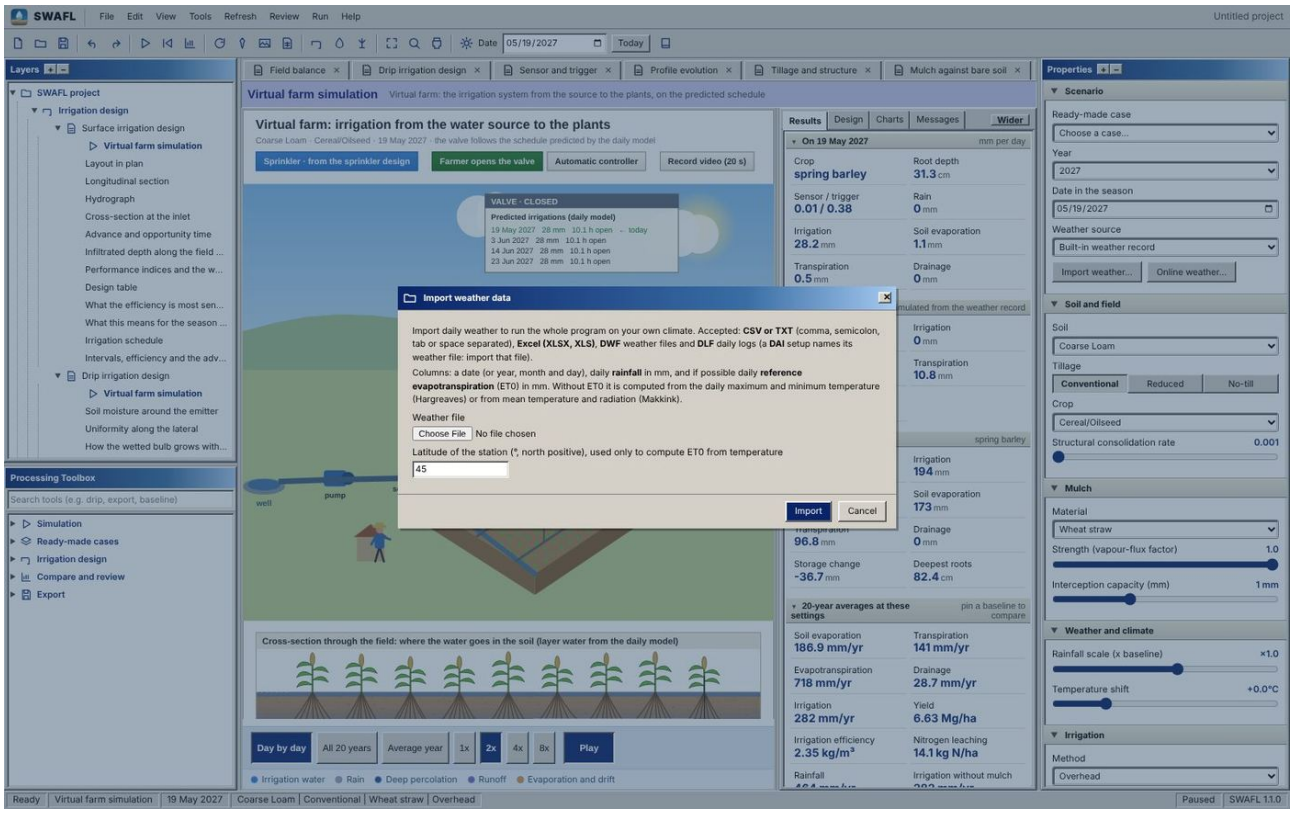


Figure 19.1: The weather import dialog.

Accepted files.

Format	Reading
CSV or TXT	comma, semicolon, tab or space separated; the separator is found from the first lines
Excel (XLSX, XLS, ODS)	the first sheet
DWF (Daisy weather file)	header lines, a line of dashes, the column names, a units line, then the data; hourly values are summed to days
DLF (Daisy daily log)	as DWF
DAI (Daisy setup file)	holds no weather; the program reads the name of the weather file it uses and asks you to import that file

Columns. The columns are found by their names, in any order and with or without units:

Quantity	Names recognised
Date	`date`, `datetime`, `time`; or `year`, `month`, `day`; or `year` and `doy` (day of year)
Rainfall (mm)	`precip…`, `rain…`, `prcp`, `ppt`, `P`; values in inches when the header says `in` or `inch`
Reference evapotranspiration (mm)	`ET0`, `ETo`, `ref…evap…`, `PET`, `potential evap…`
Temperature (°C)	`Tmax`, `Tmin`, `Tmean`, `air temp…`, `Tavg`
Global radiation	`rad`, `srad`, `globrad`, `solar` (MJ m^{-2} d^{-1}, or W m^{-2} when the values exceed 60)

Reference evapotranspiration. When the file has no ET0 column, it is computed. With daily maximum and minimum temperature, by Hargreaves:

$$ET_0 = 0.0023\,(T_{\text{mean}} + 17.8)\,\sqrt{T_{\text{max}} - T_{\text{min}}}\;R_a\;0.408 \quad [\text{mm}\,\text{d}^{-1}], \tag{19.1}$$

Provenance (Eq. 19.1): **[Literature]**: Hargreaves and Samani (1985), FAO-56 Eq. 52; R_a the extraterrestrial radiation (MJ m^{-2} d^{-1}) at the latitude typed in the dialog (FAO-56 Eq. 21).

With daily mean temperature and global radiation R_s, by Makkink:

$$ET_0 = 0.61\,\frac{\Delta}{\Delta+\gamma}\,\frac{R_s}{2.45} - 0.12, \qquad \Delta = \frac{4098\cdot 0.6108\,e^{17.27T/(T+237.3)}}{(T+237.3)^2}, \quad \gamma = 0.066\ \text{kPa K}^{-1}, \tag{19.2}$$

Provenance (Eq. 19.2): **[Literature]**: Makkink (1957), as used in the Netherlands and Denmark.

Without either, the seasonal pattern and the annual demand of the built-in climate are used, and the dialog says so.

19.3 Putting the weather on the twenty-year grid

The program works on twenty years of 365 days. The weather is put on that grid as follows:

1. The **last twenty complete years** of the file are used (a year is complete with at least 300 days). With fewer years, the available years are repeated in turn, and the dialog says how many.
2. Leap days are dropped. A missing day of rain is filled from the same day of another year of the file; a missing day of ET0 from the mean of that day of the year.
3. The mean annual rainfall $\bar{P}$ and reference evapotranspiration $\overline{ET_0}$ are computed.

The daily model (Engine 2) then uses the daily rain P_k of your weather, scaled to the rainfall scale of Properties, and the daily evaporative demand in proportion to your daily ET0:

$$P_k^{\text{model}} = P_k\,\frac{\rho}{\rho_{\text{ref}}}, \qquad e_k = \frac{ET_{0,k}}{\overline{ET_0}/365} \quad \text{(the shape of the demand through the year)}, \tag{19.3}$$

Provenance (Eq. 19.3): **[Engineered]**: the daily sequence of your weather replaces the generated rainfall (Eqs. 5.15–5.16) and the seasonal demand weight.

with ρ the rainfall scale of Properties and ρ_{ref} the step chosen for your weather by Eq. (19.4). The annual model (Engine 1) works on the simulated steps of rainfall and warming; it is set to the steps nearest your weather,

$$\rho_{\text{ref}} = \operatorname*{nearest}_{\rho\in\{0.5,\ldots,1.4\}} \frac{\bar{P}}{P_{\text{rec}}}, \tag{19.4}$$

Provenance (Eq. 19.4): **[Engineered]**: nearest simulated step; P_{rec} the mean annual rain of the built-in record (464 mm).

$$\Delta T_{\text{ref}} = \operatorname*{nearest}_{\Delta T\in\{-1,\ldots,3.5\}} \frac{\overline{ET_0}/E_{\text{rec}} - 1}{0.035}, \tag{19.5}$$

Provenance (Eq. 19.5): **[Engineered]**: the evaporative demand expressed as the warming that gives it (3.5 % per °C, Eq. 5.26); E_{rec} the mean annual reference ET of the built-in record (about 870 mm).

When your weather lies outside the simulated range (for example 1,500 mm of rain, more than 1.4 times the record), the annual results use the nearest step and the dialog says so; the daily layers use your weather as it is.

The crop water requirement (Chapter 21) uses your daily rain and ET0 directly.

The years. With your weather, the twenty years of the grid are the calendar years of your record, so a working date inside them shows that very day of your weather (Eq. 4.2).

After the import a report lists the years used, the days in the file, the mean annual rain and ET0, how ET0 was obtained, the rainfall and temperature steps of the annual model, the columns read, and any notes (repeated years, filled gaps, values outside the simulated range).

19.4 The weather of a place

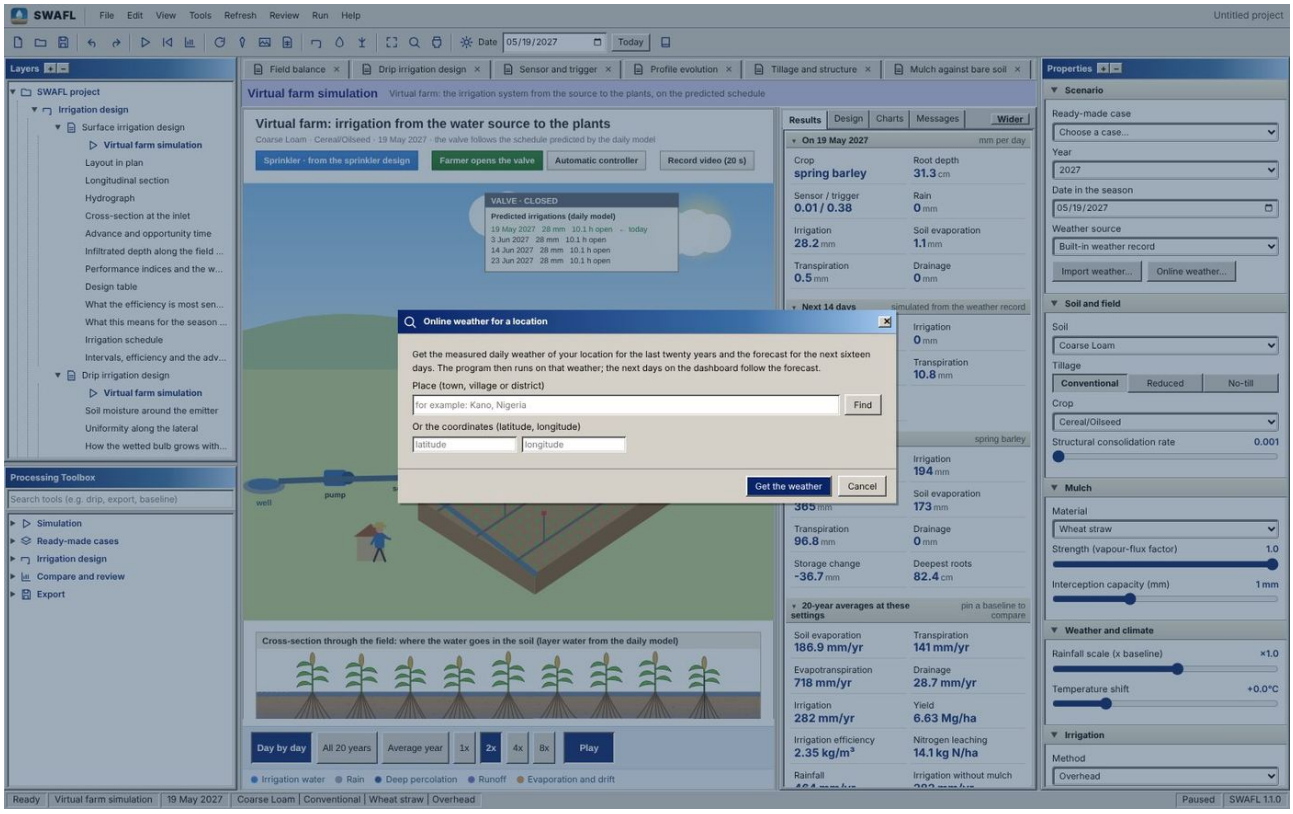


Figure 19.2: The online weather dialog.

Type the name of a town, village or district and press **Find**, then choose the place from the list; or type its latitude and longitude. **Get the weather** fetches:

- the measured daily rain, reference evapotranspiration (FAO Penman–Monteith) and maximum and minimum temperature of the place from 1 January twenty years ago to two days ago;
- the forecast for the next 16 days;
- for the rest of the current year after the forecast, the same days of the most typical year of the record (median rain).

The weather is then put on the grid as above. On the dashboard, the *Next 14 days* block says *from the weather forecast* while the days fall in the forecast.

The online weather needs the internet; everything else does not. If the place search or the weather service cannot be reached, the dialog says so and **the current weather stays in use**; nothing else changes. The service is a free public weather archive that needs no key.

19.5 Switching back

Choose *Built-in weather record* in Properties ▸ Scenario ▸ *Weather source*. The rainfall scale and temperature shift you had before the import come back. The imported file and the online weather stay available in the list until the program is closed.

Chapter 20

The virtual farm simulation

The virtual farm shows **one irrigation system working in a field** — the source, the pump or the canal gate, the pipes or the head ditch, the laterals, furrows or sprinklers, the crop and the soil — on the days the daily model irrigates. In version 1.1 the farm **follows the run of an irrigation design**: it shows the system of that design, with its inputs and results, on the real soil of the project.

20.1 Opening the farm

- In the Layers panel, under each design layer, click **Virtual farm simulation**.
- On the view bar of any layer, click ▶ **Virtual farm simulation**. From a design layer this opens the farm of that design; from any other layer, the farm of the design that belongs to the irrigation method in force (surface method: the surface design; overhead: the sprinkler design; drip methods: the drip design).
- View ▸ *Virtual farm simulation*.

While the farm is open, the irrigation method of the project is set to the method of its design (for example *Surface* for the surface design), so that the daily model irrigates as that system does. **When you leave the farm, the working date and the irrigation method of the project return to what they were.**

20.2 After the run of the design

If the design has not been run, or its inputs or a setting changed since its last run (Eq. 18.0), the farm does not draw a field. It says which design it follows and offers two buttons:

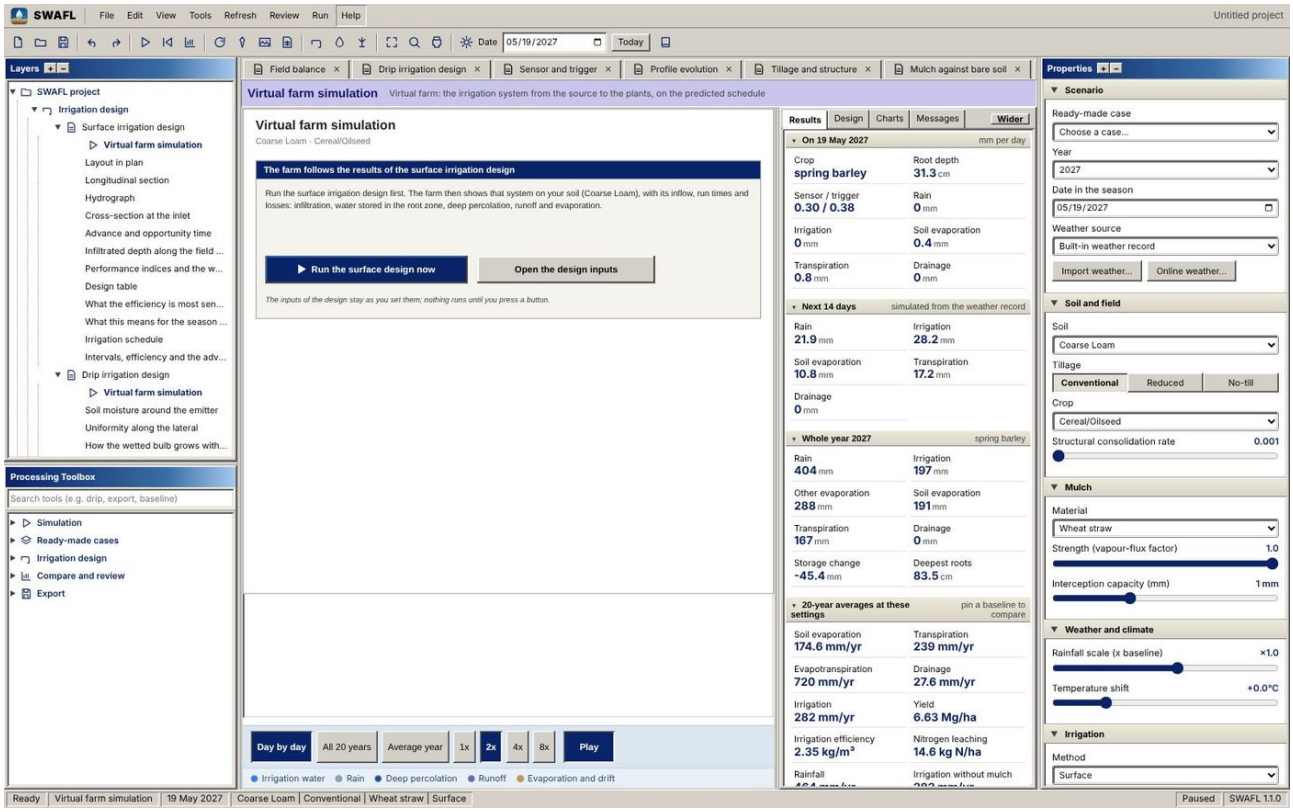


Figure 20.1: The farm before its design has run.

- **Run the … design now** runs the design with its present inputs; the farm then appears.
- **Open the design inputs** opens the design layer, to check the inputs first.

20.3 What the farm shows

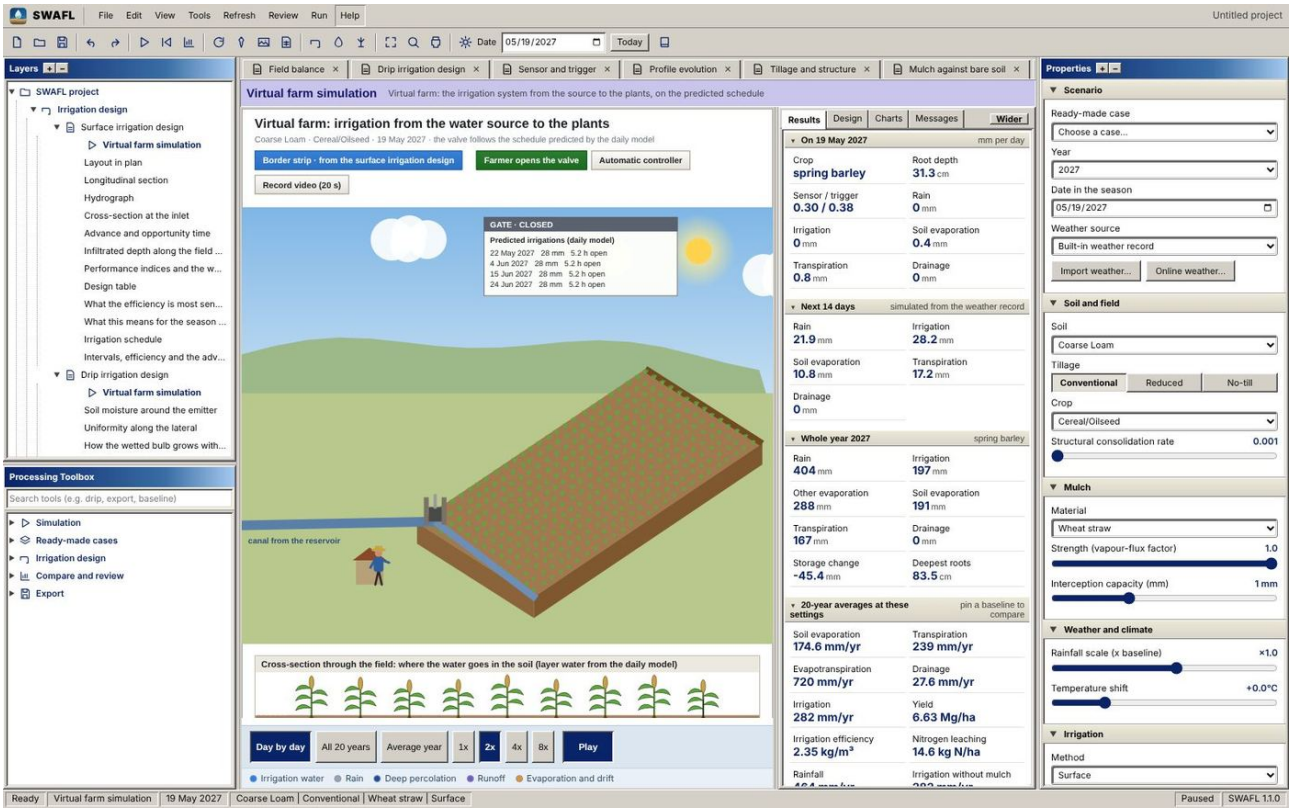


Figure 20.2: The virtual farm of the surface design.

The system button at the top names the system and the design it comes from (for example *Border strip · from the surface irrigation design*); clicking it opens the design. **Farmer opens the valve** and **Automatic controller** choose who operates the gate or valve; **Record video (20 s)** saves a film of the farm.

The farm in perspective. The field is drawn in the colour of the soil of the project, with the crop of the project in rows. For the surface systems the canal, the gate and the head ditch feed the furrows, borders or basins; the water advances along the field with the advance exponent of the design and recedes after the cut-off; runoff collects in the tail ditch. For drip and sprinklers the well, pump, filters, fertiliser tank, mainline, valve, submain and laterals (or sprinklers) carry the water. The gate or valve opens on the days the daily model irrigates — the days on which the sensor reaches the trigger — and the board at the gate lists the next four irrigations with their depths and opening times.

The cross-section through the field shows the water of each soil layer of the daily model, the wetting front as the irrigation proceeds, the roots of the crop and the sensor with its depletion against the trigger.

The predicted schedule at the gate or valve for the next 30 days: rain, irrigation, the depletion of the sensor layer and the trigger.

The irrigation in 3D. A block of the real soil, one metre deep, cut on two faces, carries the crop on its surface and shows where the water of this (or the next) irrigation goes, with numbered markers matching the legend under it:

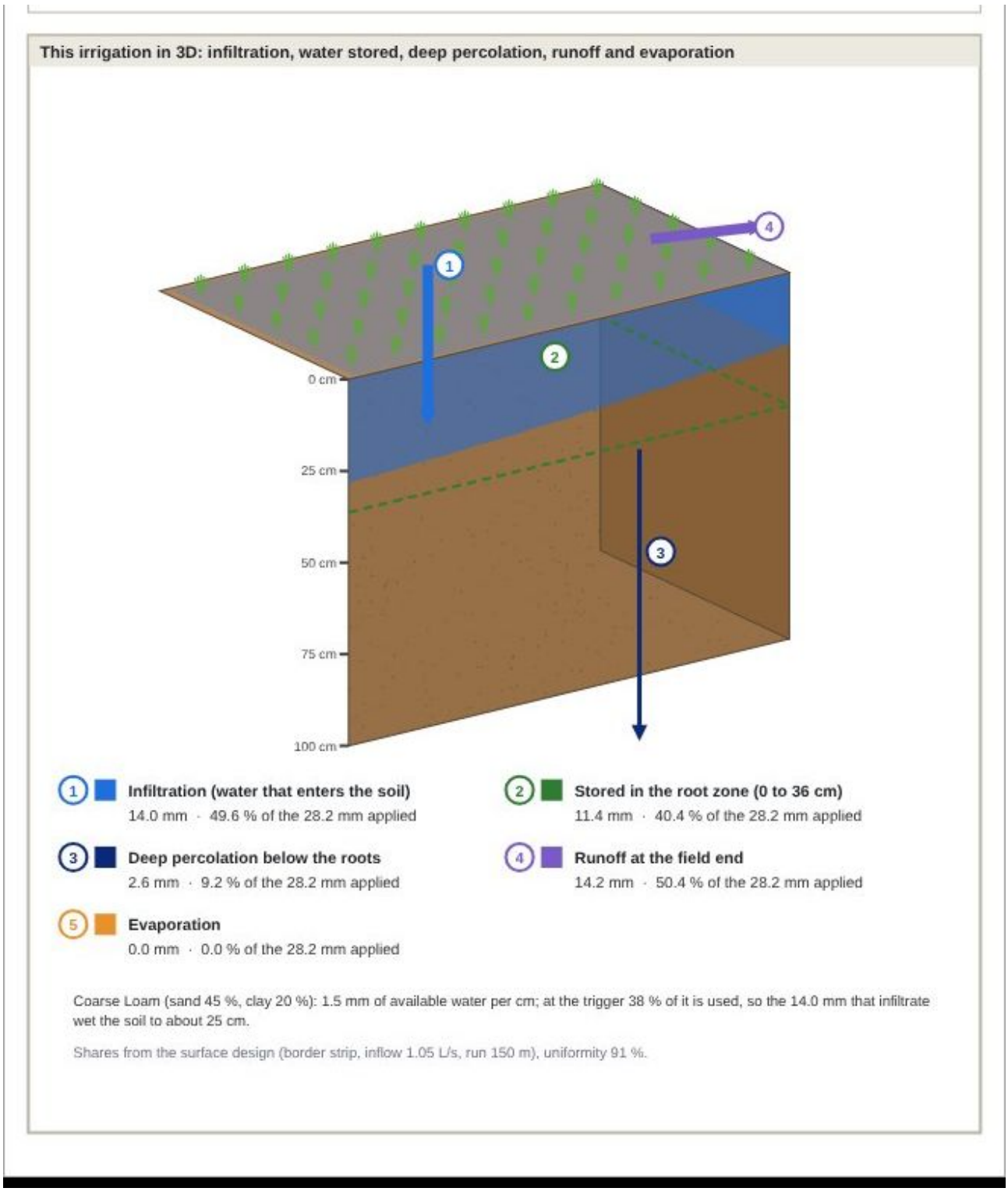


Figure 20.3: The 3D block: infiltration, water stored in the root zone, deep percolation, runoff and evaporation.

1. **Infiltration** — the water that enters the soil;
2. **Stored in the root zone** — from the surface to the depth of the roots of that day (green dashed line);
3. **Deep percolation** — the part that passes below the roots (dark blue on the cut faces);
4. **Runoff** — for surface systems, at the field end; for sprinklers, when the rate exceeds the intake;
5. **Evaporation** — and wind drift for sprinklers.

Each item gives its depth (mm) and its share of the gross depth applied. The shares are those of the last run of the design:

$$(f_{\mathrm{st}}, f_{\mathrm{dp}}, f_{\mathrm{ro}}, f_{\mathrm{ev}}) = \frac{(E_a,\ DP,\ RO,\ EV)}{E_a + DP + RO + EV}, \qquad d_x = f_x\, d_{\mathrm{gross}}, \tag{20.1}$$

Provenance (Eq. 20.1): **[Engineered]**: the partition of the design run (Eqs. 11.18, 12.19, 13.11) applied to the gross depth of the event predicted by the daily model.

with d_{gross} the depth of the irrigation of that day in the daily model. The depth the infiltrated water wets is

$$z_w = \mathrm{clip}\Big(\frac{d_{\mathrm{gross}}\,(f_{\mathrm{st}} + f_{\mathrm{dp}})}{f_T\,\mathrm{AWC}};\ 3,\ 100\Big)\ \ [\mathrm{cm}], \tag{20.2}$$

Provenance (Eq. 20.2): **[Literature]**: the infiltrated water refills the depleted share f_T of the available water AWC (mm per cm) of each centimetre in turn, from the top (the “piston” refill of the FAO-56 root-zone balance).

For the surface systems the wetted depth varies along the run as the infiltrated depth of the design does (more at the inlet, less at the tail; Eq. 11.16), scaled to the same mean. For sprinklers it varies with the uniformity of the pattern, $z_w(u) = \bar{z}_w\,[1 + 1.4(1 - \mathrm{CU})\sin 6\pi u]$. For drip the water forms bulbs around the emitters, at the depth of the placement. Below the drawing the soil is named with its sand and clay content and its available water, and the sentence states the depth wetted.

20.4 The farm and the three designs

Design	What the farm draws
Surface	The system of the design (border strips, furrows or basins), its inflow and run length, the advance and recession with the advance exponent of the run, runoff in the tail ditch.
Drip	Surface drip, or subsurface drip when the placement is 30 cm or deeper; emitters dripping and wetted bulbs at the placement depth.
Sprinkler	Sprinklers on risers, the spray, the drift plume, the overlap.

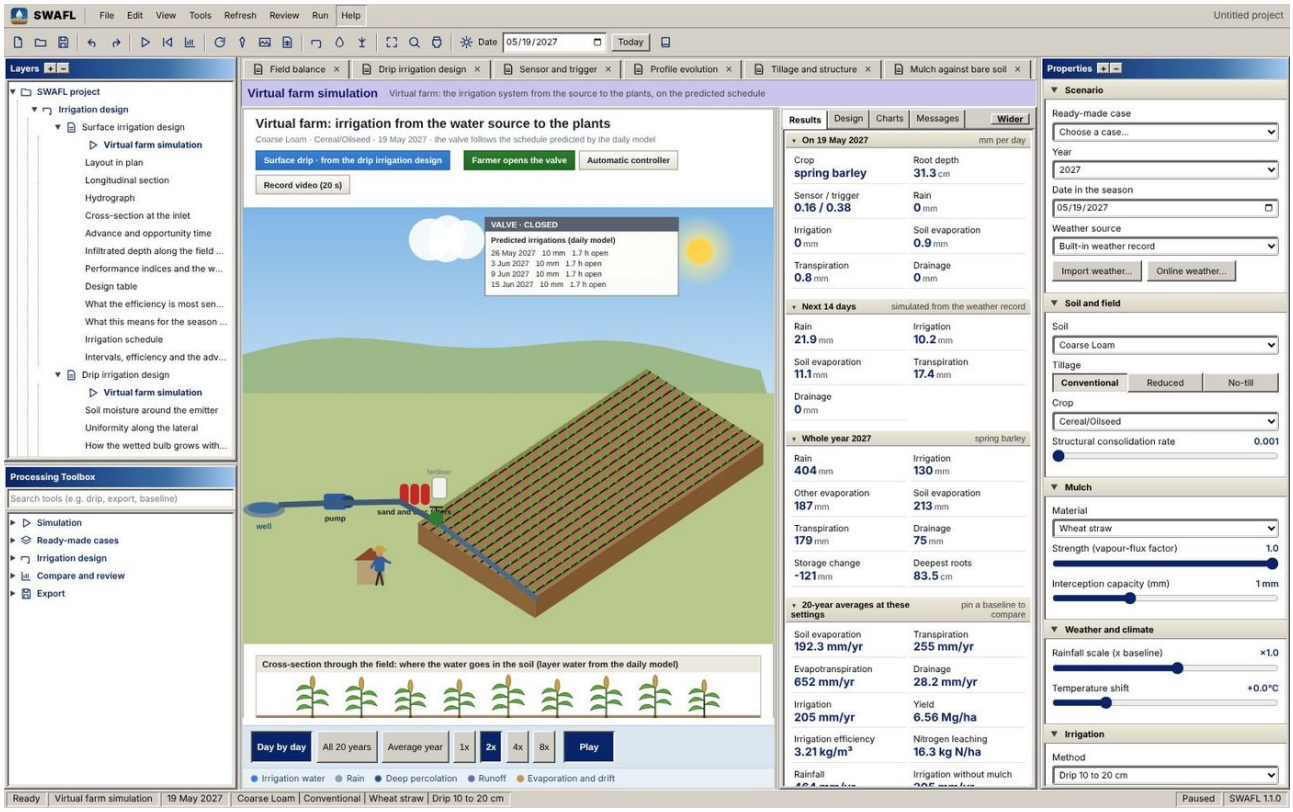


Figure 20.4: The virtual farm of the drip design.

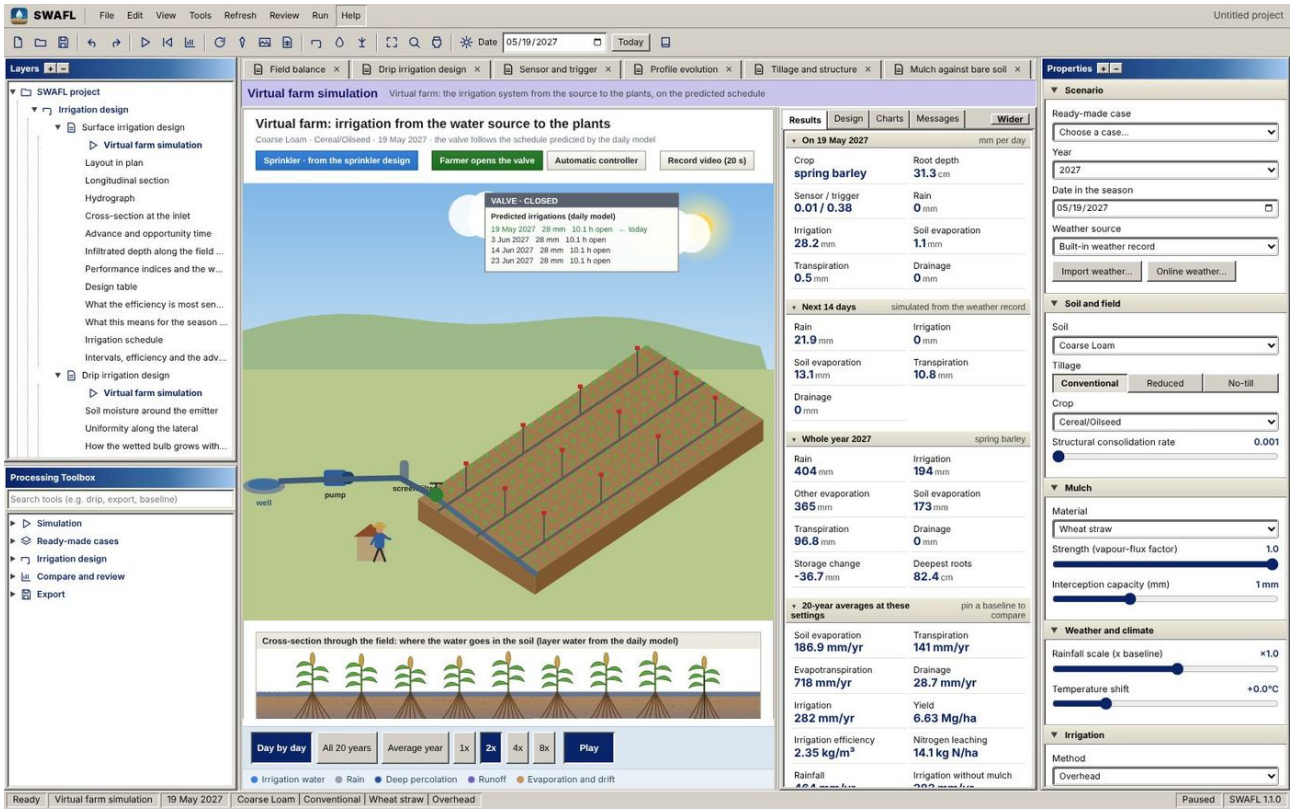


Figure 20.5: The virtual farm of the sprinkler design.

Chapter 21

The crop library and the crop water requirement

21.1 The crops

Version 1.1 carries **48 crops** besides the simulated cereal/oilseed rotation. They are chosen in Properties ▸ Soil and field ▸ **Crop**, in four groups: *Staple food crops*, *Cash and oil crops*, *Vegetables*, and *Trees, vines and palms*. For each crop the library holds the crop coefficients $K_{c,\mathrm{ini}}$, $K_{c,\mathrm{mid}}$ and $K_{c,\mathrm{end}}$, the lengths of the four growth stages (initial, development, mid-season, late season), the plant height, the maximum rooting depth, the depletion fraction p, the yield response factor K_y, the ground cover and, for trees and vines, the planting densities or training systems. Field crops take their coefficients and stages from a field study that computed the daily crop water requirement with the FAO Penman–Monteith reference evapotranspiration; trees, vines and berries from the updated FAO-56 coefficients of fruit trees and vines. Rooting depth, p and K_y are the FAO-56 and FAO-33 tabulated values; where a crop is not tabulated the group value is used and the table says *group default*.

The staple and cash crops asked for in this release carry these coefficients:

Crop	$K_{c,\mathrm{ini}}$	$K_{c,\mathrm{mid}}$	$K_{c,\mathrm{end}}$	Note
Cassava	0.30	0.80	0.30	
Yam	0.50	1.10	0.85	stages and rooting depth: group default
Rice	1.05	1.20	0.90	bunded: no runoff in the season
Maize	0.30	1.20	0.60	
Sorghum	0.30	1.00	0.55	
Millet	0.30	1.00	0.30	
Cowpeas	0.40	1.05	0.50	
Groundnut	0.40	1.15	0.60	
Soybean	0.40	1.15	0.50	
Oil palm	0.90	0.95	0.95	perennial
Cocoa	1.00	1.05	1.05	perennial, high canopy shade

The complete library is listed in Table 21.2 at the end of this chapter.

21.2 The Crop water requirement layer

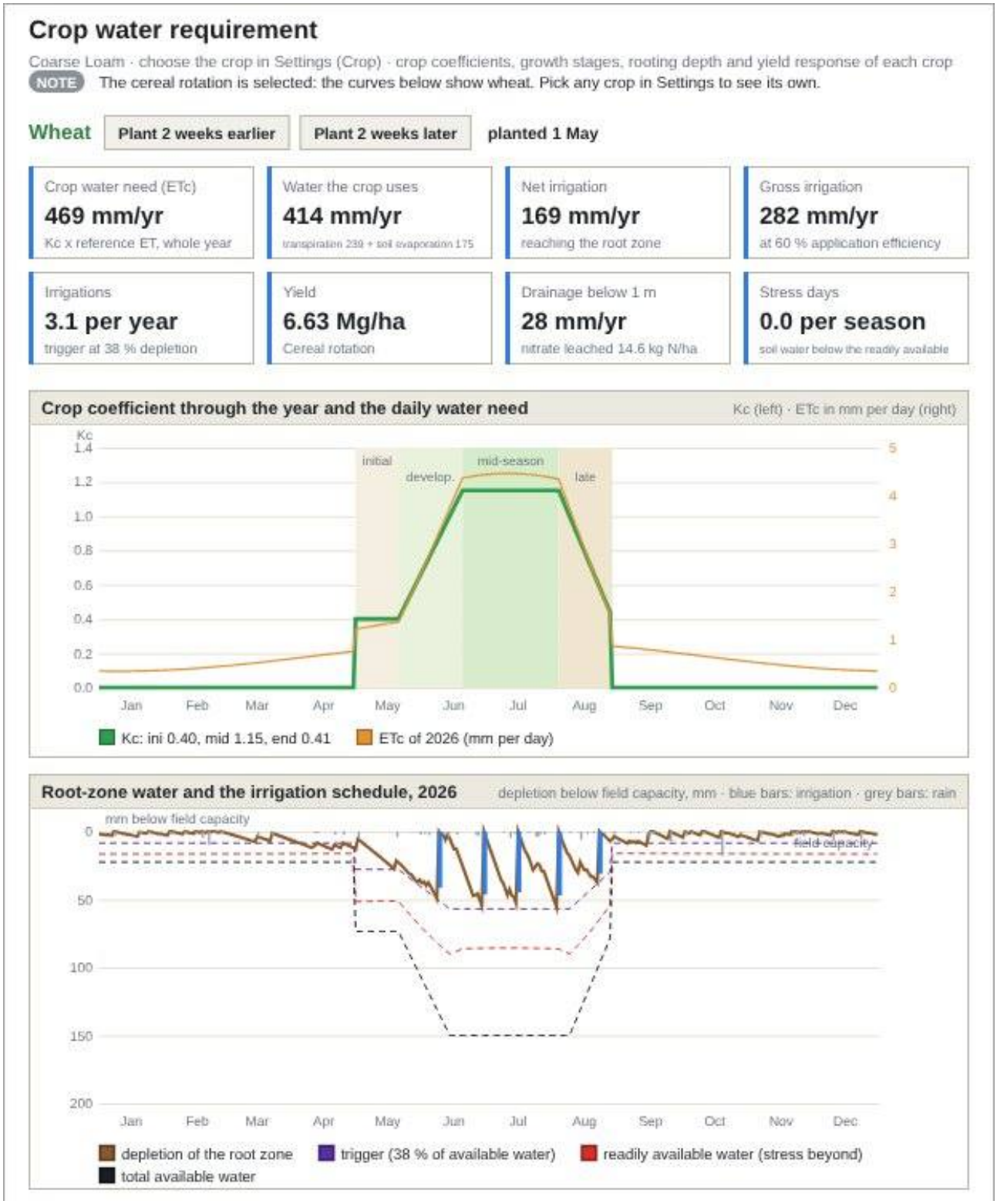


Figure 21.1: The Crop water requirement layer.

The layer *Crops* ▸ *Crop water requirement* shows, for the chosen crop on the soil, weather, trigger and method of the project:

- the **crop coefficient through the year** and the daily water need ET_c of the working year (the label reads *ETc of 2026*, following the year in Properties);
- the **root-zone water and the irrigation schedule** of the working year: the depletion below field capacity against the readily available water and the trigger, with every irrigation

of the year;
- buttons to plant two weeks earlier or later (field crops) or to choose the planting density or training system (trees and vines);
- **every crop of the library on this soil**: the yearly crop water need, net irrigation, irrigations per year and relative yield, so that crops can be compared on the same field.

21.3 The daily balance of a crop

The layer computes, for each of the twenty years of the weather in use, the FAO-56 daily balance of the root zone. The crop coefficient follows the four stages,

$$K_c(t) = \begin{cases} K_{c,\mathrm{ini}} & t < L_1 \\ K_{c,\mathrm{ini}} + (K_{c,\mathrm{mid}} - K_{c,\mathrm{ini}})\frac{t-L_1}{L_2} & L_1 \le t < L_1 + L_2 \\ K_{c,\mathrm{mid}} & L_1 + L_2 \le t < L_1 + L_2 + L_3 \\ K_{c,\mathrm{mid}} + (K_{c,\mathrm{end}} - K_{c,\mathrm{mid}})\,\min\!\left(1, \frac{t-L_1-L_2-L_3}{L_4}\right) & \text{after} \end{cases} \tag{21.1}$$

Provenance (Eq. 21.1): **[Literature]**: FAO-56 crop coefficient curve (Allen et al., 1998, Fig. 25); t days after planting; evergreen crops repeat it every year.

and outside the season the bare soil has $K_c = 0.25$. The daily crop water need is

$$ET_{c,d} = K_c(d)\, ET_{0,d}, \tag{21.2}$$

Provenance (Eq. 21.2): **[Literature]**: FAO-56 single crop coefficient approach.

with $ET_{0,d}$ the daily reference evapotranspiration of the weather in use (your own weather when imported; otherwise the annual reference ET of each recorded year shaped by the seasonal weight, adjusted by 3.5 % per °C of the temperature shift). The roots of an annual crop grow with the canopy, perennial roots are at full depth:

$$z_r = \mathrm{clip}\!\left(15 + (R_{\max} - 15)\,\min\!\left(1, \frac{1.15\,K_c}{\max(0.3, K_{c,\mathrm{mid}})}\right);\ 15,\ R_{\max}\right)\ \ [\mathrm{cm}], \qquad R_{\max} \le 100\ \mathrm{cm}, \tag{21.3}$$

Provenance (Eq. 21.3): **[Engineered]**: root growth tied to the crop coefficient, limited to the 1 m of the model.

The total and readily available water and the depletion D_r of the root zone follow FAO-56,

$$\begin{aligned} &\mathrm{TAW} = \mathrm{AWC}\, z_r, \qquad p_d = \mathrm{clip}\big(p + 0.04\,(5 - K_c\, ET_0);\ 0.1,\ 0.8\big), \qquad \mathrm{RAW} = p_d\,\mathrm{TAW}, \\ &K_s = \begin{cases} 1 & D_r \le \mathrm{RAW} \\ \dfrac{\mathrm{TAW} - D_r}{(1-p_d)\,\mathrm{TAW}} & D_r > \mathrm{RAW} \end{cases}, \qquad ET_a = K_s\, ET_c, \\ &D_{r,d} = D_{r,d-1} - (P_d - RO_d) + ET_{a,d}, \qquad RO_d = 0.3\,\max(0, P_d - 35), \end{aligned} \tag{21.4}$$

Provenance (Eq. 21.4): **[Literature]**: FAO-56 Eqs. 82–85 (water stress coefficient, adjustment of p with the demand) and the root-zone balance Eq. 85; the runoff of heavy rain is **[Engineered]**.

with any negative D_r draining below the roots. Irrigation refills the root zone to field capacity on the day the depletion reaches the trigger of the project, $D_r \ge f_T\,\mathrm{TAW}$, with f_T the depletion at the trigger suction (Eq. 5.3): the net depth is $I_n = D_r$ and the gross depth $I = I_n / E_{a,m}$, with the method efficiency $E_{a,m}$ = 0.60 (surface), 0.75 (overhead) or 0.90 (drip). The relative yield of a year is

$$Y_{\mathrm{rel}} = \mathrm{clip}\!\left(1 - K_y\!\left(1 - \frac{\sum ET_a}{\sum ET_c}\right);\ 0,\ 1\right), \tag{21.5}$$

Provenance (Eq. 21.5): **[Literature]**: FAO-33 yield response (Doorenbos and Kassam, 1979); $K_y = 1$ when not tabulated, as the table says.

21.4 How a library crop changes every other layer

The annual model (Engine 1) is the result of the simulations of the cereal rotation. A library crop is transferred from it by the ratio of two runs of the balance above on the same soil, weather, trigger and method: one for the crop (c) and one for the cereal reference (r, FAO-56 small grains, K_c 0.30/1.15/0.25). With m the annual-model result of the cereal,

$$I = \begin{cases} I_m \,\mathrm{clip}\big(I_c/I_r;\ 0,\ 4\big) & I_r > 5 \text{ mm} \\ I_c & \text{otherwise} \end{cases} \tag{21.6}$$

$$\begin{aligned} T &= \min\Big(0.9\,ET,\ \ T_m \,\mathrm{clip}\Big(\frac{ET_{a,c}}{ET_{c,c}} \cdot \frac{ET_{c,c}}{ET_{c,r}} \cdot \Big(\frac{f_{c,c}}{f_{c,r}}\Big)^{0.3};\ 0.2,\ 3\Big)\Big), \\ E^* &= E_m \,\mathrm{clip}\Big(\frac{1-0.7\,f_{c,c}}{1-0.7\,f_{c,r}};\ 0.6,\ 1.6\Big), \end{aligned} \tag{21.7}$$

$$D = \max\big(0.2\,D_m,\ D_m + 0.5\,(DP_c - DP_r)\big), \qquad ET = \mathrm{clip}\big(P + I - D - \Delta S_m;\ 50,\ 5000\big), \tag{21.8}$$

$$(E,\ E_{\text{other}}) = (E^*,\ E^*_{\text{other}})\,\frac{ET - T}{E^* + E^*_{\text{other}}}, \qquad N_L = N_{L,m}\,\frac{D+5}{D_m+5}, \qquad Y = 100\,Y_{\text{rel},c}\,[\%], \tag{21.9}$$

Provenance (Eqs. 21.6–21.9): **[Engineered]**: transfer by the ratio of two FAO-56 balances; the clips keep the transfer within the range in which the cereal result is informative; the remainder of the balance is shared between soil evaporation and the other evaporation terms in proportion to their targets, so that the balance closes without inflating either.

with f_c the ground cover, $E^*_{\text{other}} = E_{\text{other},m}\,ET/ET_m$ and ΔS_m the storage change of the cereal result. The yield of a library crop is therefore shown as the relative yield in per cent, and the differences of yield in the sensitivity and feedback layers are in *% points*.

21.5 Table 21.2. The crop library

Table 21.2. The 48 crops of the library: crop coefficients, stage lengths (initial/development/mid/late, days), plant height (m), maximum rooting depth (cm), depletion fraction p and yield response factor K_y (— when not tabulated).

Crop	Group	$K_{c,ini}$	$K_{c,mid}$	$K_{c,end}$	Stages (d)	Height (m)	Root (cm)	p	K_y
Rice	field crop	1.05	1.20	0.90	15/30/45/30	0.8	75	0.20	—
Maize (grain)	field crop	0.30	1.20	0.60	15/30/40/20	1.5	135	0.55	1.25
Millet	field crop	0.30	1.00	0.30	15/25/40/25	1.5	150	0.55	—
Sorghum (grain)	field crop	0.30	1.00	0.55	20/35/40/30	1.5	150	0.55	0.90
Cassava	field crop	0.30	0.80	0.30	20/40/90/60	1.0	65	0.35	—
Yam	field crop	0.50	1.10	0.85	30/60/90/60	1.5	60	0.40	—
Cowpea (beans)	field crop	0.40	1.05	0.50	20/30/30/20	0.4	80	0.45	—

Crop	Group	$K_{c,ini}$	$K_{c,mid}$	$K_{c,end}$	Stages (d)	Height (m)	Root (cm)	p	K_y
Green gram (mung bean)	field crop	0.40	0.96	0.51	17/23/34/16	0.4	75	0.45	—
Soybean	field crop	0.40	1.15	0.50	20/30/45/25	0.6	95	0.50	0.85
Groundnut	field crop	0.40	1.15	0.60	20/35/40/25	0.4	75	0.50	0.70
Cotton	field crop	0.35	1.15	0.50	30/45/65/50	1.1	135	0.65	0.85
Wheat	field crop	0.40	1.15	0.41	20/30/45/25	0.8	165	0.55	1.05
Chickpea	field crop	0.40	1.00	0.40	15/35/45/25	0.4	80	0.50	—
Mustard	field crop	0.35	1.15	0.35	15/35/45/25	1.3	125	0.60	—
Tomato	field crop	0.60	1.15	0.80	30/40/40/25	0.6	100	0.40	1.05
Potato	field crop	0.50	1.15	0.75	25/30/45/30	0.6	50	0.35	1.10
Onion (dry)	field crop	0.70	1.05	0.75	15/25/70/40	0.4	45	0.30	1.10
Cabbage	field crop	0.70	1.05	0.95	40/60/50/15	0.4	65	0.45	0.95
Sweet pepper	field crop	0.60	1.05	0.90	25/35/40/20	0.7	75	0.30	1.10
Table grapes	tree, vine or palm	—	—	—	20/40/120/60	—	150	0.35	—
Wine grapes	tree, vine or palm	—	—	—	20/40/120/60	—	150	0.45	—
Kiwi	tree, vine or palm	—	—	—	20/40/120/60	—	100	0.35	—
Highbush blue-berry	tree, vine or palm	—	—	—	20/50/90/20	—	90	0.50	—
Hop	tree, vine or palm	—	—	—	25/40/80/10	—	110	0.50	—
Citrus	tree, vine or palm	—	—	—	60/90/120/95	—	135	0.50	—
Olive	tree, vine or palm	—	—	—	60/90/120/95	—	145	0.65	—
Apple	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Pear	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Cherry	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Peach	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Apricot	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Almond	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.40	—
Hazelnut	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Walnut	tree, vine or palm	—	—	—	20/70/90/30	—	205	0.50	—

Crop	Group	$K_{c,ini}$	$K_{c,mid}$	$K_{c,end}$	Stages (d)	Height (m)	Root (cm)	p	K_y
Pecan	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Pistachio	tree, vine or palm	—	—	—	20/70/90/30	—	125	0.40	—
Fig tree	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Persimmon	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Pomegranate	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Avocado	tree, vine or palm	—	—	—	60/90/120/95	—	75	0.70	—
Mango	tree, vine or palm	—	—	—	60/90/120/95	—	120	0.50	—
Date palm	tree, vine or palm	—	—	—	60/90/120/95	—	200	0.50	—
Coconut	tree, vine or palm	—	—	—	60/90/120/95	—	120	0.50	—
Papaya	tree, vine or palm	—	—	—	60/90/120/95	—	90	0.50	—
Guava	tree, vine or palm	—	—	—	60/90/120/95	—	120	0.50	—
Oil palm	tree, vine or palm	0.90	—	—	60/90/120/95	—	90	0.65	—
Cocoa	tree, vine or palm	1.00	—	—	60/90/120/95	—	85	0.30	—
Cherimoya	tree, vine or palm	—	—	—	60/90/120/95	—	120	0.50	—

Chapter 22

Seasonal data: twenty seasons, year by year

The group *Seasonal data* of the Layers panel holds three layers built on the **year-by-year results of the simulations**: the twenty growing seasons of each run, not only their mean. They show how reliable a setting is from one year to the next. In version 1.0 the group was called *Year by year (Daisy data)*; the name of the simulation model has gone from the program and is given here.

22.1 Where the yearly values come from

For each soil the simulations hold the twenty yearly values of rain, irrigation, evaporation, transpiration, drainage, storage change, yield and nitrogen of the reference runs (irrigated and rainfed). A layer reads the run nearest your settings and moves each year by the response of the annual model to your settings,

$$P'_y = \rho\, P_y, \qquad X'_y = X_y \,\mathrm{clip}\Big(\frac{X^{\mathrm{model}}(\text{your settings})}{X^{\mathrm{model}}(\text{reference})};\ 0,\ 6\Big)\ \ (X \in \{I, T, E, E_{\mathrm{other}}, N_L, Y\}),$$

$$D'_y = \begin{cases} D_y \dfrac{D^{\mathrm{model}}}{D^{\mathrm{model}}_{\mathrm{ref}}} & \bar{D} > 5,\ D^{\mathrm{model}}_{\mathrm{ref}} > 5 \\ D_y + \big(D^{\mathrm{model}} - D^{\mathrm{model}}_{\mathrm{ref}}\big) & \text{otherwise} \end{cases}, \qquad ET'_0 = ET_0\,(1 + 0.035\,\Delta T), \quad (22.1)$$

Provenance (Eq. 22.1): **[Data-driven]**: the yearly values of the simulations, moved by the ratio (or, for small drainage, the difference) of Engine 1 between your settings and the reference of the run (your mulch, rain ×1.0, no warming, nitrogen 100 %); for a library crop the yield is the relative yield of Eq. (21.5) year by year and the irrigation follows the ratio of the two crop balances of Section 21.4.

so that every year follows the crop, rain, temperature, nitrogen, mulch, trigger and method you set, while keeping the weather sequence of the record. A sentence under each layer names the run used and the adjustment made. The calendar years on the axes end in the working year.

22.2 Reliability over 20 seasons

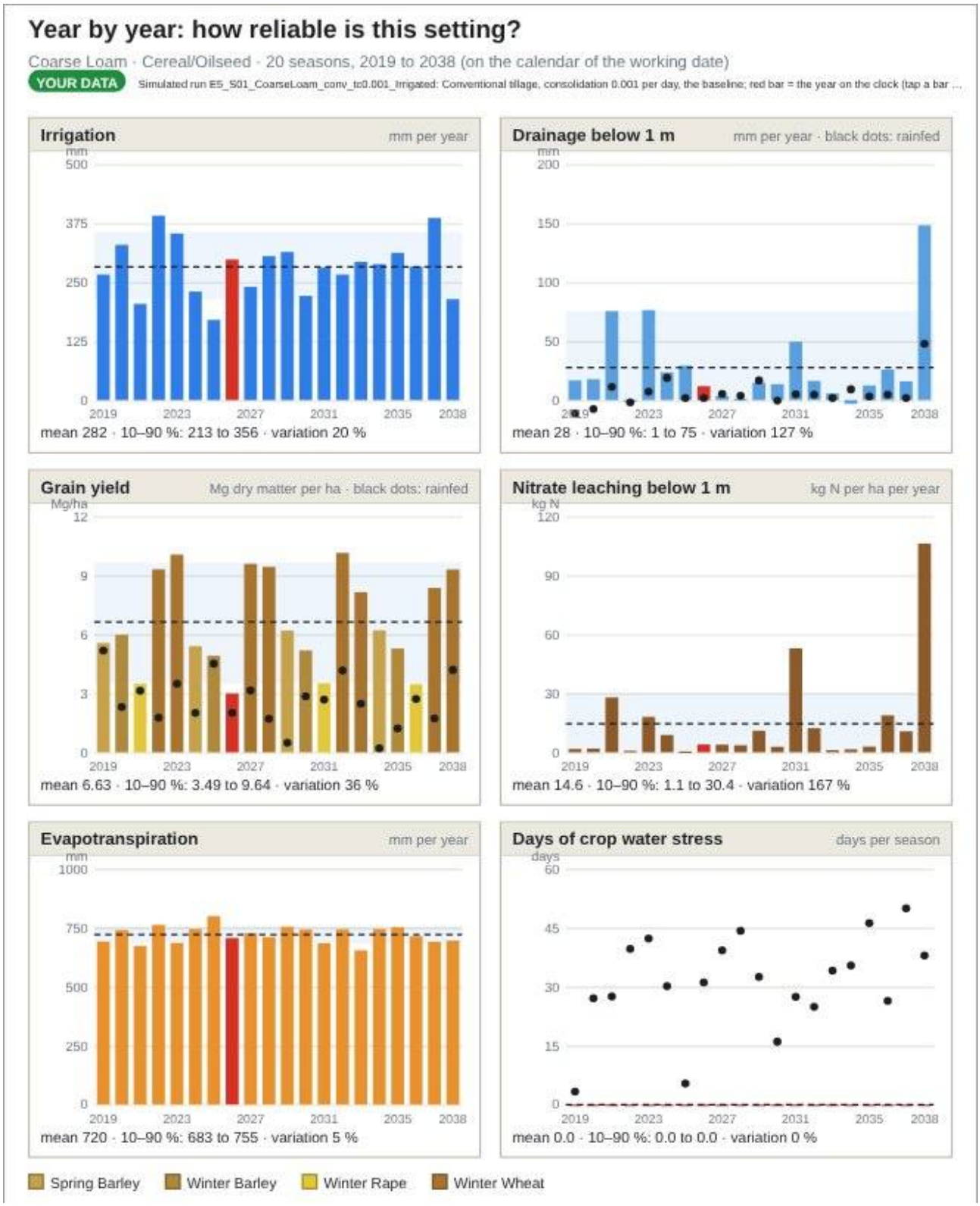


Figure 22.1: Reliability over 20 seasons.

Reliability of this setting over the 20 seasons: the irrigation, drainage, yield (or relative yield) and water stress of each season, with their mean and spread. *How often a given irrigation is exceeded*: the twenty seasonal irrigation amounts sorted into an exceedance curve, so that the amount needed in four years out of five can be read off. Click a year to set the working year to it.

22.3 Full water balance

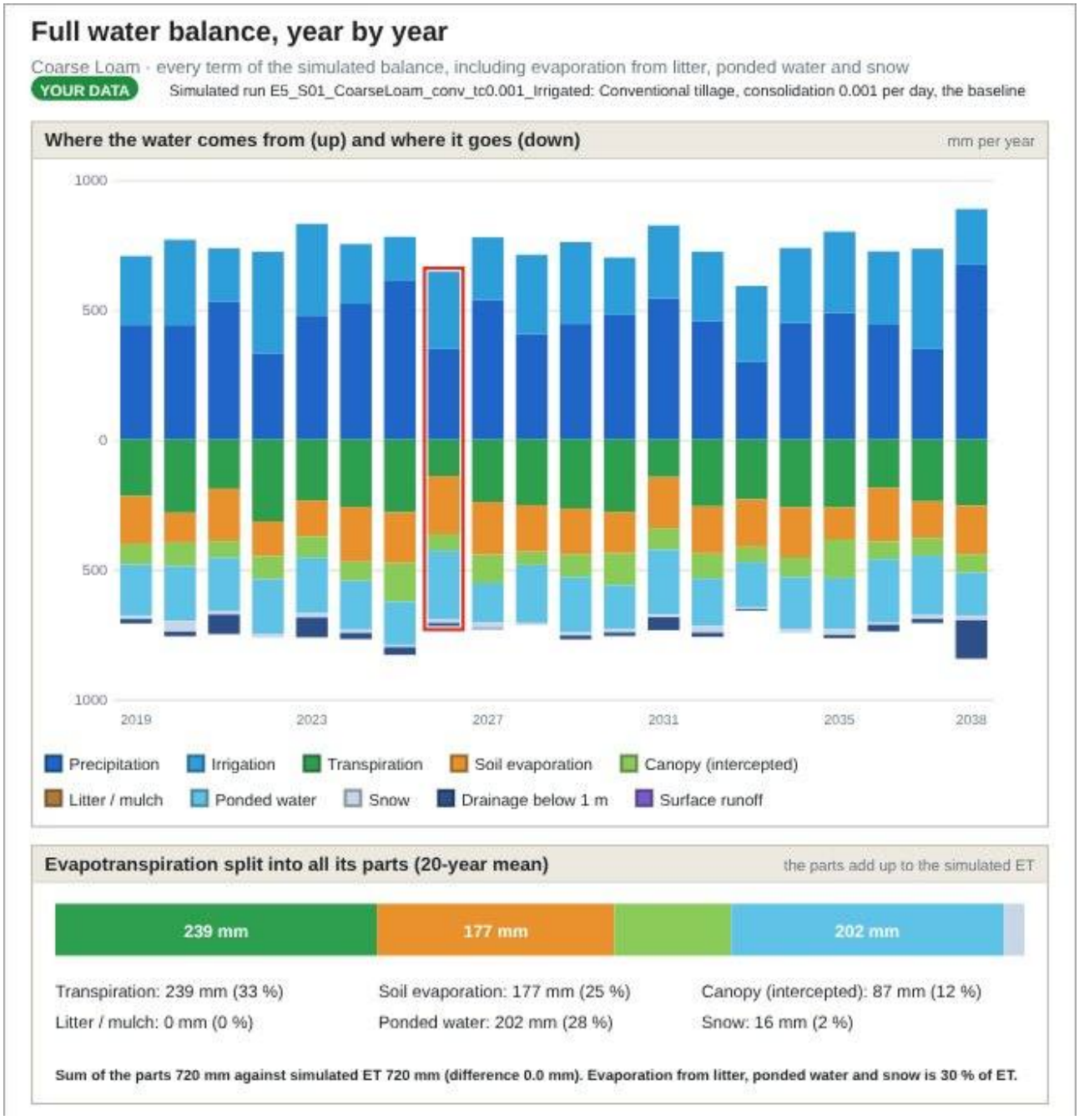


Figure 22.2: Full water balance.

Where the water comes from (up) and where it goes (down): the complete balance of each season, $P + I = E + T + E_{\text{other}} + D + \Delta S$, stacked year by year, so that a dry season (high irrigation, little drainage) and a wet season can be told apart at a glance. *Evapotranspiration split into all its parts*: transpiration, soil evaporation, canopy, litter, ponded and snow evaporation (20-year mean). *The atmosphere*: reference demand, potential and actual evapotranspiration, year by year. *How the water enters and leaves the soil*: infiltration through the soil matrix and the macropores, runoff and drainage.

22.4 Nitrogen budget



Figure 22.3: Nitrogen budget.

Supply (up) against uptake and losses (down): fertiliser and mineralisation against crop uptake, leaching, denitrification and volatilisation, season by season; *Nitrogen use (20-year mean)*; and *Nitrate goes where the water goes*: the leaching of every season against its drainage, with the regression line. The slope of that line is printed under the chart: *each extra millimetre of drainage carries … kg N/ha of nitrate*.

Chapter 23

Reports

23.1 Choosing the layers

File ▸ *Save report as PDF (choose the layers)…* (Ctrl+Shift+P, or the *Report* button of the toolbar) opens a window with every layer of the Layers panel, in its groups, and the *Virtual farm simulation*. **Every layer is ticked to start with.** Untick a layer, or a whole group with its box, or everything with *All layers*; the count at the bottom tells how many layers are ticked. **Make the PDF** builds the report and asks where to save it.

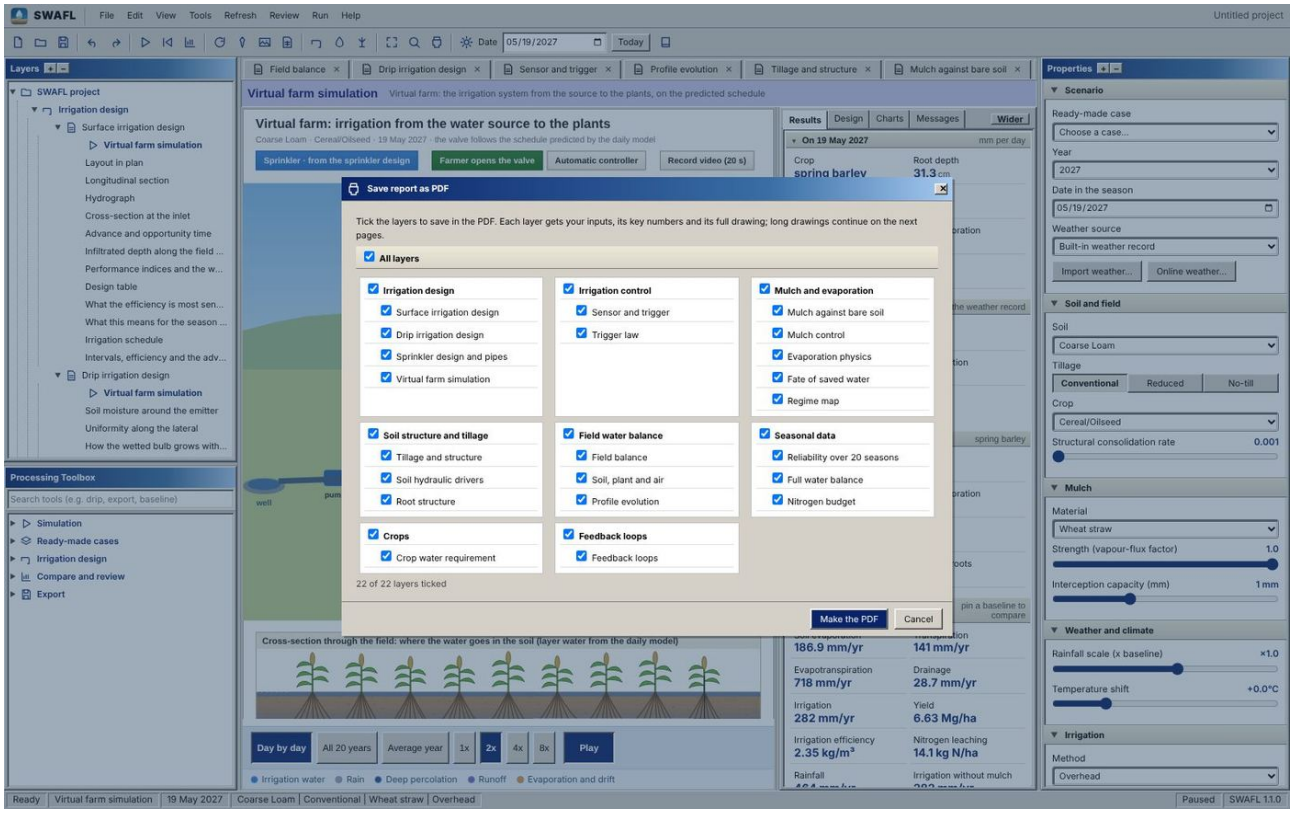


Figure 23.1: The report window: every layer ticked.

23.2 What the report holds

- **The first page**: your inputs (soil, tillage, crop, consolidation rate, mulch, rain, temperature, trigger, sensor depth, method, nitrogen), the results at these inputs (the 20-year averages), the water balance of the year, and the list of the layers included.
- **For each layer**: its name and purpose, the sentence that sums it up, the key numbers at your inputs, and **the whole drawing from top to bottom**. When the key numbers run past the end of the page they continue on the next page. When the drawing is longer than

the page it continues on the next pages, cut at its section headings (or, where a layer has none, at a blank band between two parts), so no part of a picture or of its text is hidden or cut through.
- **The design layers** are drawn at the end of the irrigation, so every result is complete; their virtual farm shows the irrigation of the next irrigation day with its 3D water fate.
- **The field balance** appears three times: day by day, all twenty years, and the average year.

A report of every layer has about 60–70 pages and takes a few seconds to build.

23.3 The page and its numbers

The pages are A4. The drawing of a layer is placed at the full text width (515 points) and cut into parts of at most

$$h_{\max} = 748 \text{ pt} \times \frac{W}{515} \ \text{[px of the drawing]}, \tag{23.1}$$

Provenance (Eq. 23.1): **[Engineered]**: page geometry; W is the width of the drawing in pixels, so a wide drawing is cut into taller parts than a narrow one.

and the first part fills the room left under the key numbers when it is at least 220 points.

Chapter 24

What changed from version 1.0 to version 1.1

24.1 The window

- The program, its title bar, the status bar and the About window show **1.1.0**. The build removes any older program before building, and the program clears its cache on start, so an old version cannot be shown by mistake.
- The **Layers panel** has a new order: Irrigation design, Irrigation control, Mulch and evaporation, Soil structure and tillage, Field water balance, Seasonal data, Crops, Feedback loops. The short codes of the layers (S2, V1, N4, …) are gone; layers are named by their names only.
- The **virtual farm** is no longer a layer of its own: it sits under each design layer and behind the *Virtual farm simulation* button of every view bar (Chapter 20).
- The **year strip** and the date bar under the drawings are gone; the year (1980–2100) and the date are set in Properties and on the toolbar.
- The **dashboard** is laid out like the Properties panel and every block folds on its own; the Charts tab draws the hydrograph, the times of the irrigation and where the water goes for each design (Chapter 14).
- **Properties**: the *Weather of years after 1999* selector and the explanatory notes under it, under the consolidation rate and under the design shortcuts have gone; their content is in this manual. *Weather source* chooses the built-in record, an imported file or the online weather.
- The name of the simulation model is no longer shown anywhere in the program; badges that said so now say *Simulated*.
- Labels are drawn after the pictures and moved apart when they would overlap; text near the edge of a drawing is moved inwards instead of being cut; long drawings take their full height at once.

24.2 The designs

- Inputs are **value boxes** with exact values, units and a list of the allowed range (Section 18.2).
- **Nothing runs while you type**: results appear after **Run simulation** and are withdrawn when an input or a setting changes (Section 18.1).
- **Find the most efficient inputs**, with your present inputs competing (Section 18.4); the surface search also tries shorter runs.
- **Field size** as an area or as a length and a width, for drip and sprinklers (Section 18.3).
- **Field sketch** and **Compare the irrigation systems** (Sections 18.5–18.6).
- The surface layer iterates the advance exponent in the steady-intake term (Chapter 11); the slow-advance cases of the textbook example are now within 2 %.

24.3 The data and the models

- **Weather**: import of daily weather files and the online weather of a place, with every layer following (Chapter 19).
- **Crops**: 48 crops with their FAO-56 coefficients; the *Crop water requirement* layer (Chapter 21).
- **Seasonal data**: the twenty seasons of the simulations, following every setting (Chapter 22).
- **Sensor records** follow every setting and every soil by the delta method (Eq. 6.4a).
- **Sand and loamy sand** now respond to the trigger, the sensor depth and the method (Eq. 5.12a).
- The water balance of every crop closes (the limit on evapotranspiration was raised from 2,000 to 5,000 mm).
- Yield is in Mg/ha for the simulated cereal and in per cent of the potential for a library crop, with differences in *% points*.

24.4 Reports

- The report asks which layers to include, all ticked to start with; notes and drawings continue on the next pages; designs are drawn complete (Chapter 23).

Symbols and units

Table A.1. Symbols used in this manual, grouped by chapter of first use. Units are SI-derived as shown; $[-]$ is dimensionless.

Symbol	Meaning	Unit
θ, θ_r, θ_s	Volumetric water content; residual; saturated	$m^3\ m^{-3}$
θ_{fc}, θ_{wp}	Water content at field capacity; at wilting point	$m^3\ m^{-3}$
h, ψ	Soil-water suction (positive); matric potential	cm; kPa
α, n, m	van Genuchten parameters ($m = 1 - 1/n$)	cm^{-1}, [–], [–]
K_s, $K(h)$	Saturated and unsaturated hydraulic conductivity	$mm\ h^{-1}$
AWC	Plant-available water capacity	$mm\ cm^{-1}$
f, f_T	Relative depletion of the sensor layer; trigger depletion	[–]
$R_{\max}$, R_d	Maximum and present rooting depth	cm
P, I, E, T, D	Rain, irrigation, soil evaporation, transpiration, drainage	$mm\ d^{-1}$ or $mm\ a^{-1}$
E_0	Atmospheric evaporative demand	$mm\ d^{-1}$
$\mathrm{ET}_{\mathrm{pk}}$	Peak 5-day mean of $E + T$	$mm\ d^{-1}$
k_t	Calibrated transpiration coefficient of Engine 2	[–]
Z_n	Required (net) depth per irrigation	mm
$Z(\tau)$, $F(t)$	Cumulative infiltration (Kostiakov; Green–Ampt)	mm
k, A	Kostiakov intake constants	$mm\ min^{-A}$; [–]
ψ_f	Wetting-front suction	mm
L, S, w, n	Field length; slope; spacing or width; Manning roughness	m; %; m; $s\ m^{-1/3}$
q, Q	Unit and total inflow	$L\ s^{-1}\ m^{-1}$, $L\ s^{-1}$
t_x, t_L, t_{co}, τ	Advance time; time to end; cut-off time; opportunity time	min
σ_y, σ_z	Surface and subsurface shape factors	[–]
E_a, E_s	Application and storage efficiency	[–] or %
DP, RO, WD	Deep percolation, runoff, wind drift and interception fractions	[–]
CU, DU, E_u	Christiansen uniformity; distribution uniformity; emission uniformity	[–] or %
S_e, S_l, L_L, H_0, D	Emitter spacing; lateral spacing; lateral length; inlet head; inside diameter	m, m, m, m, mm
c_v, x	Emitter coefficient of variation; flow exponent	[–]
r, a, u	Bulb radius; aspect ratio; upper semi-axis ratio	mm, [–], [–]
d, P, h_s, R	Nozzle diameter; pressure; pressure head; wetted radius	mm, kPa, m, m
S_s, S_l	Sprinkler spacing along and between laterals	m
u	Wind speed	$m\ s^{-1}$
AR	Mean gross application rate	$mm\ h^{-1}$
C	Hazen–Williams roughness coefficient	[–]

Symbol	Meaning	Unit
J	Hydraulic gradient	m m^{-1}
TDH, H_{man}	Total dynamic head; manometric head	m
A_f, Q_s	Irrigated area; farm supply	ha; L s^{-1}

Control value lists

Table B.1. Index lists behind the Properties controls (defaults in bold). All other controls are described with their ranges in Chapter 3.

Control	Values
Trigger (sensor suction)	−50, −75, −100, −150, −200, −250, **−300**, −350, −400, −500, −600, −700, −800, −1000, −1200, −1500, −2000, −3000, −5000, −10000 cm
Sensor depth	5, 10, 15, …, 100 cm (20 values), default **20** cm
Nitrogen rate	10, 25, 40, 55, 70, 85, 100, 115, 130, 150
Mulch storage capacity (CAP)	0.5, 1, 2, 4 mm
Rainfall factor	0.5 to 1.4
Temperature offset	−1 to +3.5 °C
Mulch strength (vapour-flux factor)	0.1 to 1.0 (1.0 = bare soil)
Structural consolidation rate	0.001, 0.0025, 0.005, 0.01, 0.02 d^{-1}
Crop	the simulated cereal/oilseed rotation and the 48 library crops of Table 21.2

Table B.2. Mulch materials (thermal weight / storage capacity in mm): husk 0.5 / 2, straw 0.4 / 4, chips 0.3 / 2, sawdust 0.6 / 1, grass 0.7 / 4, plastic 0.1 / 0.5.

Simulation data sets

The annual meta-model (Engine 1) is built on the following Daisy 7.1 run sets of the study (Chapter 5, Table 5.1): E2, E3, E4, E5 (300 runs), E7, E11, E13 and E14. The study as a whole comprises 10,190 simulation runs in 17 experiments, available from the project repository (`Results/` and `Software/` folders). The 20-year weather record (1980–1999) is the baseline of the study; the projections to 2100 are analogue-year scenarios (Chapter 4).

Provenance ledger of the numbered equations

All 134 numbered equations are labelled in the text and listed here: 38 **[Literature]**, 18 **[Data-driven]** and 78 **[Engineered]**. A **[Literature]** label means the relation is the published one; it does not mean that SWAFL has been validated in the field. The surface, drip and sprinkler design relations marked in Paper 3 were tested against published worked examples; nothing has been tested against field measurements.

Eq.	Label	Basis
4.1	Engineered	calendar bookkeeping and choice of analogue year; no physical law
4.2	Engineered	calendar bookkeeping and choice of analogue year; no physical law
5.1	Literature	van Genuchten (1980)
5.2	Literature	Mualem (1976) with van Genuchten (1980)
5.3	Literature	standard definition of plant-available water between field capacity and wilting point
5.4	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.5	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.6	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.7	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.8	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.9	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.10	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.11	Data-driven	set-specific corrections from the Daisy run sets; clip bounds engineered
5.12	Engineered	closure constraint and bounds chosen for the workbench
5.13	Engineered	canopy curves, parameters chosen for the workbench; not a measured leaf-area record
5.14	Engineered	canopy curves, parameters chosen for the workbench; not a measured leaf-area record
5.15	Engineered	stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen

Eq.	Label	Basis
5.16	Engineered	stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen
5.17	Engineered	tipping-bucket redistribution with a chosen coefficient
5.18	Engineered	rooting rules with constants chosen for the workbench
5.19	Engineered	rooting rules with constants chosen for the workbench
5.20	Engineered	rooting rules with constants chosen for the workbench
5.21	Engineered	demand partitioning with chosen weights and a calibration factor
5.22	Engineered	demand partitioning with chosen weights and a calibration factor
5.23	Engineered	demand partitioning with chosen weights and a calibration factor
5.24	Data-driven	scaling of the daily series to the Engine 1 twenty-year means
5.25	Literature	water-balance identity
5.26	Engineered	atmosphere factor with chosen constants
5.27	Literature	readily available water as in FAO-56 practice, with an engineered clip
6.1	Data-driven	counts over the simulated daily series
6.2	Data-driven	counts over the simulated daily series
6.3	Literature	inverse of the van Genuchten curve, solved by bisection
6.4	Literature	same relation as Eq. (5.27) without the clip
6.5	Data-driven	ratio of Daisy run-set E2 yields
7.1	Literature	inverse of Eq. (5.3); the 30 cm floor is engineered
7.2	Engineered	series-resistance chain after van den Honert (1948); resistance values chosen for the workbench
7.3	Engineered	series-resistance chain after van den Honert (1948); resistance values chosen for the workbench
7.4	Engineered	series-resistance chain after van den Honert (1948); resistance values chosen for the workbench
7.5	Literature	Nobel (2009)
7.6	Engineered	indicator defined for the workbench
8.1	Engineered	exponential relaxation with chosen constants
8.2	Engineered	porosity relation with chosen coefficients
8.3	Literature	van Genuchten (1980)
8.4	Literature	Young–Laplace (Jurin) capillary law
8.5	Data-driven	differences of Daisy run-set E5 results
8.6	Data-driven	ordinary least squares applied to Daisy results
8.7	Engineered	diagnostic defined for the workbench
9.1	Data-driven	differences and ratios of Daisy run-set E14 and E11 results
9.2	Engineered	two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench
9.3	Engineered	two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench
9.4	Engineered	drying-front geometry chosen for drawing
9.5	Engineered	drying-front geometry chosen for drawing

Eq.	Label	Basis
9.6	Data-driven	differences and ratios of Daisy run-set E14 and E11 results
9.7	Data-driven	differences and ratios of Daisy run-set E14 and E11 results
9.8	Engineered	index defined for the workbench
10.1	Engineered	drawing rule
10.2	Data-driven	one-step differences of Daisy results
11.1	Literature	Saxton and Rawls (2006), implemented as written
11.2	Literature	Saxton and Rawls (2006), implemented as written
11.3	Engineered	rational-polynomial approximation of a Rawls-type suction
11.4	Engineered	initial and wetted water contents chosen for the workbench
11.5	Literature	Green and Ampt (1911)
11.6	Literature	Kostiakov (1932)
11.7	Literature	ordinary least squares on the log–log plot
11.8	Literature	Manning (1891)
11.9	Literature	volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3)
11.10	Literature	volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3)
11.11	Literature	volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3)
11.12	Engineered	alternative advance models set by worked examples and course notes
11.13	Engineered	alternative advance models set by worked examples and course notes
11.15	Literature	inversion of the Kostiakov law
11.16	Engineered	cut-off rules chosen as design options
11.17	Literature	standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997)
11.18	Literature	standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997)
11.19	Literature	standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997)
11.20	Engineered	hydrograph shapes chosen for drawing; volume conserving
11.21	Engineered	hydrograph shapes chosen for drawing; volume conserving
11.22	Engineered	search limit; the furrow stream limit follows Criddle et al. (1956)
11.23	Engineered	optimisation score and seasonal bookkeeping chosen for the workbench
11.24	Engineered	optimisation score and seasonal bookkeeping chosen for the workbench
12.1	Engineered	depth and demand rules chosen for the workbench
12.2	Engineered	depth and demand rules chosen for the workbench
12.3	Literature	Hazen and Williams (1905)
12.4	Literature	Keller and Karmeli (1974) emitter law
12.5	Engineered	pressure-compensating emitter model with chosen constants

Eq.	Label	Basis
12.6	Literature	definitions of mean and minimum discharge
12.7	Literature	Keller and Karmeli (1974)
12.8	Engineered	combined spread chosen to drive the bulb quantiles
12.9	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.10	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.11	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.12	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.13	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.14	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.15	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.16	Engineered	five-quantile delivery and run-time rules chosen for the workbench
12.17	Literature	definition of the requirement over the area served by one emitter
12.18	Engineered	five-quantile delivery and run-time rules chosen for the workbench
12.19	Engineered	five-quantile delivery and run-time rules chosen for the workbench
12.20	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
12.21	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
12.22	Engineered	Christiansen-type reduction with a chosen factor 0.4
12.23	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
12.24	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
12.25	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
12.26	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
13.1	Engineered	depression storage reduction with slope
13.2	Literature	Torricelli orifice equation
13.3	Engineered	range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen
13.4	Engineered	pattern and wind shapes chosen to reproduce published qualitative patterns
13.5	Engineered	pattern and wind shapes chosen to reproduce published qualitative patterns
13.6	Literature	overlap by superposition of single-sprinkler patterns

Eq.	Label	Basis
13.7	Literature	Christiansen (1942)
13.8	Engineered	loss and ponding rules chosen for the workbench
13.9	Engineered	loss and ponding rules chosen for the workbench
13.10	Engineered	loss and ponding rules chosen for the workbench
13.11	Literature	efficiency and distribution uniformity of Merriam and Keller (1978)
13.12	Engineered	run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified
13.13	Engineered	run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified
13.14	Engineered	run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified
13.15	Engineered	run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified
13.16	Engineered	run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified
13.17	Literature	Blasius friction law for smooth pipe
13.18	Literature	20 % rule of Keller and Bliesner (1990)
13.19	Engineered	head and sets bookkeeping
13.20	Literature	hydraulic power definition
13.21	Engineered	head and sets bookkeeping
13.22	Literature	Christiansen reduction factor with Hazen–Williams exponent
13.23	Engineered	head and sets bookkeeping
14.1	Engineered	bookkeeping of Engine 2 series
14.2	Engineered	bookkeeping of Engine 2 series
14.3	Literature	water-balance identity
14.4	Engineered	bookkeeping of Engine 2 series
14.5	Engineered	bookkeeping of Engine 2 series

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Equation reference

This appendix prints every numbered equation of the program in full, in the order of the chapters, with its provenance and the layers of the program that draw on it. The equations added in version 1.1 (Eqs. 5.12a, 6.4a and those of Chapters 14, 18–23) are printed in their chapters.

Eq. 4.1 — dates; used by: Surface irrigation design

$$k_{\text{start}} = 365\,a + d_{\text{oy}}, \qquad \text{date}(k) = 1\ \text{Jan of year}\ \left(Y_0 + \lfloor k/365 \rfloor - a\right) + \left(k - 365\lfloor k/365 \rfloor\right)\ \text{days}, \tag{4.1}$$

Provenance (Eq. 4.1): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

Eq. 4.2 — dates

$$a = \begin{cases} \operatorname{argsort}_y \left(P_{\text{thr},y}\right)_{[10]} & \text{typical (median, 11th smallest of 20)} \\ \arg\min_y P_{\text{thr},y} & \text{dry} \\ \arg\max_y P_{\text{thr},y} & \text{wet} \\ Y_0 - 1980 & 1980 \le Y_0 \le 1999 \end{cases} \tag{4.2}$$

Provenance (Eq. 4.2): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

Eq. 5.1 — engine; used by: Full water balance, Nitrogen budget, Reliability over 20 seasons, Soil, plant and air, Surface irrigation design, Tillage and structure

$$S_e(h) = \Big[1 + (\alpha|h|)^n\Big]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\,S_e(h) \tag{5.1}$$

Provenance (Eq. 5.1): **[Literature]**: van Genuchten (1980).

Eq. 5.2 — engine; used by: Full water balance, Nitrogen budget, Reliability over 20 seasons, Soil, plant and air, Surface irrigation design

$$K_r(S_e) = S_e^{1/2}\Big[1 - \left(1 - S_e^{1/m}\right)^m\Big]^2. \tag{5.2}$$

Provenance (Eq. 5.2): **[Literature]**: Mualem (1976) with van Genuchten (1980).

Eq. 5.3 — engine; used by: Crop water requirement, Drip irrigation design, Field balance, Full water balance, Nitrogen budget, Reliability over 20 seasons, Soil, plant and air, Sprinkler design and pipes, Surface irrigation design, Trigger law, Virtual farm simulation

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.4 — engine; used by: Evaporation physics, Feedback loops, Field balance, Full water balance, Mulch against bare soil, Mulch control, Nitrogen budget, Regime map, Reliability over 20 seasons, Sensor and trigger, Soil hydraulic drivers, Surface irrigation design

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5 — engine; used by: Evaporation physics, Feedback loops, Field balance, Full water balance, Mulch against bare soil, Mulch control, Nitrogen budget, Reliability over 20 seasons, Sensor and trigger, Soil hydraulic drivers, Surface irrigation design

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6 — engine; used by: Evaporation physics, Feedback loops, Field balance, Full water balance, Mulch against bare soil, Mulch control, Nitrogen budget, Reliability over 20 seasons, Sensor and trigger, Soil hydraulic drivers, Surface irrigation design

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7 — engine; used by: Evaporation physics, Feedback loops, Field balance, Full water balance, Mulch against bare soil, Mulch control, Nitrogen budget, Reliability over 20 seasons, Sensor and trigger, Soil hydraulic drivers, Surface irrigation design

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8 — engine; used by: Evaporation physics, Feedback loops, Field balance, Full water balance, Mulch against bare soil, Mulch control, Nitrogen budget, Reliability over 20 seasons, Sensor and trigger, Soil hydraulic drivers, Surface irrigation design

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9 — engine; used by: Evaporation physics, Feedback loops, Field balance, Full water balance, Mulch against bare soil, Mulch control, Nitrogen budget, Reliability over 20 seasons, Sensor and trigger, Soil hydraulic drivers, Surface irrigation design

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10 — engine; used by: Evaporation physics, Feedback loops, Field balance, Full water balance, Mulch against bare soil, Mulch control, Nitrogen budget, Reliability over 20 seasons, Sensor and trigger, Soil hydraulic drivers, Surface irrigation design

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11 — engine; used by: Feedback loops, Field balance, Full water balance, Nitrogen budget, Reliability over 20 seasons, Sensor and trigger, Soil hydraulic drivers, Surface irrigation design

$$Q \leftarrow Q\,\text{clip}\Big(\frac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_\kappa \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12 — engine; used by: Feedback loops, Field balance, Full water balance, Nitrogen budget, Reliability over 20 seasons, Sensor and trigger, Soil hydraulic drivers, Surface irrigation design

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 5.13 — engine; used by: Field balance

$$c_{\text{spring}}(d) = \frac{1}{1 + \exp(-(d - d_{\text{mid}})/s)}, \quad s = 13\ (\text{vegetable } 8), \tag{5.13}$$

Provenance (Eq. 5.13): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

Eq. 5.14 — engine; used by: Field balance

$$c_{\text{winter}}(d) = \begin{cases} 0.08 + 0.12\ \min\{1, (d - d_{\text{sow}})/60\} & d < d_{\text{pk}} - 90 \\ 0.2 + \dfrac{0.8}{1 + \exp(-(d - d_{\text{pk}} + 45)/12)} & \text{otherwise} \end{cases} \tag{5.14}$$

Provenance (Eq. 5.14): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

Eq. 5.15 — engine; used by: Field balance, Profile evolution

$$R = \min\{70,\ \mu_0\,\gamma_y\,(-\ln U)\}, \qquad U \sim \text{Uniform}(0, 1), \tag{5.15}$$

Provenance (Eq. 5.15): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

Eq. 5.16 — engine; used by: Field balance

$$I_c = \mu_0 \ln\frac{1}{r}, \qquad r = \frac{\max\{40,\ P - P_{\text{other}}\}}{P}, \qquad T_f = \max\{0,\ R - I_c\}. \tag{5.16}$$

Provenance (Eq. 5.16): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

Eq. 5.17 — engine; used by: Field balance, Profile evolution

$$q_i = 0.08\,(w_{i+1} - w_i), \qquad w_i \leftarrow w_i + q_i,\ \ w_{i+1} \leftarrow w_{i+1} - q_i, \tag{5.17}$$

Provenance (Eq. 5.17): **[Engineered]**: tipping-bucket redistribution with a chosen coefficient.

Eq. 5.18 — engine; used by: Field balance, Root structure

$$R_{\max} = \text{clip}\Big(R_c\,(0.85{+}0.3\,\text{sand})\big[1{+}0.12\log_{10}(|h_T|/300)\big]\big[1{-}0.18\,\tfrac{N-100}{100}\big];\ 25,\ 100\Big)\ \text{cm}, \tag{5.18}$$

Provenance (Eq. 5.18): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.19 — engine; used by: Field balance, Root structure

$$R_d \leftarrow \max\Big\{R_d,\ \min\big\{R_{\max}\,\text{clip}\big(\tfrac{c-0.04}{0.96};0,1\big)^{0.8},\ R_{\max},\ R_d + 1.7\,(0.35 + \bar{s}_{\text{wet}})\,\kappa\big\}\Big\}, \tag{5.19}$$

Provenance (Eq. 5.19): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.20 — engine; used by: Field balance, Root structure

$$\rho_i \propto \exp\Big(-\frac{z_i}{\lambda R_d + 8}\Big)(0.45 + 0.9\,s_i)\Big[1 + 3\,e^{-((z_i - z_d)/12)^2}\Big]_{\text{drip only}}, \quad z_i \le R_d, \qquad \sum_i \rho_i = 1, \tag{5.20}$$

Provenance (Eq. 5.20): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.21 — engine; used by: Field balance, Root structure

$$E_d = E_{\text{yr}}\,\frac{w(\text{doy})}{\sum_{k=0}^{364} w(k)}\,(1 - 0.3\,c), \qquad w(\text{doy}) = 0.35 + 0.65 \cdot \tfrac{1}{2}\Big[1 + \sin\tfrac{2\pi(\text{doy}-100)}{365}\Big], \tag{5.21}$$

Provenance (Eq. 5.21): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.22 — engine; used by: Field balance, Root structure

$$T_d = k_t\,T_{\text{yr}}\,\frac{c(d)}{\bar{C}}, \qquad \bar{C} = \tfrac{1}{20}\sum_{20\ \text{yr}} c, \tag{5.22}$$

Provenance (Eq. 5.22): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.23 — engine; used by: Field balance, Root structure

$$q_i^T = T_d\,\frac{\rho_i\,a_i}{\sum_j \rho_j a_j}, \qquad a_i = \text{clip}\Big(\frac{s_i - 0.15}{0.6};\ 0,\ 1\Big), \tag{5.23}$$

Provenance (Eq. 5.23): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.24 — engine; used by: Field balance

$$S_P = \frac{P}{\overline{P_{\text{sim}}}}, \quad S_{\text{thr}} = \frac{P - P_{\text{other}}}{\overline{T_{f,\text{sim}}}}, \quad S_E = \frac{E}{\overline{E_{\text{sim}}}}, \quad S_T = \frac{T}{\overline{T_{\text{sim}}}}, \quad S_D = \frac{D}{\overline{D_{\text{sim}}}}, \tag{5.24}$$

Provenance (Eq. 5.24): **[Data-driven]**: scaling of the daily series to the Engine 1 twenty-year means.

Eq. 5.25 — engine; used by: Field balance, Soil, plant and air

$$\Delta S_d = T_f + I - E - T - D. \tag{5.25}$$

Provenance (Eq. 5.25): **[Literature]**: water-balance identity.

Eq. 5.26 — engine; used by: Crop water requirement, Evaporation physics, Field balance

$$E_{0,\text{bare}} = 4.2\,(1{+}0.07\,\Delta T)\,\frac{0.35 + 0.65\cdot\frac{1}{2}[1+\sin(2\pi(\text{doy}-100)/365)]}{0.675}\,\text{clip}(1{-}0.12(\rho{-}1);\,0.85,\,1.12)\sqrt{\gamma_y} \tag{5.26}$$

Provenance (Eq. 5.26): **[Engineered]**: atmosphere factor with chosen constants.

Eq. 5.27 — engine; used by: Crop water requirement, Drip irrigation design, Sprinkler design and pipes, Surface irrigation design, Trigger law, Virtual farm simulation

$$Z_n = \text{clip}\big(f_T \cdot \text{AWC} \cdot R_{\max};\,25,\,150\big)\ \text{mm} \qquad (\text{surface; drip and sprinkler use their own limits}) \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 6.1 — control; used by: Sensor and trigger

$$N_y(h_T) = \#\big\{\, d \in \text{growing season of year } y : h_d > h_T \,\big\} \quad (183 \text{ days from 1 April}), \tag{6.1}$$

Provenance (Eq. 6.1): **[Data-driven]**: counts over the simulated daily series.

Eq. 6.2 — control; used by: Sensor and trigger

$$X = 100\Big(1 - \frac{\bar{N}_{\text{irr}}}{\bar{N}_{\text{rainfed}}}\Big), \qquad \bar{N} = \tfrac{1}{20}\sum_y N_y. \tag{6.2}$$

Provenance (Eq. 6.2): **[Data-driven]**: counts over the simulated daily series.

Eq. 6.3 — control; used by: Trigger law

$$h(f) = \Big\{\, h : f(h) = f \,\Big\}, \tag{6.3}$$

Provenance (Eq. 6.3): **[Literature]**: inverse of the van Genuchten curve, solved by bisection.

Eq. 6.4 — control; used by: Trigger law

$$Z_n = f \cdot \text{AWC} \cdot R_{\max} \quad [\text{mm}] \tag{6.4}$$

Provenance (Eq. 6.4): **[Literature]**: same relation as Eq. (5.27) without the clip.

Eq. 6.5 — control; used by: Trigger law

$$Y_{\text{rel}} = \frac{Y(h_T)}{\max_h\, Y(h)}, \tag{6.5}$$

Provenance (Eq. 6.5): **[Data-driven]**: ratio of Daisy run-set E2 yields.

Eq. 7.1 — field; used by: Soil, plant and air

$$h_i = \max\{30,\ h(f_i)\}, \qquad f_i = 1 - s_i, \qquad \psi_{\text{soil},i} = -h_i \cdot 9.807 \times 10^{-5}\ \text{MPa cm}^{-1}. \tag{7.1}$$

Provenance (Eq. 7.1): **[Literature]**: inverse of Eq. (5.3); the 30 cm floor is engineered.

Eq. 7.2 — field; used by: Soil, plant and air

$$R_{\text{soil}} = 0.03\ \max\{K_r,\,0.002\}^{-0.55},\quad R_{\text{root}} = 0.10,\quad R_{\text{xyl}} = 0.05,\quad R_{\text{leaf}} = 0.08 \tag{7.2}$$
$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}},\qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$
$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

Eq. 7.3 — field; used by: Soil, plant and air

$$R_{\text{soil}} = 0.03\ \max\{K_r,\,0.002\}^{-0.55},\quad R_{\text{root}} = 0.10,\quad R_{\text{xyl}} = 0.05,\quad R_{\text{leaf}} = 0.08 \tag{7.2}$$
$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}},\qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$
$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

Eq. 7.4 — field; used by: Soil, plant and air

$$R_{\text{soil}} = 0.03\ \max\{K_r,\,0.002\}^{-0.55},\quad R_{\text{root}} = 0.10,\quad R_{\text{xyl}} = 0.05,\quad R_{\text{leaf}} = 0.08 \tag{7.2}$$
$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}},\qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$
$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

Eq. 7.5 — field; used by: Soil, plant and air

$$\psi_{\text{air}} = \frac{R\,T}{V_w}\,\ln \text{RH}\ \approx\ 135.5\,\ln \text{RH}\ \ \text{MPa} \tag{7.5}$$

Provenance (Eq. 7.5): **[Literature]**: Nobel (2009).

Eq. 7.6 — field; used by: Profile evolution

$$\text{share} = \frac{\sum_{i:\ z_i \le 60\,\text{cm},\ h_i > h_T} \Delta z_i}{\sum_{i:\ z_i \le 60\,\text{cm}} \Delta z_i}, \tag{7.6}$$

Provenance (Eq. 7.6): **[Engineered]**: indicator defined for the workbench.

Eq. 8.1 — structure; used by: Field balance, Soil hydraulic drivers, Tillage and structure

$$p(t) = p_c + (p_0 - p_c)\,e^{-\tau_c t},\qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

Eq. 8.2 — structure; used by: Tillage and structure

$$\phi = \text{clip}\big(0.36 + 0.0016\,\text{clay} + 0.0009\,\text{silt};\ 0.34,\ 0.50\big),\qquad \theta_r = 0.01 + 0.0006\,\text{clay}, \tag{8.2}$$

Provenance (Eq. 8.2): **[Engineered]**: porosity relation with chosen coefficients.

Eq. 8.3 — structure; used by: Evaporation physics, Tillage and structure

$$\theta(h) = \theta_r + (\phi - \theta_r)\,S_e(h). \tag{8.3}$$

Provenance (Eq. 8.3): **[Literature]**: van Genuchten (1980).

Eq. 8.4 — structure; used by: Evaporation physics, Tillage and structure

$$r_c = \frac{2\,\sigma\cos\gamma}{\rho_w\,g\,h}\ \approx\ \frac{0.149\ \text{cm}^2}{h}\quad (\text{or } r_c[\text{mm}] \approx 1.49/h[\text{cm}]), \tag{8.4}$$

Provenance (Eq. 8.4): **[Literature]**: Young–Laplace (Jurin) capillary law.

Eq. 8.5 — structure; used by: Soil hydraulic drivers

$$\Delta_{\text{till}}(s) = D_s(\text{conv}, \tau^*) - D_s(\text{notill}, \tau^*), \qquad \Delta_{\text{cons}}(s) = D_s(\text{till}^*, \tau_{\text{fast}}) - D_s(\text{till}^*, \tau_{\text{slow}}), \tag{8.5}$$

Provenance (Eq. 8.5): **[Data-driven]**: differences of Daisy run-set E5 results.

Eq. 8.6 — structure; used by: Soil hydraulic drivers

$$\hat{\Delta} = a + b\,c, \qquad b = \frac{\sum_s (c_s - \bar{c})(\Delta_s - \bar{\Delta})}{\sum_s (c_s - \bar{c})^2}, \quad a = \bar{\Delta} - b\,\bar{c}. \tag{8.6}$$

Provenance (Eq. 8.6): **[Data-driven]**: ordinary least squares applied to Daisy results.

Eq. 8.7 — structure; used by: Root structure

$$z_p = k_p\,\Delta z, \qquad k_p = \min\Big\{k : \frac{\sum_{i \le k} U_i}{\sum_i U_i} \ge p\Big\}, \qquad U_i = \sum_{d \in \text{year}} u_i(d), \quad p = 0.5,\ 0.9, \tag{8.7}$$

Provenance (Eq. 8.7): **[Engineered]**: diagnostic defined for the workbench.

Eq. 9.1 — mulch; used by: Mulch against bare soil, Mulch control

$$\Delta E = E_{1.0} - E_{\varphi}, \qquad \Delta I = I_{1.0} - I_{\varphi}, \qquad \Delta D = D_{\varphi} - D_{1.0}, \tag{9.1}$$

Provenance (Eq. 9.1): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

Eq. 9.2 — mulch; used by: Evaporation physics

$$t_1 = \frac{Q_1 - q_0}{e_0}, \qquad e(t) = \begin{cases} e_0, & t \le t_1 \\ \dfrac{e_0}{\sqrt{1 + (t - t_1)/\tau}}, & t > t_1 \end{cases} \tag{9.2}$$

Provenance (Eq. 9.2): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

Eq. 9.3 — mulch; used by: Evaporation physics

$$Q(t) = q_0 + \begin{cases} e_0\,t, & t \le t_1 \\ (Q_1 - q_0) + 2\,e_0\,\tau\Big(\sqrt{1 + (t - t_1)/\tau} - 1\Big), & t > t_1 \end{cases} \tag{9.3}$$

Provenance (Eq. 9.3): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

Eq. 9.4 — mulch; used by: Evaporation physics

$$\ell(t) = \min\Big\{60,\ \frac{Q(t) - Q_1}{\Delta\theta}\Big\}\ \text{mm} \quad (Q > Q_1), \tag{9.4}$$

Provenance (Eq. 9.4): **[Engineered]**: drying-front geometry chosen for drawing.

Eq. 9.5 — mulch; used by: Evaporation physics

$$\log_{10} h(z) = \log_{10} h_s + \big(\log_{10} h_i - \log_{10} h_s\big)\big(1 - e^{-(z-\ell)/\lambda}\big) \quad (z \ge \ell), \qquad h = 10^5\ \text{cm for } z < \ell. \tag{9.5}$$

Provenance (Eq. 9.5): **[Engineered]**: drying-front geometry chosen for drawing.

Eq. 9.6 — mulch; used by: Fate of saved water

$$\begin{aligned} \Delta E(\rho) &= E_{1.0}(\rho) - E_{\varphi^*}(\rho), \quad \Delta I(\rho) = I_{1.0}(\rho) - I_{\varphi^*}(\rho), \\ \Delta D(\rho) &= D_{\varphi^*}(\rho) - D_{1.0}(\rho), \quad \Delta T(\rho) = T_{\varphi^*}(\rho) - T_{1.0}(\rho), \end{aligned} \tag{9.6}$$

Provenance (Eq. 9.6): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

Eq. 9.7 — mulch; used by: Fate of saved water

$$f_I = \frac{\max\{\Delta I, 0\}}{\max\{\Delta E, 0.1\}},\; f_D = \frac{\max\{\Delta D, 0\}}{\max\{\Delta E, 0.1\}},\; f_T = \frac{\max\{\Delta T, 0\}}{\max\{\Delta E, 0.1\}},\; f_{\text{rest}} = \max\{0,\, 1 - f_I - f_D - f_T\}. \tag{9.7}$$

Provenance (Eq. 9.7): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

Eq. 9.8 — mulch; used by: Regime map

$$\mathcal{J} = \frac{E - D}{\max\{1,\; E + D\}}, \qquad \text{regime} = \begin{cases} \text{evaporation-dominated} & \mathcal{J} > +0.05 \\ \text{drainage-dominated} & \mathcal{J} < -0.05 \\ \text{on the boundary} & |\mathcal{J}| \le 0.05 \end{cases} \tag{9.8}$$

Provenance (Eq. 9.8): **[Engineered]**: index defined for the workbench.

Eq. 10.1 — loops; used by: Feedback loops

$$w_k = 5 + \text{clip}(\mu_k;\, 0,\, 4). \tag{10.1}$$

Provenance (Eq. 10.1): **[Engineered]**: drawing rule.

Eq. 10.2 — loops; used by: Feedback loops

$$\kappa_N = \frac{N_L^{(\text{rain step})} - N_L}{D^{(\text{rain step})} - D} \quad [\text{kg N ha}^{-1}\ \text{mm}^{-1}], \tag{10.2}$$

Provenance (Eq. 10.2): **[Data-driven]**: one-step differences of Daisy results.

Eq. 11.1 — surface; used by: Surface irrigation design

$$\begin{aligned} \theta_{1500,t} &= -0.024\,S + 0.487\,C + 0.006\,\text{OM} + 0.005\,S\,\text{OM} - 0.013\,C\,\text{OM} + 0.068\,S\,C + 0.031, \\ \theta_{1500} &= \theta_{1500,t} + (0.14\,\theta_{1500,t} - 0.02), \\ \theta_{33,t} &= -0.251\,S + 0.195\,C + 0.011\,\text{OM} + 0.006\,S\,\text{OM} - 0.027\,C\,\text{OM} + 0.452\,S\,C + 0.299, \\ \theta_{33} &= \theta_{33,t} + (1.283\,\theta_{33,t}^2 - 0.374\,\theta_{33,t} - 0.015), \end{aligned} \tag{11.1}$$

Provenance (Eq. 11.1): **[Literature]**: Saxton and Rawls (2006), implemented as written.

Eq. 11.2 — surface; used by: Surface irrigation design

$$\begin{aligned} \theta_{(S-33),t} &= 0.278\,S + 0.034\,C + 0.022\,\text{OM} - 0.018\,S\,\text{OM} - 0.027\,C\,\text{OM} - 0.584\,S\,C + 0.078, \\ \theta_{(S-33)} &= \theta_{(S-33),t} + (0.636\,\theta_{(S-33),t} - 0.107), \qquad \theta_s = \theta_{33} + \theta_{(S-33)} - 0.097\,S + 0.043, \\ B &= \frac{\ln 1500 - \ln 33}{\ln \theta_{33} - \ln \theta_{1500}}, \qquad \lambda = \frac{1}{B}, \qquad K_s = 1930\,(\theta_s - \theta_{33})^{\,3-\lambda}. \end{aligned} \tag{11.2}$$

Provenance (Eq. 11.2): **[Literature]**: Saxton and Rawls (2006), implemented as written.

Eq. 11.3 — surface; used by: Drip irrigation design, Surface irrigation design

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

Eq. 11.4 — surface; used by: Surface irrigation design

$$\theta_0 = \theta_{fc} - f_T\,(\theta_{fc} - \theta_{wp}), \qquad \Delta\theta = \max\{0.05,\ 0.95\,\theta_s - \theta_0\}. \tag{11.4}$$

Provenance (Eq. 11.4): **[Engineered]**: initial and wetted water contents chosen for the workbench.

Eq. 11.5 — surface; used by: Surface irrigation design

$$F - \psi_f\,\Delta\theta\,\ln\!\Big(1 + \frac{F}{\psi_f\,\Delta\theta}\Big) = K_s\,\frac{t}{60} \quad (t \text{ in min}), \tag{11.5}$$

Provenance (Eq. 11.5): **[Literature]**: Green and Ampt (1911).

Eq. 11.6 — surface; used by: Sprinkler design and pipes, Surface irrigation design

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 11.7 — surface; used by: Surface irrigation design

$$A = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \ln k = \bar{y} - A\,\bar{x}, \qquad x_i = \ln \tau_i,\ y_i = \ln\!\big(F(\tau_i)/1000\big), \tag{11.7}$$

Provenance (Eq. 11.7): **[Literature]**: ordinary least squares on the log–log plot.

Eq. 11.8 — surface; used by: Surface irrigation design

$$Q = \frac{1}{n}\,A\,R_h^{2/3}\,S^{1/2}, \qquad R_h = \frac{A}{P}. \tag{11.8}$$

Provenance (Eq. 11.8): **[Literature]**: Manning (1891).

Eq. 11.9 — surface; used by: Surface irrigation design

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^{A}\,w\,x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^{A}\,w}. \tag{11.9}$$

Provenance (Eq. 11.9): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.10 — surface; used by: Surface irrigation design

$$\sigma_z = \frac{A + r\,(1 - A) + 1}{(1 + A)(1 + r)}, \tag{11.10}$$

Provenance (Eq. 11.10): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.11 — surface; used by: Surface irrigation design

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

Provenance (Eq. 11.11): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.12 — surface; used by: Surface irrigation design

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

Provenance (Eq. 11.12): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.13 — surface; used by: Surface irrigation design

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\,k\,w\,L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

Provenance (Eq. 11.13): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.15 — surface; used by: Surface irrigation design

$$t_{st} = \Big(\frac{Z_n}{k}\Big)^{1/A}. \tag{11.15}$$

Provenance (Eq. 11.15): **[Literature]**: inversion of the Kostiakov law.

Eq. 11.16 — surface; used by: Surface irrigation design, Virtual farm simulation

$$t_{co} = \begin{cases} t_L + t_{st} & \text{“}E_s = 1\text{”: the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{“85 \% of L”} \\ t_{co}^{\text{set}} & \text{“Manual”} \end{cases} \tag{11.16}$$

Provenance (Eq. 11.16): **[Engineered]**: cut-off rules chosen as design options.

Eq. 11.17 — surface; used by: Surface irrigation design

$$Z(x) = k\,\big[t_{co} - t_x(x)\big]^A, \quad Z_0 = k\,t_{co}^A \text{ (inlet)}, \qquad Z_L = k\,(t_{co} - t_L)^A \text{ (tail)}. \tag{11.17}$$

Provenance (Eq. 11.17): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.18 — surface; used by: Surface irrigation design, Virtual farm simulation

$$\begin{aligned} &V_{\text{app}} = Q\,t_{co} \quad \big(\text{with cutback } c_b < 1:\ Q\,t_L + c_b\,Q\,(t_{co} - t_L)\big), \quad V_{\text{req}} = Z_n\,L\,w, \\ &V_{\text{inf}} = w\int_0^L Z\,dx, && V_{\text{st}} = w\int_0^L \min(Z, Z_n)\,dx, \\ &V_{\text{dp}} = w\int_0^L \max(Z - Z_n, 0)\,dx, && V_{\text{df}} = w\int_0^L \max(Z_n - Z, 0)\,dx, \\ &V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}},\ 0). \end{aligned} \tag{11.18}$$

Provenance (Eq. 11.18): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.19 — surface; used by: Surface irrigation design, Virtual farm simulation

$$E_a = \frac{V_{\text{st}}}{V_{\text{app}}}, \quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}}, \quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}}, \quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}}, \quad \text{CU} = 100\Big(1 - \frac{\frac{1}{N+1}\sum_{i=0}^{N} |Z_i - \bar{Z}|}{\bar{Z}}\Big)\ \%. \tag{11.19}$$

Provenance (Eq. 11.19): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.20 — surface; used by: Surface irrigation design

$$V_c(t) = w\sum_i k\,\big[\min(t, t_{co}) - \bar{t}_{x,i}\big]_+^A\,\Delta x\ \cdot s_c, \qquad s_c = \frac{V_{\text{inf}}}{V_c(t_{co})}, \qquad I(t) = \frac{dV_c}{dt}, \tag{11.20}$$

Provenance (Eq. 11.20): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

Eq. 11.21 — surface; used by: Surface irrigation design

$$R(t) = \frac{2V_s}{t_r}\Big(1 - \frac{t - t_{co}}{t_r}\Big), \quad t_{co} < t < t_{co} + t_r. \tag{11.21}$$

Provenance (Eq. 11.21): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

Eq. 11.22 — surface; used by: Surface irrigation design

$$q_{\max} = \min\Big\{ q_{\max}^{\text{sys}},\ \frac{0.04}{S} \cdot \frac{1000}{60} \Big\}\ \mathrm{L\,s^{-1}} \quad (S \text{ in } \%), \tag{11.22}$$

Provenance (Eq. 11.22): **[Engineered]**: search limit; the furrow stream limit follows Criddle et al. (1956).

Eq. 11.23 — surface; used by: Surface irrigation design

$$\text{score} = \min(E_a, 1) - 3\max(0,\ 0.8 - E_s) - 3\,[c_b < 1]\ \max(0,\ 0.05 - \text{RO}) - 5\max(0,\ E_a - 1). \tag{11.23}$$

Provenance (Eq. 11.23): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

Eq. 11.24 — surface; used by: Surface irrigation design

$$\begin{aligned} n_{\text{ev}} &= \sum I \,/\, Z_n, \qquad Z_g = \frac{V_{\text{app}}}{L\,w} \times 1000\ \ (\text{mm}), \qquad Z_{\text{season}} = n_{\text{ev}}\, Z_g, \\ N_{0M} &= \frac{Q_s}{q}, \qquad N_{\text{units}} = \frac{W_f}{w}, \quad N_{\text{sets}} = \max\Big(1, \frac{N_{\text{units}}}{N_{0M}}\Big), \qquad T_{\text{tot}} = \frac{N_{\text{sets}}\, t_{co}}{60}\ \ (\text{h}), \end{aligned} \tag{11.24}$$

Provenance (Eq. 11.24): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

Eq. 12.1 — drip; used by: Drip irrigation design

$$Z_n = \min\big\{ Z_n^{\text{DSS}},\ \max(4,\ \text{ET}_{\text{pk}}\,\Delta t) \big\}\ \ \text{mm}, \tag{12.1}$$

Provenance (Eq. 12.1): **[Engineered]**: depth and demand rules chosen for the workbench.

Eq. 12.2 — drip; used by: Drip irrigation design

$$\text{ET}_{\text{pk}} = \text{clip}\Big\{ \max_{d}\ \tfrac{1}{5} \sum_{j=0}^{4} (E_{d+j} + T_{d+j});\ 2.5,\ 9 \Big\}. \tag{12.2}$$

Provenance (Eq. 12.2): **[Engineered]**: depth and demand rules chosen for the workbench.

Eq. 12.3 — drip; used by: Drip irrigation design, Sprinkler design and pipes

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \ge j+1} q_i\ \ (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m}. \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

Eq. 12.4 — drip; used by: Drip irrigation design

$$q_j = q\Big(\frac{h_j}{H_0}\Big)^{x} \quad (\text{non-compensating};\ h_j/H_0 \ge 0.01), \tag{12.4}$$

Provenance (Eq. 12.4): **[Literature]**: Keller and Karmeli (1974) emitter law.

Eq. 12.5 — drip; used by: Drip irrigation design

$$q_j = \begin{cases} q[1 + 0.04\,(h_j - H_0)/H_0], & h_j \ge H_{\min}, \\ q\,(\max(h_j, 0.1)/H_{\min})^{0.5}, & h_j < H_{\min}, \end{cases} \tag{12.5}$$

Provenance (Eq. 12.5): **[Engineered]**: pressure-compensating emitter model with chosen constants.

Eq. 12.6 — drip; used by: Drip irrigation design

$$\bar{q} = \frac{1}{N+1}\sum_{j=0}^{N} q_j, \qquad q_{\min} = \min_j q_j, \qquad Q_{\text{lat}} = \frac{N\,\bar{q}}{3600}\ \ (\mathrm{L\,s^{-1}}), \tag{12.6}$$

Provenance (Eq. 12.6): **[Literature]**: definitions of mean and minimum discharge.

Eq. 12.7 — drip; used by: Drip irrigation design

$$E_u = \text{clip}\Big\{\Big(1 - \frac{1.27\,c_v}{\sqrt{n_e}}\Big)\frac{q_{\min}}{\bar{q}};\ 0.3,\ 1\Big\}. \tag{12.7}$$

Provenance (Eq. 12.7): **[Literature]**: Keller and Karmeli (1974).

Eq. 12.8 — drip; used by: Drip irrigation design

$$\sigma_q = \sqrt{c_v^2 + \sigma_P^2}, \qquad \sigma_P^2 = \frac{1}{N+1}\sum_{j=0}^{N}\Big(\frac{q_j}{\bar{q}} - 1\Big)^2. \tag{12.8}$$

Provenance (Eq. 12.8): **[Engineered]**: combined spread chosen to drive the bulb quantiles.

Eq. 12.9 — drip; used by: Drip irrigation design

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

Provenance (Eq. 12.9): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.10 — drip; used by: Drip irrigation design

$$\rho(z) = r\,\sqrt{1 - \zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \ge z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.11 — drip; used by: Drip irrigation design

$$a = \text{clip}\Big\{1 + 0.6\,\log_{10}\Big(1 + \frac{K_s}{q\cdot 10^6/(2\pi\lambda^2)}\Big);\ 1,\ 1.9\Big\}, \qquad \lambda = \max(40, \psi_f), \tag{12.11}$$

Provenance (Eq. 12.11): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.12 — drip; used by: Drip irrigation design

$$V_w = \int \pi\,\rho(z)^2\,dz, \tag{12.12}$$

Provenance (Eq. 12.12): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.13 — drip; used by: Drip irrigation design

$$P_w = \text{clip}\Big\{\frac{2r}{1000\,S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

Provenance (Eq. 12.13): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.14 — drip; used by: Drip irrigation design

$$V_{\mathrm{dp}}^{(1)} = V\,\frac{\int_{z>R_{\max}} \pi\rho^2\,dz}{V_w}, \tag{12.14}$$

Provenance (Eq. 12.14): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.15 — drip; used by: Drip irrigation design

$$V_{\mathrm{ev}} = \min\!\left\{0.5\,V,\ \pi\rho(0)^2\cdot 10^{-6}\cdot 2E_0\right\}\quad (\mathrm{L},\ \text{area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.16 — drip; used by: Drip irrigation design

$$V_i = \max\!\left\{0.02\,\bar{V},\ \bar{V}\,(1+\sigma_q z_i)\right\},\qquad \bar{V} = \bar{q}\,t, \tag{12.16}$$

Provenance (Eq. 12.16): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.17 — drip; used by: Drip irrigation design

$$V_{\mathrm{req}} = Z_n\,S_e\,S_l\ \ (\mathrm{L}),\qquad V_{\mathrm{st},i} = \min\{V_{\mathrm{root},i},\ V_{\mathrm{req}}\},\qquad V_{\mathrm{ex},i} = V_{\mathrm{root},i} - V_{\mathrm{st},i}. \tag{12.17}$$

Provenance (Eq. 12.17): **[Literature]**: definition of the requirement over the area served by one emitter.

Eq. 12.18 — drip; used by: Drip irrigation design

$$t = \mathrm{clip}\!\left\{\frac{V_{\mathrm{req}}}{\bar{q}\,E_u};\ 0.05,\ 96\right\}\ \mathrm{h}. \tag{12.18}$$

Provenance (Eq. 12.18): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.19 — drip; used by: Drip irrigation design, Virtual farm simulation

$$E_a = \frac{\langle V_{\mathrm{st}}\rangle}{\langle V\rangle},\quad E_s = \frac{\langle V_{\mathrm{st}}\rangle}{V_{\mathrm{req}}},\quad \mathrm{DP} = \frac{\langle V_{\mathrm{dp}}^{(1)} + V_{\mathrm{ex}}\rangle}{\langle V\rangle},\quad \mathrm{EV} = \frac{\langle V_{\mathrm{ev}}\rangle}{\langle V\rangle},\quad Z_{\mathrm{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

Provenance (Eq. 12.19): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.20 — drip; used by: Drip irrigation design

$$W_f = \frac{10^4 A_f}{L_L},\quad N_L = \left\lfloor \frac{W_f}{S_l} \right\rfloor,\quad n_{\mathrm{zone}} = \left\lceil \frac{N_L\,Q_{\mathrm{lat}}}{Q_s} \right\rceil,\quad n_l = \left\lceil \frac{N_L}{n_{\mathrm{zone}}} \right\rceil,\quad Q_z = n_l\,Q_{\mathrm{lat}}, \tag{12.20}$$

Provenance (Eq. 12.20): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.21 — drip; used by: Drip irrigation design, Sprinkler design and pipes

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\}\ \mathrm{mm} \tag{12.21}$$

Provenance (Eq. 12.21): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.22 — drip; used by: Drip irrigation design, Sprinkler design and pipes

$$h_{f,sm} = 0.4\,\frac{10.67\,L_{sm}\,Q_z^{1.852}}{C^{1.852}D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\,L_m\,Q_z^{1.852}}{C^{1.852}D_m^{4.871}}. \tag{12.22}$$

Provenance (Eq. 12.22): **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

Eq. 12.23 — drip; used by: Drip irrigation design, Sprinkler design and pipes

$$\mathrm{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9\ \text{m}, \qquad P = \frac{\rho g\, Q_z\, \mathrm{TDH}}{\eta} = \frac{9.81\,(Q_z/1000)\,\mathrm{TDH}}{0.65}\ \text{kW}. \tag{12.23}$$

Provenance (Eq. 12.23): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.24 — drip; used by: Drip irrigation design

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\,Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\text{L s}^{-1}\,\text{ha}^{-1}), \tag{12.24}$$

Provenance (Eq. 12.24): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.25 — drip; used by: Drip irrigation design

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\,t, \qquad H_{\text{pump}} = n_{\text{ev}}\,t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\,P. \tag{12.25}$$

Provenance (Eq. 12.25): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.26 — drip; used by: Drip irrigation design, Sprinkler design and pipes

$$\text{score} = \min(E_a, 1) - 2.5\,\max(0,\ 0.9 - E_s), \tag{12.26}$$

Provenance (Eq. 12.26): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 13.1 — sprinkler; used by: Sprinkler design and pipes

$$S_d^{\text{eff}} = \frac{\max(0, S_d)}{1 + 0.25\,S} \qquad (S \text{ in \%}). \tag{13.1}$$

Provenance (Eq. 13.1): **[Engineered]**: depression storage reduction with slope.

Eq. 13.2 — sprinkler; used by: Sprinkler design and pipes

$$q = 0.95\,A_n\sqrt{2g\,h} \times 6 \times 10^4\ \ (\text{L min}^{-1}), \qquad A_n = \pi\Big(\frac{d}{2000}\Big)^2\ \text{m}^2, \tag{13.2}$$

Provenance (Eq. 13.2): **[Literature]**: Torricelli orifice equation.

Eq. 13.3 — sprinkler; used by: Sprinkler design and pipes

$$R = c_R\sqrt{d\,h}\,f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \Big[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\Big]^{0.45}, \tag{13.3}$$

Provenance (Eq. 13.3): **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

Eq. 13.4 — sprinkler; used by: Sprinkler design and pipes

$$\widehat{p}(x) = \begin{cases} 1 - x^m, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^6), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^5), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \widehat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.5 — sprinkler; used by: Sprinkler design and pipes

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.6 — sprinkler; used by: Sprinkler design and pipes

$$a(x,y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_k a_k, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

Eq. 13.7 — sprinkler; used by: Sprinkler design and pipes

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

Eq. 13.8 — sprinkler; used by: Sprinkler design and pipes

$$w_d = \frac{1}{100}\,\mathrm{clip}\Big\{(1.2{+}0.5u{+}0.04u^2)\,F\Big(\frac{P}{250}\Big)^{0.7}[1{+}0.025\,(T_a{-}20)]\Big[0.6{+}0.4\,\frac{\sin\alpha}{0.4226}\Big];\ 0.5,\ 45\Big\}. \tag{13.8}$$

Provenance (Eq. 13.8): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.9 — sprinkler; used by: Sprinkler design and pipes

$$t_p = \Big(\frac{a}{K}\Big)^{1/(A-1)}, \qquad F = \begin{cases} \sup, & T \le t_p \\ K\,T^A, & T > t_p \end{cases} \qquad R = \sup{-}F, \quad \mathrm{RO} = \max(0, R{-}S_d^{\mathrm{eff}}), \quad I = \sup{-}\mathrm{RO}. \tag{13.9}$$

Provenance (Eq. 13.9): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.10 — sprinkler; used by: Sprinkler design and pipes

$$I_c = \min\{I,\ I_{\mathrm{int}}\}, \qquad I_{\mathrm{int}} = 0.6\,\mathrm{CAP}\,[\mathrm{VFF} < 0.95] + 1.2\,\mathrm{LAI}_{\mathrm{live}}, \tag{13.10}$$

Provenance (Eq. 13.10): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.11 — sprinkler; used by: Sprinkler design and pipes, Virtual farm simulation

$$\begin{aligned} E_a &= \frac{\overline{\min(Z_k, Z_n)}}{\mathrm{gross}}, & E_s &= \frac{\overline{\min(Z_k, Z_n)}}{\overline{Z}_n}, & \mathrm{DU} &= \frac{\overline{Z}_{25}}{\overline{Z}}, \\ \mathrm{DP} &= \frac{\overline{\max(Z_k - Z_n, 0)}}{\mathrm{gross}}, & \mathrm{RO} &= \frac{\overline{\mathrm{RO}_k}}{\mathrm{gross}}, & \mathrm{WD} &= w_d + \frac{\overline{I_c}}{\mathrm{gross}}, \end{aligned} \tag{13.11}$$

Provenance (Eq. 13.11): **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

Eq. 13.12 — sprinkler; used by: Sprinkler design and pipes

$$T = \min\{T :\ \overline{Z}_{25}(T) \ge Z_n\}, \qquad T \in [1,\ 2400]\ \mathrm{min}, \tag{13.12}$$

Provenance (Eq. 13.12): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.13 — sprinkler; used by: Sprinkler design and pipes

$$k = \begin{cases} 1, & W \le 5.5 \\ 0.9, & 5.5 < W \le 13 \\ 0.8, & 13 < W \le 22 \\ 0.7, & W > 22. \end{cases} \tag{13.13}$$

Provenance (Eq. 13.13): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.14 — sprinkler; used by: Sprinkler design and pipes

$$S_s \le k\,R_w, \qquad S_l \le 1.5\,R_w, \qquad L_y = \frac{10^4 A_f}{L_x}, \quad N = \lfloor L_x / S_s \rfloor, \quad N_m = \lfloor L_y / S_l \rfloor, \tag{13.14}$$

Provenance (Eq. 13.14): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.15 — sprinkler; used by: Sprinkler design and pipes

$$t_{\text{set}} = \frac{Z_{\text{gross}}}{\text{AR}} \ \text{(h)}, \qquad n_{\text{sets/day}} = \left\lfloor \frac{h_{\text{op}}}{t_{\text{set}}} \right\rfloor, \qquad T_{\text{int}} = \max\!\left(1, \left\lfloor \frac{Z_n}{\text{ET}_{\text{pk}}} \right\rfloor\right) \text{ days}, \tag{13.15}$$

Provenance (Eq. 13.15): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.16 — sprinkler; used by: Sprinkler design and pipes

$$N_d(n) = \left\lceil \frac{N_m}{n\, n_{\text{sets/day}}} \right\rceil, \qquad Q_{\text{in}}(n) = n\, N\, q_h, \tag{13.16}$$

Provenance (Eq. 13.16): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.17 — sprinkler; used by: Sprinkler design and pipes

$$J = \frac{0.472\, Q^{1.75}}{D^{4.75}} \ \ (\text{m m}^{-1};\ Q \text{ in L h}^{-1},\ D \text{ in mm}), \tag{13.17}$$

Provenance (Eq. 13.17): **[Literature]**: Blasius friction law for smooth pipe.

Eq. 13.18 — sprinkler; used by: Sprinkler design and pipes

$$h_{f,\text{lat}} = 1.1\, F\, J(Q_L, D)\,(L_x - S_s/2) \le 0.2\, h_s, \qquad Q_L = 1000\, N\, q_h, \tag{13.18}$$

Provenance (Eq. 13.18): **[Literature]**: 20 % rule of Keller and Bliesner (1990).

Eq. 13.19 — sprinkler; used by: Sprinkler design and pipes

$$H_{\text{man}} = z_{\text{lift}} + J\, L_{\text{conv}} + J\, L_{\text{main}} + h_s + 0.75\, h_{f,\text{lat}}, \tag{13.19}$$

Provenance (Eq. 13.19): **[Engineered]**: head and sets bookkeeping.

Eq. 13.20 — sprinkler; used by: Sprinkler design and pipes

$$P_{\text{CV}} = \frac{1000\,(Q_m / 3.6\times 10^6)\, H_{\text{man}}}{75\,\eta}, \quad \eta = 0.75, \qquad P_{\text{kW}} = 0.7355\, P_{\text{CV}}, \tag{13.20}$$

Provenance (Eq. 13.20): **[Literature]**: hydraulic power definition.

Eq. 13.21 — sprinkler; used by: Sprinkler design and pipes

$$N_L = \left\lfloor \frac{10^4 A_f / L_L}{S_l} \right\rfloor, \quad n_l = \min\!\left(N_L,\ \left\lfloor \frac{Q_s}{Q_{\text{lat}}} \right\rfloor\right), \quad n_{\text{set}} = \left\lceil \frac{N_L}{n_l} \right\rceil, \quad Q_z = n_l Q_{\text{lat}}. \tag{13.21}$$

Provenance (Eq. 13.21): **[Engineered]**: head and sets bookkeeping.

Eq. 13.22 — sprinkler; used by: Sprinkler design and pipes

$$F_m = \frac{1}{2.852} + \frac{1}{2N_s} + \frac{\sqrt{0.852}}{6N_s^2}, \qquad h_{f,\mathrm{lat}} = F_m\, h_{HW}(Q_{\mathrm{lat}}, D, L_L), \tag{13.22}$$

Provenance (Eq. 13.22): **[Literature]**: Christiansen reduction factor with Hazen–Williams exponent.

Eq. 13.23 — sprinkler; used by: Sprinkler design and pipes

$$n_{\mathrm{ev}} = \frac{y_I}{Z_n}, \qquad n_{\mathrm{sp}} = \mathrm{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\mathrm{tot}} = \frac{n_{\mathrm{sp}}\, q}{60}, \qquad n_{\mathrm{set}} = \Big\lceil \frac{Q_{\mathrm{tot}}}{Q_s} \Big\rceil, \qquad t_{\mathrm{round}} = \frac{n_{\mathrm{set}}\, T}{60}\ \mathrm{h}. \tag{13.23}$$

Provenance (Eq. 13.23): **[Engineered]**: head and sets bookkeeping.

Eq. 14.1 — dashboard

$$\mathrm{Rain} = P_k, \quad \mathrm{Irr} = I_k, \quad \mathrm{Evap} = E_k, \quad \mathrm{Transp} = T_k, \quad \mathrm{Drain} = D_k \ \ (\mathrm{mm\,d^{-1}}). \tag{14.1}$$

Provenance (Eq. 14.1): **[Engineered]**: bookkeeping of Engine 2 series.

Eq. 14.2 — dashboard

$$\Sigma X = \sum_{j=1}^{N} X_{k+j}, \qquad X \in \{P, I, E, T, D\}. \tag{14.2}$$

Provenance (Eq. 14.2): **[Engineered]**: bookkeeping of Engine 2 series.

Eq. 14.3 — dashboard

$$\Delta S = P + I - E_{\mathrm{int}} - E - T - D, \tag{14.3}$$

Provenance (Eq. 14.3): **[Literature]**: water-balance identity.

Eq. 14.4 — dashboard

$$\Delta_i = x_i - x_i^{\mathrm{pin}}, \tag{14.4}$$

Provenance (Eq. 14.4): **[Engineered]**: bookkeeping of Engine 2 series.

Eq. 14.5 — dashboard

$$Y^{(y)} = \sum_{d \in y} X_d, \qquad X \in \{P, I, D, E, T\}. \tag{14.5}$$

Provenance (Eq. 14.5): **[Engineered]**: bookkeeping of Engine 2 series.